VALUATION-PARITY EQUIDISTRIBUTION, QUADRATIC PELL CAPACITY, AND SUPPORT-REDUCED PADÉ–WRONSKIAN BOUNDS FOR THE BROCARD–RAMANUJAN EQUATION

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ABSTRACT. Let $n!+1=m^2$ be the Brocard–Ramanujan Diophantine equation, and write $n!=q_n s_n^2$ with q_n square-free and $m+s_n\sqrt{q_n}=\varepsilon_n^k$ in $\mathbb{Z}[\sqrt{q_n}]$. We prove that $k\in\{1,2\}$ for all sufficiently large solutions (and $n\in\{4,5,7\}$), and that the Iwasawa invariants of $K_n=\mathbb{Q}(\sqrt{q_n})$ satisfy $\lambda_p(K_n)\geq 1$ for a positive proportion of unramified primes $p\leq n$; consequently, Greenberg's Iwasawa conjecture implies that $(4,5)$, $(5,11)$, and $(7,71)$ are the only solutions to $n!+1=m^2$. Unconditionally, we prove that the number of solutions $n\leq N$ with $k=2$ is $\ll_\varepsilon N^\varepsilon$, and that for any polynomial $P(x)\in\mathbb{Z}[x]$ of degree $r\geq 2$ and $s\in\mathbb{Z}\setminus\{0\}$, the number of solutions $n\leq N$ to $s\cdot n!=P(x)$ is $\ll_{P,s,\varepsilon} N^{6/7+\varepsilon}$ (improving to $N^{2/3+\varepsilon}$ for larger degree gaps).

1. INTRODUCTION

1.1. **Background and State of the Problem.** In 1876 and again in 1885, Brocard [9, 10] asked for all positive integer solutions (n,m) to the Diophantine equation

$$n!+1 \;=\; m^2. \tag{1.1}$$

In 1913, Ramanujan [45] independently posed the same problem and noted the three solutions $(n,m)\in\{(4,5),(5,11),(7,71)\}$. While quadratic-residue sieves have verified that no further solutions exist for $n\leq 10^9$ (Berndt and Galway [7]), $n<4\times 10^{11}$ (Matson [37]), and $n\leq 10^{15}$ (Epstein and Glickman [20]), determining whether (1.1) has only finitely many solutions remains a longstanding open problem in Diophantine number theory.

For higher-degree and shifted variants, several finiteness theorems are known. Erdős and Obláth [21] proved in 1937 that $n!=x^p\pm y^p$ $(\gcd(x,y)=1)$ and $n!\pm m!=x^p$ have only finitely many integer solutions for prime exponents $p\geq 3$, and Pollack and Shapiro [44] showed that $n!+1=x^4$ has no solutions. Conditionally, Overholt [42] observed in 1993 that the weak *abc* conjecture implies the finiteness of solutions to (1.1). Dąbrowski [16] studied $n!+A=m^2$, proving via the Chebotarev density theorem that it has only finitely many solutions when A is not a square and is conditionally finite under the *abc* conjecture when $A=u^2$. Luca [36] extended Overholt's conditional theorem to general polynomial-factorial equations $P(x)=n!$ of degree $r\geq 2$ under the *abc* conjecture, and further multiplicative and additive variants were studied by Kihel and Luca [32], Dąbrowski and Ulas [17], and Krasniqi [33]. Unconditionally, Berend and Harmse [5] proved that $P(x)=n!$ has only finitely many solutions whenever $P(x)\in\mathbb{Z}[x]$ is irreducible of degree ≥ 2 or violates p-adic valuation-balance constraints. However, when $P(x)$ splits over $\mathbb{Q}$ into linear factors of equal multiplicity—as in $x^2-1=(x-1)(x+1)$—local p-adic obstructions alone do not rule out infinitely many solutions.

A complementary line of inquiry seeks unconditional upper bounds on the counting function

$$\mathcal{N}_{P,s}(N) \;:=\; \#\{1\leq n\leq N : \exists\, x\in\mathbb{Z} \text{ such that } s\cdot n!=P(x)\} \tag{1.2}$$

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for a polynomial $P(x) \in \mathbb{Z}[x]$ of degree $r \geq 2$ and $s \in \mathbb{Z} \setminus \{0\}$, and in particular on $\mathcal{N}_{\text{Brocard}}(N) := \mathcal{N}_{x^2-1,1}(N)$. Answering a question of Erdős, Berend and Osgood [6] proved in 1992 via Osgood's effective bounds on linear forms in Siegel G-functions that $\mathcal{N}_{P,s}(N) = o(N)$ as $N \to \infty$. Recently, Bui, Pratt, and Zaharescu [12] introduced a three-variable Padé approximation method for the algebraic functions $\omega_i(x) = \prod_{j=1}^{\beta_i-1}(1-jx)^{-1/r}$, establishing the first unconditional power-saving bound

$$\mathcal{N}_{P,s}(N) \ll_{P,s} N^{33/34}. \tag{1.3}$$

1.2. **Main Results.** To state our principal theorems, let $n \geq 4$ be an integer and write the canonical square-free / square factorization of $n!$ as

$$n! = q_n s_n^2, \qquad q_n := \operatorname{core}(n!) = \prod_{p \in \mathcal{P}_{\text{odd}}(n)} p, \qquad s_n := \prod_{p \leq n} p^{\lfloor v_p(n!)/2 \rfloor}, \tag{1.4}$$

where $\mathcal{P}_{\text{odd}}(n) := \{p \leq n : 2 \nmid v_p(n!)\}$ and $\mathcal{P}_{\text{even}}(n) := \{p \leq n : 2 \mid v_p(n!)\}$ denote the sets of primes $p \leq n$ dividing $n!$ to odd and even powers, respectively. For every integer $n \geq 4$ with $n \neq 6$ (note that $6! + 1 = 721 = 7 \times 103$ is not a square), $q_n \geq 6$ is square-free and divisible by a prime $p \equiv 3 \pmod 4$, so the negative Pell equation $x^2 - q_n y^2 = -1$ is insoluble, and the group of positive units of the quadratic order $\mathbb{Z}[\sqrt{q_n}] \subset K_n := \mathbb{Q}(\sqrt{q_n})$ is generated by a unique fundamental unit $\varepsilon_n = X_1(q_n) + Y_1(q_n)\sqrt{q_n} > 1$ of norm $X_1(q_n)^2 - q_n Y_1(q_n)^2 = +1$. Every positive integer solution (n, m) to (1.1) satisfies $m^2 - q_n s_n^2 = 1$ and therefore corresponds to a unique positive integer Pell index $k \geq 1$ such that

$$m + s_n\sqrt{q_n} = \varepsilon_n^k = \big(X_1(q_n) + Y_1(q_n)\sqrt{q_n}\big)^k. \tag{1.5}$$

A priori, lower bounds on the fundamental regulator $\log \varepsilon_n > 0.2\, n$ imply $1 \leq k < 2.5 \log n$ for all $n \geq 8$. Our first main theorem proves that k collapses unconditionally to $\{1, 2\}$ for all sufficiently large n (and for $n \in \{4, 5, 7\}$).

Theorem 1.1 (Unconditional Collapse of the Pell Index)**.** *Let (n, m) be a positive integer solution to $n! + 1 = m^2$ with $n \geq 4$, and let $k \geq 1$ be the Pell index defined by (1.5). For the known solutions we have $k(4) = 1$, $k(5) = 1$, and $k(7) = 2$, and for all sufficiently large solutions (n, m),*

$$k \in \{1, 2\}. \tag{1.6}$$

Moreover, as $n \to \infty$, the fundamental regulator $\log \varepsilon_n$ satisfies the quantization law

$$\frac{\log \varepsilon_n}{n \log n} = \frac{1}{2k} + O\Big(\frac{1}{\log n}\Big), \tag{1.7}$$

and no two solutions $4 \leq n_1 < n_2$ share the same square-free factorial core ($q_{n_1} \neq q_{n_2}$).

This Pell embedding has a direct consequence for the p-adic arithmetic and Iwasawa theory of the totally real quadratic field $K_n = \mathbb{Q}(\sqrt{q_n})$. For every $n \geq 116$, every prime $p \in (n/5, n/4]$ (a non-empty set by Nagura's theorem [41]) satisfies $p > n/5 > 2.5 \log n > k \geq 1$, so $p \nmid k$ unconditionally; for sufficiently large n, Theorem 1.1 ($k \in \{1, 2\}$) ensures $p \nmid k$ for all primes $p \geq 5$. Taking the p-adic logarithm of $\varepsilon_n^{2k} = (1 + 2q_n s_n^2) + 2m s_n \sqrt{q_n}$ transfers the p-adic valuation $v_p(s_n) = \lfloor v_p(n!)/2 \rfloor$ of s_n directly to the Kubota–Leopoldt p-adic regulator $R_p(K_n)$ and p-adic L-function $L_p(1, \chi_{D_n})$. Recall that Greenberg's Iwasawa conjecture [27] asserts that for every totally real number field K and every prime p, the Iwasawa invariants $\mu_p(K)$ and $\lambda_p(K)$ of the cyclotomic $\mathbb{Z}_p$-extension K_∞/K vanish identically. Our second main theorem shows that any solution to (1.1) with $n \geq 116$ forces $\lambda_p(K_n) \geq 1$ for every prime $p \in (n/5, n/4]$ (and for a positive-density set of unramified primes $p \leq n$ as $n \to \infty$), thereby resolving Brocard's problem conditionally under Greenberg's conjecture.

Theorem 1.2 (Non-Vanishing of Iwasawa Invariants and Conditional Finiteness under Greenberg's Conjecture)**.** *Let (n, m) be a positive integer solution to $n! + 1 = m^2$ with $n \geq 116$, and let $K_n = \mathbb{Q}(\sqrt{q_n})$ be the associated real quadratic field of fundamental discriminant $D_n \in \{q_n, 4q_n\}$ and primitive quadratic character χ_{D_n}. For every prime $p \in (n/5, n/4]$—a non-empty set for all $n \geq 116$, and more generally for every prime $p \in \mathcal{P}_{\text{even}}(n)$ with $p \geq 5$, $p \nmid k$, and $v_p(n!) \geq 4$ (a set of primes of*

asymptotic natural density $\frac{5}{6} - \log 2 \approx 0.14019$ *among all primes* $p \le n$*)—the prime* p *is unramified in* K_n*, and the Kubota–Leopoldt* p*-adic* L*-function and Iwasawa* λ_p*-invariant of* K_n *satisfy*

$$v_p(L_p(1,\chi_{D_n})) \;\ge\; 1 \qquad \text{and} \qquad \lambda_p(K_n) \;=\; \lambda_p(\chi_{D_n}) \;\ge\; 1. \tag{1.8}$$

Consequently, if Greenberg's conjecture ($\lambda_p(K_n) = 0$*) holds for* $K_n = \mathbb{Q}(\sqrt{q_n})$ *at even a single prime* $p \in (n/5, n/4]$*, then* $n! + 1 = m^2$ *has no solutions with* $n \ge 116$*, and* $(4,5)$*,* $(5,11)$*, and* $(7,71)$ *are the only solutions to* (1.1)*.*

Our third main theorem establishes two unconditional counting bounds: an $O_\varepsilon(N^\varepsilon)$ sparsity bound for solutions to (1.1) in the critical regime $k = 2$, and an $O_{P,s,\varepsilon}(N^{6/7+\varepsilon})$ upper bound for the general polynomial-factorial counting function $\mathcal{N}_{P,s}(N)$ (improving (1.3), with still stronger exponents up to $O_{P,s,\varepsilon}(N^{2/3+\varepsilon})$ when $P(x)$ has degree gap $\nu \ge 3$).

Theorem 1.3 (Unconditional N^ε and $N^{6/7+\varepsilon}$ Counting Bounds)**.** *For every* $\varepsilon > 0$ *and all* $N \ge 1$*, the following unconditional bounds hold:*

(i) ***Extreme Sparsity of the*** $k = 2$ ***Regime:*** *Any two solutions* $4 \le n_1 < n_2$ *to* $n! + 1 = m^2$ *in the* $k = 2$ *regime satisfy* $n_2 - n_1 > n_1^{1-\varepsilon}$ *for* n_1 *sufficiently large in terms of* ε*, and*

$$\#\{1 \le n \le N : \exists\, m \in \mathbb{N},\ n! + 1 = m^2 \text{ with } k = 2\} \;\ll_\varepsilon\; N^\varepsilon. \tag{1.9}$$

(ii) ***Global*** $N^{6/7+\varepsilon}$ ***and Monomial-Product Bounds for*** $s \cdot n! = P(x)$***:*** *For any polynomial* $P(x) \in \mathbb{Z}[x]$ *of degree* $r \ge 2$ *and any* $s \in \mathbb{Z} \setminus \{0\}$*,*

$$\mathcal{N}_{P,s}(N) \;\ll_{P,s,\varepsilon}\; N^{6/7+\varepsilon}, \qquad \text{and in particular} \qquad \mathcal{N}_{\text{Brocard}}(N) \;\ll_\varepsilon\; N^{6/7+\varepsilon}. \tag{1.10}$$

Furthermore, if $P(x) = a_r x^r + a_{r-\nu} x^{r-\nu} + \cdots + a_0 \in \mathbb{Z}[x]$ *(*$a_r \ne 0, r \ge 3$*) has degree gap* $\nu := r - \deg(P - a_r x^r) \ge 3$ *after depressing the* x^{r-1} *term, then* $\mathcal{N}_{P,s}(N) \ll_{P,s,\varepsilon} N^{5/6+\varepsilon}$ *for* $\nu = 3$*,* $N^{4/5+\varepsilon}$ *for* $\nu = 4$*,* $N^{3/4+\varepsilon}$ *for* $\nu \ge 6$*, and* $N^{2/3+\varepsilon}$ *as* $\nu \to \infty$*.*

Beyond Theorems 1.1–1.3, we establish in the body of the paper several unconditional structural results governing the $k = 2$ and $k = 1$ regimes, the harmonic analysis of $n! + 1 = m^2$, and the associated Frey–Hellegouarch elliptic curves:

- **Two-stage Euclidean quartic law in the** $k = 2$ **regime (Section 4, Theorem 4.2):** When $k = 2$, Euclidean division of the even-valuation product A_n by the odd-valuation product B_n obeys the exact quartic identity $A_n = B_n Z^4 + Z^2$ ($R_n^2 = Q_n$), a second-stage quotient law $\lfloor B_+ R_n / B_- \rfloor = Z_-^4$, and stretched-exponential midpoint core bounds $\delta_i(n) := \operatorname{core}(X_1(q_n) \mp 1) \ge \exp(n^{1/2-o(1)})$.
- **Effective Landau–Siegel bounds and Rédei–Reichardt** 4**-rank cohomology in the** $k = 1$ **regime (Section 5, Theorems 5.2 and 5.5):** The midpoint cores satisfy the effective unconditional lower bound $\delta_i^{\text{mid}}(n) > \exp(c_0\sqrt{\log n})$, the norm equations $x^2 - q_n y^2 = \pm 2$ are insoluble, and the kernel of the ramified Rédei symbol matrix R_n^{ram} over $\mathbb{F}_2$ satisfies $\dim_{\mathbb{F}_2} \ker(R_n^{\text{ram}}) = 1 + \operatorname{rank}_4 \operatorname{Cl}^+(q_n)$ with an explicit $\mathbb{F}_2$-nullvector locking the Legendre symbols of $\delta_1(n)$ and $\delta_2(n)$.
- **CRT torus partition of unity and resonance-free Riesz product moments (Section 8, Theorems 8.1 and 8.2):** Every integer $1 \le m \le M_n - 1$ satisfying $m^2 \equiv 1 \pmod{n!}$ maps into a union of two intervals on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ of total measure $(4 + o(1))(n!)^{-1/2}$, and the associated Riesz product satisfies the exact Parseval moment identity $M_n^{-1} \sum_{h=0}^{M_n-1} |\widehat{\mu}_n(h)|^{2k} = (2^{-2k}\binom{2k}{k})^{\pi(n)}$ for all $1 \le k < \frac{1}{2} P_{\min}(n)$.
- **Level-2 Frey curve CM reduction and effective Tatuzawa class number explosion (Section 9, Theorems 9.1 and 9.2):** The 2-isogenous Frey curve $F_n^{(2)}$ satisfies $j(F_n^{(2)}) \equiv 1728 \pmod{X_1(q_n)^2}$, and the non-genus factor of the class group $\operatorname{Cl}(q_n)$ satisfies $h_{\text{non-gen}}(q_n) \ge q_n^{1/2-\varepsilon-o(1)}$ with at most one exceptional index n_*.

1.3. **Principal Obstacles and Overview of the Method.** We now describe the main obstacles encountered in proving Theorems 1.1–1.3 and the ideas used to overcome them.

1. Valuation-parity cancellation and the collapse of the Pell index. In (1.5), the Chebyshev polynomial factorization $s_n = Y_k(q_n) = Y_1(q_n)U_{k-1}(X_1(q_n))$ splits $n! = q_n s_n^2$ into the fundamental Pell factor $q_n Y_1(q_n)^2$ and the Chebyshev factor $U_{k-1}(X_1(q_n))^2$. By a p-adic Lifting-the-Exponent calculation (Lemma 3.2), every prime $p \in \mathcal{P}_{\mathrm{odd}}(n)$ divides q_n and satisfies $v_p(U_{k-1}(X_1(q_n))) = v_p(k)$, so p is absorbed into $q_n Y_1(q_n)^2$ with valuation $v_p(n!) - 2v_p(k)$. Consequently,

$$\log(q_n Y_1(q_n)^2) \;\geq\; S_{\mathrm{odd}}(n) - 2\log k, \qquad \text{where} \qquad S_{\mathrm{odd}}(n) \;:=\; \sum_{\substack{3\leq p\leq n\\ p\in\mathcal{P}_{\mathrm{odd}}(n)}} v_p(n!)\log p.$$

On the other hand, $q_n Y_1(q_n)^2 < \varepsilon_n^2 = (m + s_n\sqrt{q_n})^{2/k} \asymp (n!)^{1/k}$, so $\log(q_n Y_1(q_n)^2) \leq \frac{1}{k} n\log n + O(n)$. Thus, proving $k \in \{1,2\}$ reduces to proving the asymptotic valuation-parity equidistribution $S_{\mathrm{odd}}(n) = (\frac{1}{2} + o(1))n\log n$, or equivalently $\Delta_{\mathrm{parity}}(n) := S_{\mathrm{even}}(n) - S_{\mathrm{odd}}(n) = \sum_{3\leq p\leq n}(-1)^{v_p(n!)} v_p(n!)\log p = o(n\log n)$.

The obstacle in estimating $\Delta_{\mathrm{parity}}(n)$ is that Legendre's valuation formula $v_p(n!) = \sum_{r=1}^{R}\lfloor n/p^r\rfloor$ on a dyadic block $p \asymp P$ involves $R = \lfloor \frac{\log n}{\log P}\rfloor$ floor functions; expanding $(-1)^{v_p(n!)}$ via an R-fold product of Vaaler trigonometric polynomials of degree H produces an L^1 coefficient norm $(\frac{2}{\pi}\log H)^R$ that blows up whenever $R \to \infty$, while on the transition region $p \approx n^{1/r}$ the highest active floor $\lfloor n/p^r\rfloor$ has derivative too small for van der Corput bounds. In Section 2, we overcome both difficulties simultaneously. First, on $3 \leq p \leq \sqrt{n}$, Legendre's base-p digit formula $v_p(n!) = \frac{n - s_p(n)}{p-1}$ replaces $v_p(n!)\log p$ by $n\frac{\log p}{p-1}$ with an exact error of $O(\sqrt{n})$ without using exponential sums. Second, on $(n^{1/3}, n]$, we have $p^3 > n$, so $v_p(n!) = \lfloor n/p\rfloor + \lfloor n/p^2\rfloor$ involves at most *two* floor functions ($R \leq 2$). By excising the collar $\mathcal{C}(n) = (\frac{\sqrt{n}}{(\log n)^{10}}, \sqrt{n}]$ and partitioning $\mathcal{C}(n)$ into $M(n) \leq (\log n)^{20}$ subintervals J_m on each of which $\lfloor n/p^2\rfloor = m$ is constant, $(-1)^{v_p(n!)}$ reduces to the 1-variable phase $(-1)^m(-1)^{\lfloor n/p\rfloor}$ on each J_m, while below the collar on $(n^{1/3}, \frac{\sqrt{n}}{(\log n)^{10}}]$ both phases n/p and n/p^2 oscillate with derivative parameter $T \geq (\log n)^{14}$. This establishes $S_{\mathrm{odd}}(n) = (\frac{1}{2} + o(1))n\log n$ and completes the proof of Theorem 1.1 and Theorem 1.2 in Section 3.

2. Quartic Euclidean rigidity, Rédei cohomology, and the pairwise capacity diamond. With $k \in \{1,2\}$ established, Sections 4 and 5 analyze the fine arithmetic structure of the two regimes $k = 2$ and $k = 1$. When $k = 2$, $n! = 4(X_1(q_n) - 1)X_1(q_n)^2(X_1(q_n) + 1)$, and we prove that Euclidean division of the even-valuation product A_n by the odd-valuation product B_n obeys the exact two-stage quartic laws $A_n = B_n Z^4 + Z^2$ ($Q_n = Z^4, R_n = Z^2$) and $\lfloor B_+ R_n / B_-\rfloor = Z_-^4$, forcing the midpoint cores $\delta_i(n) = \mathrm{core}(X_1(q_n) \mp 1)$ to grow at least as fast as $\exp(n^{1/2-o(1)})$. When $k = 1$, writing $\frac{m-1}{2} = \delta_1(n)z_1^2$ and $\frac{m+1}{2} = \delta_2(n)z_2^2$ ($\delta_1(n)\delta_2(n) = q_n$), we prove effective Landau–Siegel sub-exponential lower bounds $\delta_i(n) > \exp(c_0\sqrt{\log n})$, an exact Galois cohomology sequence identifying the kernel of the ramified Rédei matrix over $\mathbb{F}_2$ with $1 + \mathrm{rank}_4\,\mathrm{Cl}^+(q_n)$, and a reciprocal cross-symmetry law equating the odd-valuation weight of $\frac{m+1}{2}$ with the even-valuation weight of $\frac{m-1}{2}$ up to $O(n)$.

In Section 6, we combine this cross-symmetry law with a universal $\mathrm{SL}_2(\mathbb{Z})$ four-GCD decomposition $XYZW = n_1!/4$ for any pair of solutions $n_1 < n_2$. Because Vaaler's trigonometric polynomial for $(-1)^{\lfloor x\rfloor}$ is supported exclusively on *odd harmonics* $2m + 1 \neq 0$, the correlation sum $\sum_{p\leq n_1}(-1)^{v_p(n_1!)+v_p(n_2!)} v_p(n_1!)\log p$ has non-vanishing diagonal phase $\frac{(2m_2+1)(n_2-n_1)}{2p} \neq 0$. This forces the odd-core polarization weights $(\theta(n_1), \theta(n_2))$ of any two solutions with $n_2 - n_1 = n_1^{\alpha}$ to lie inside the *Capacity Diamond* $\frac{1-\alpha}{2} - o(1) \leq \theta(n_1) + \theta(n_2) \leq \frac{3+\alpha}{2} + o(1)$. Since $\theta(n) = 0$ whenever $k = 2$, this immediately forces $\alpha \geq 1 - o(1)$, proving $n_2 - n_1 > n_1^{1-\varepsilon}$ and $\#\mathcal{B}_{k=2}(N) \ll_\varepsilon N^{\varepsilon}$ (Theorem 1.3(i)).

3. The support-reduced Wronskian and monomial-product Padé approximations. To prove the global counting bound $\mathcal{N}_{P,s}(N) \ll_{P,s,\varepsilon} N^{6/7+\varepsilon}$ of Theorem 1.3(ii) in Section 7, we generalize

the Padé approximation method of Bui, Pratt, and Zaharescu [12] from $m = 2$ shifts to $m+1$ variables $\omega_0(x), \omega_1(x), \dots, \omega_m(x)$. In [12], the Padé vanishing-slack parameter ε_0 in $O = \lfloor (m + 1 - \varepsilon_0)D \rfloor$ was restricted to $\varepsilon_0 \in (0, 1)$; for $\varepsilon_0 < 1$, the maximum saving over all $m \geq 2$ is capped at $\theta < 1/14$ ($O(N^{13/14})$). The reason for restricting $\varepsilon_0 < 1$ in [12] was that when $\varepsilon_0 \geq 1$, Siegel's lemma may produce a Padé tuple $(P_0, \dots, P_m)$ in which one or more polynomials $P_i(x)$ vanish identically, causing the full $(m + 1) \times (m + 1)$ Wronskian determinant to vanish identically. We resolve this by passing to the non-zero support $S := \{0 \leq i \leq m : P_i(x) \not\equiv 0\}$ ($s := |S|$) and forming the $s \times s$ *support-reduced Wronskian* $\Delta_S(x) := \det(P_{i_a}^{[k]}(x))_{0 \leq k < s,\, a \in S}$. Because the subfamily $\{\omega_i : i \in S\}$ is linearly independent over $\mathbb{C}(x)$ and every P_i ($i \in S$) is non-zero, $\Delta_S(x) \not\equiv 0$ is guaranteed for all $\varepsilon_0 \in (0, m - 1)$ with $s \geq \lceil m + 1 - \varepsilon_0 \rceil \geq 2$, and because the solution vector $p_i = n_0^{\beta_i/r} x_i > 0$ has strictly positive coordinates at every index $0 \leq i \leq m$, its restriction to S is never zero. Optimizing over $\varepsilon_0 \in (0, m - 1)$ at $m = 5$ and $\varepsilon_0 = 5 - \sqrt{5} \in (2, 3)$ yields $\theta_5(5 - \sqrt{5}) = \frac{7-3\sqrt{5}}{2} \approx 0.14590 > \frac{1}{7}$, reaching the $m = 5$ Pigeonhole cap $\theta_5^* = \frac{1}{7}$ and proving $\mathcal{N}_{P,s}(N) \ll_{P,s,\varepsilon} N^{6/7+\varepsilon}$. For polynomials with degree gap $\nu \geq 3$, we further replace the linear shift basis by the monomial-product basis $\Omega_{\mathbf{e}}(x) = \prod_{i=1}^{d} \omega_i(x)^{e_i}$ ($|\mathbf{e}| \leq E \leq \min(r - 1, \nu - 1)$), generating $\binom{d+E}{d}$ basis functions from only d shifts and raising the Pigeonhole cap from $\frac{1}{m+2}$ to $\frac{1}{d+2}$.

Finally, Section 8 develops the harmonic analysis of (1.1) on the Chinese Remainder Theorem torus $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, and Section 9 establishes the properties of the associated Frey–Hellegouarch elliptic curves, the Tatuzawa class-group bounds, and the analytic thresholds for level-lowering and linear forms in logarithms.

1.4. **Notation.** Throughout the paper, $\mathbb{N} = \{1, 2, 3, \dots\}$ denotes the set of positive integers, $\mathcal{P}$ the set of primes, $\pi(x) = \sum_{p \leq x} 1$, and $\vartheta(x) = \sum_{p \leq x} \log p$. For a prime p and a non-zero integer a, $v_p(a)$ denotes the exact p-adic valuation of a, and $p^e \parallel a$ means $v_p(a) = e$. For a positive integer $A = \prod_p p^{v_p(A)}$, we write $\mathrm{core}(A) := \prod_{2 \nmid v_p(A)} p$ for its square-free core, $\mathrm{rad}(A) := \prod_{p \mid A} p$ for its radical, $\omega(A) := \sum_{p \mid A} 1$ for its number of distinct prime factors, $P^+(A)$ for its largest prime factor, and $A^* := A/2^{v_2(A)}$ for its odd part. We write $e(t) := e^{2\pi i t}$, and we use Vinogradov's notation $f \ll g$ interchangeably with Landau's notation $f = O(g)$, indicating dependence of implied constants on auxiliary parameters by subscripts.

2. Valuation-Parity Equidistribution of Factorials

For an integer $n \geq 4$, define the odd- and even-valuation prime weights of $\log(n!)$ over odd primes $3 \leq p \leq n$ together with their parity discrepancy:

$$S_{\mathrm{odd}}(n) := \sum_{\substack{3\leq p\leq n\\ p\in\mathcal{P}_{\mathrm{odd}}(n)}} v_p(n!)\log p, \qquad S_{\mathrm{even}}(n) := \sum_{\substack{3\leq p\leq n\\ p\in\mathcal{P}_{\mathrm{even}}(n)}} v_p(n!)\log p, \tag{2.1}$$

and

$$\Delta_{\mathrm{parity}}(n) := S_{\mathrm{even}}(n) - S_{\mathrm{odd}}(n) = \sum_{3\leq p\leq n} (-1)^{v_p(n!)} v_p(n!)\log p. \tag{2.2}$$

For any cutoff $B \geq 3$, define the small-prime valuation-parity series

$$\Sigma_{\mathrm{parity}}(n;B) := \sum_{3\leq p\leq B} (-1)^{v_p(n!)-1}\frac{\log p}{p-1}, \qquad \Sigma^*_{\mathrm{parity}}(n) := \Sigma_{\mathrm{parity}}(n;\sqrt{n}). \tag{2.3}$$

In this section we establish the asymptotic equidistribution of $S_{\mathrm{odd}}(n)$ and $S_{\mathrm{even}}(n)$, the second-order formula for $\Delta_{\mathrm{parity}}(n)$, the multi-modulus power-band equidistribution law, and the Jessen–Wintner limit distribution of $\Sigma_{\mathrm{parity}}(n;n^{1/3})$.

2.1. Harmonic Prime Layers and the Square-Free Factorial Core.

We begin by partitioning the upper prime range $(\sqrt{n}, n]$ into harmonic layers on which $v_p(n!)$ is constant.

Lemma 2.1 (Harmonic Layer Stratification and Asymptotics of $\mathcal{P}_{\mathrm{odd}}(n)$ and $\log q_n$ [4, 49, 11])**.** *Let $n \geq 4$ be an integer, and define the harmonic intervals $I_j := (\frac{n}{j+1}, \frac{n}{j}]$ for $j \geq 1$.*

(i) *For every integer $1 \leq j \leq \lfloor\sqrt{n}\rfloor - 1$ and every prime $p \in I_j = (\frac{n}{j+1}, \frac{n}{j}]$, we have $v_p(n!) = j$. In particular, on odd layers I_{2r-1} ($1 \leq 2r-1 \leq \lfloor\sqrt{n}\rfloor - 1$), every prime $p \in I_{2r-1}$ belongs to $\mathcal{P}_{\mathrm{odd}}(n)$ with $v_p(q_n) = 1$ and $v_p(s_n) = r-1$; on even layers I_{2r} ($2 \leq 2r \leq \lfloor\sqrt{n}\rfloor - 1$), every prime $p \in I_{2r}$ belongs to $\mathcal{P}_{\mathrm{even}}(n)$ with $v_p(q_n) = 0$ and $v_p(s_n) = r$.*

(ii) *There exists an absolute effective constant $c_0 > 0$ such that as $n \to \infty$,*

$$\log q_n = (\log 2)\, n + O\Big(n\exp\big(-c_0\sqrt{\log n}\big)\Big), \qquad |\mathcal{P}_{\mathrm{odd}}(n)| = (\log 2 + o(1))\,\pi(n). \tag{2.4}$$

Proof. (i) For $1 \leq j \leq \lfloor\sqrt{n}\rfloor - 1$, we have $j+1 \leq \lfloor\sqrt{n}\rfloor \leq \sqrt{n}$, so every prime $p \in I_j = (\frac{n}{j+1}, \frac{n}{j}]$ strictly satisfies $p > \frac{n}{j+1} \geq \sqrt{n}$. Consequently, $p^2 > n$, so $\lfloor n/p^m\rfloor = 0$ for all $m \geq 2$. Since $\frac{n}{j+1} < p \leq \frac{n}{j} \iff j \leq \frac{n}{p} < j+1$, Legendre's formula [35] gives

$$v_p(n!) = \sum_{m=1}^{\infty}\left\lfloor\frac{n}{p^m}\right\rfloor = \left\lfloor\frac{n}{p}\right\rfloor = j.$$

Writing $v_p(n!) = v_p(q_n) + 2v_p(s_n)$ with $v_p(q_n) \in \{0,1\}$ yields the stated formulas on I_{2r-1} and I_{2r}.

(ii) Split the prime range $p \leq n$ at $\sqrt{n}$. The primes $p \leq \sqrt{n}$ contribute at most $\sum_{p\leq\sqrt{n}}\log p = \vartheta(\sqrt{n}) \ll \sqrt{n}$ to $\log q_n$ and at most $\pi(\sqrt{n}) \ll \frac{\sqrt{n}}{\log n}$ to $|\mathcal{P}_{\mathrm{odd}}(n)|$. On the upper range $(\sqrt{n}, n]$, set $J := \lfloor\exp(\frac{1}{2}c_1\sqrt{\log n})\rfloor$, where $c_1 > 0$ is the constant in the de la Vallée Poussin Prime Number Theorem [18, Chapter 18]:

$$\vartheta(x) = x + O\Big(x\exp\big(-c_1\sqrt{\log x}\big)\Big), \qquad \pi(x) = \mathrm{li}(x) + O\Big(x\exp\big(-c_1\sqrt{\log x}\big)\Big). \tag{2.5}$$

The primes in $(\sqrt{n}, n/J]$ contribute at most $\vartheta(n/J) \ll \frac{n}{J} = n\exp(-\frac{1}{2}c_1\sqrt{\log n})$ to $\log q_n$. On $(n/J, n]$, part (i) shows that a prime $p \in (n/J, n]$ belongs to $\mathcal{P}_{\mathrm{odd}}(n)$ if and only if $p \in I_{2r-1} = (\frac{n}{2r}, \frac{n}{2r-1}]$ for some $1 \leq 2r-1 \leq J-1$. Summing $\vartheta(\frac{n}{2r-1}) - \vartheta(\frac{n}{2r})$ over $1 \leq 2r-1 \leq J-1$ via (2.5) gives

$$\begin{aligned}\log q_n &= \sum_{1\leq 2r-1\leq J-1}\left(\vartheta\left(\frac{n}{2r-1}\right) - \vartheta\left(\frac{n}{2r}\right)\right) + O\left(\frac{n}{J}\right)\\ &= n\sum_{1\leq 2r-1\leq J-1}\left(\frac{1}{2r-1} - \frac{1}{2r}\right) + O\left(n(\log J)\exp\big(-c_1\sqrt{\log(n/J)}\big) + \frac{n}{J}\right).\end{aligned}$$

Since $\sum_{r=1}^{\infty}(\frac{1}{2r-1}-\frac{1}{2r})=\sum_{j=1}^{\infty}\frac{(-1)^{j-1}}{j}=\log 2$ and $\sum_{2r>J}(\frac{1}{2r-1}-\frac{1}{2r})=O(J^{-1})$, setting $c_0:=c_1/3$ yields the first relation in (2.4). Similarly, summing $\pi(\frac{n}{2r-1})-\pi(\frac{n}{2r})=(\frac{1}{2r-1}-\frac{1}{2r}+o(1))\frac{n}{\log n}$ over $1\le 2r-1\le\lfloor\log n\rfloor$ yields $|\mathcal{P}_{\mathrm{odd}}(n)|=(\log 2+o(1))\pi(n)$. □

2.2. **Exact Base-p Digit Reduction Below $\sqrt{n}$.** Whereas primes $p>\sqrt{n}$ satisfy $v_p(n!)=\lfloor n/p\rfloor$, primes $p\le\sqrt{n}$ have higher powers $p^2,p^3,\cdots\le n$ contributing to $v_p(n!)$. The next lemma shows that Legendre's base-p digit identity replaces the weighted sum $\sum_{3\le p\le\sqrt{n}}(-1)^{v_p(n!)}v_p(n!)\log p$ by $-n\Sigma^*_{\mathrm{parity}}(n)$ up to a deterministic error of $O(\sqrt{n})$, avoiding any Fourier expansion on $p\le\sqrt{n}$ when relating $\Delta_{\mathrm{parity}}(n)$ to $\Sigma^*_{\mathrm{parity}}(n)$.

Lemma 2.2 (Exact Legendre Base-p Reduction on $3\le p\le\sqrt{n}$)**.** *For every integer $n\ge 4$, write the base-p expansion of n as $n=\sum_{j=0}^{K_p}d_j^{(p)}(n)p^j$ ($0\le d_j^{(p)}(n)\le p-1$, $K_p=\lfloor\frac{\log n}{\log p}\rfloor$), and let $s_p(n):=\sum_{j=0}^{K_p}d_j^{(p)}(n)$. Then:*

(i) *For every odd prime p, the valuation parity of $n!$ is determined solely by the sum of the* odd-indexed *base-p digits of n:*

$$v_p(n!)\ \equiv\ \sum_{m=0}^{\lfloor(K_p-1)/2\rfloor}d_{2m+1}^{(p)}(n)\pmod 2. \tag{2.6}$$

(ii) *Across $3\le p\le\sqrt{n}$, we have*

$$\sum_{3\le p\le\sqrt{n}}(-1)^{v_p(n!)}v_p(n!)\log p\ =\ -n\,\Sigma^*_{\mathrm{parity}}(n)+E_{\mathrm{digit}}(n), \tag{2.7}$$

where the digit-sum remainder $E_{\mathrm{digit}}(n):=\sum_{3\le p\le\sqrt{n}}(-1)^{v_p(n!)-1}\frac{s_p(n)\log p}{p-1}$ satisfies the explicit bound

$$\big|E_{\mathrm{digit}}(n)\big|\ \le\ 2\,\pi(\sqrt{n})\log n\ =\ O(\sqrt{n}). \tag{2.8}$$

Proof. (i) By Legendre's identity,

$$v_p(n!)\ =\ \frac{n-s_p(n)}{p-1}\ =\ \sum_{j=1}^{K_p}d_j^{(p)}(n)\,\frac{p^j-1}{p-1}. \tag{2.9}$$

Since p is odd, $p\equiv 1\pmod 2$, so $\frac{p^j-1}{p-1}=1+p+p^2+\cdots+p^{j-1}\equiv j\pmod 2$. Thus $\frac{p^j-1}{p-1}$ is even when $j=2m$ is even and odd when $j=2m+1$ is odd, reducing (2.9) modulo 2 to (2.6).

(ii) Multiplying $v_p(n!)=\frac{n-s_p(n)}{p-1}$ by $(-1)^{v_p(n!)}\log p=-(-1)^{v_p(n!)-1}\log p$ and summing over all primes $3\le p\le\sqrt{n}$ yields (2.7). For each prime $3\le p\le\sqrt{n}$, since $0\le d_j^{(p)}(n)\le p-1$ for every $0\le j\le K_p$, we have

$$0\ \le\ \frac{s_p(n)}{p-1}\ \le\ K_p+1\ =\ \left\lfloor\frac{\log n}{\log p}\right\rfloor+1\ \le\ \frac{\log n}{\log p}+1.$$

Multiplying by $\log p\le\frac{1}{2}\log n$ gives $0\le\frac{s_p(n)\log p}{p-1}\le\log n+\log p\le\frac{3}{2}\log n<2\log n$. Summing over the $\pi(\sqrt{n})$ primes in $[3,\sqrt{n}]$ and applying Chebyshev's bound $\pi(\sqrt{n})\ll\frac{\sqrt{n}}{\log n}$ yields (2.8). □

2.3. **Collar-Decomposed Two-Variable Vaaler Sums and Equidistribution Theorems.** To evaluate the upper range $(\sqrt{n},n]$ and cancel $\Sigma^*_{\mathrm{parity}}(n)$ on $(n^\delta,\sqrt{n}]$, we approximate the square-wave parity function $(-1)^{\lfloor x\rfloor}=2\psi(x)-4\psi(x/2)$ ($x\notin\mathbb{Z}$, where $\psi(t):=t-\lfloor t\rfloor-\frac{1}{2}$) by Vaaler's extremal trigonometric polynomials [60, Theorem 18] with matched cutoff degrees H and $2H+1$.

Lemma 2.3 (Vaaler's Trigonometric Polynomial Approximation [60, Theorem 18])**.** *For every integer $H\ge 1$, let*

$$\Psi_H(t)\ =\ \sum_{1\le|h|\le H}c_h(H)\,e(ht),\qquad B_H(t)\ =\ \sum_{|h|\le H}b_h(H)\,e(ht),$$

be Vaaler's trigonometric polynomials with $c_h(H) = -\frac{1}{2\pi i h} W\big(\frac{|h|}{H+1}\big)$ *(*$|c_h(H)| \le \frac{1}{2\pi|h|}$*),* $b_0(H) = \frac{1}{2H+2}$*, and* $|b_h(H)| \le \frac{1}{2H+2}$ *satisfying* $|\psi(t) - \Psi_H(t)| \le B_H(t)$ *for all* $t \in \mathbb{R}$*. Define*

$$\widetilde{\Psi}_H(x) \;:=\; 2\Psi_H(x) - 4\Psi_{2H+1}\Big(\frac{x}{2}\Big), \qquad \widetilde{B}_H(x) \;:=\; 2B_H(x) + 4B_{2H+1}\Big(\frac{x}{2}\Big). \tag{2.10}$$

Then every even harmonic $h = 2m$ *(*$1 \le |m| \le H$*) cancels identically in* $\widetilde{\Psi}_H(x)$*, so that*

$$\widetilde{\Psi}_H(x) \;=\; \sum_{\substack{1\le|h|\le 2H+1\\ 2\nmid h}} \tilde{c}_h(H)\, e\Big(\frac{hx}{2}\Big), \qquad \widetilde{B}_H(x) \;=\; \frac{2}{H+1} + \sum_{1\le|h|\le 2H+1} \tilde{b}_h(H)\, e\Big(\frac{hx}{2}\Big),$$

with $|\tilde{c}_h(H)| \le \frac{2}{\pi|h|}$ *and* $|\tilde{b}_h(H)| < \frac{3}{H+1}$*, and for all* $x \in \mathbb{R}$*:*

$$\big|(-1)^{\lfloor x\rfloor} - \widetilde{\Psi}_H(x)\big| \;\le\; \widetilde{B}_H(x), \qquad \sum_{\substack{1\le|h|\le 2H+1\\ 2\nmid h}} |\tilde{c}_h(H)| \;\le\; \frac{4}{\pi}\big(\log(2H+1)+1\big). \tag{2.11}$$

Proof. For every $x \notin \mathbb{Z}$, writing $\lfloor x\rfloor = 2\lfloor x/2\rfloor + r$ with $r \in \{0,1\}$ shows that $2\psi(x) - 4\psi(x/2) = 1 - 2r = (-1)^{\lfloor x\rfloor}$. By the triangle inequality, $|(-1)^{\lfloor x\rfloor} - \widetilde{\Psi}_H(x)| \le 2|\psi(x) - \Psi_H(x)| + 4|\psi(x/2) - \Psi_{2H+1}(x/2)| \le \widetilde{B}_H(x)$ for all $x \notin \mathbb{Z}$, and at integer points $x \in \mathbb{Z}$ (where $B_H(0) = \frac{1}{2}$ and $|\Psi_H(m)| = 0$) we have $\widetilde{B}_H(m) \ge 2(\frac{1}{2}) = 1 = |(-1)^m - \widetilde{\Psi}_H(m)|$. In $2\Psi_H(x) - 4\Psi_{2H+1}(x/2)$, the coefficient of $e(mx) = e((2m)x/2)$ for $1 \le |m| \le H$ is $2c_m(H) - 4c_{2m}(2H+1) = -\frac{1}{\pi i m}\big(W(\frac{|m|}{H+1}) - W(\frac{2|m|}{2H+2})\big) = 0$ because $\frac{2|m|}{(2H+1)+1} = \frac{|m|}{H+1}$. Thus $\widetilde{\Psi}_H(x)$ is supported strictly on odd frequencies $h = 2m+1$ $(1 \le |h| \le 2H+1)$ with $\tilde{c}_h(H) = -4c_h(2H+1)$ $(|\tilde{c}_h(H)| \le \frac{2}{\pi|h|})$, while the zero-frequency term of $\widetilde{B}_H(x)$ is $2b_0(H) + 4b_0(2H+1) = \frac{2}{2H+2} + \frac{4}{4H+4} = \frac{2}{H+1} < \frac{3}{H+1}$. □

Fix any $\delta \in (0, 1/3]$ and set $R := \lfloor 1/\delta\rfloor$ and $A_R := 6R + 16$. On the range $p > n^\delta$, we have $p^{R+1} > n$, so Legendre's formula has at most R non-zero terms:

$$v_p(n!) \;=\; \sum_{r=1}^{R} \Big\lfloor \frac{n}{p^r} \Big\rfloor \qquad \forall\, p > n^\delta. \tag{2.12}$$

Near each root transition $p = n^{1/r}$ $(2 \le r \le R)$, we define the transition collar

$$\mathcal{C}_r(n) \;:=\; \left(\frac{n^{1/r}}{(\log n)^{A_R/r}},\, n^{1/r}\right] \qquad (2 \le r \le R), \tag{2.13}$$

on which $1 \le \frac{n}{p^r} < (\log n)^{A_R}$, so $\lfloor n/p^r\rfloor = m_r$ takes at most $M_r(n) := \lfloor (\log n)^{A_R}\rfloor$ constant integer values on the disjoint subintervals

$$J_{r,m_r} \;:=\; \left(\left(\frac{n}{m_r+1}\right)^{1/r}, \left(\frac{n}{m_r}\right)^{1/r}\right] \cap \mathcal{C}_r(n), \qquad 1 \le m_r \le M_r(n). \tag{2.14}$$

On each J_{r,m_r}, $(-1)^{\lfloor n/p^r\rfloor} = (-1)^{m_r}$ is constant, reducing the active floors to $1 \le s \le r-1$ (each with $n/p^s \ge n^{1-s/r} \ge n^{1/r} \ge n^\delta$), whereas on the complementary inter-collar intervals $\mathcal{D}_r(n) := \big(n^{1/(r+1)}, \frac{n^{1/r}}{(\log n)^{A_R/r}}\big]$ every active phase n/p^s $(1 \le s \le r)$ on a dyadic block $(P, 2P] \subset \mathcal{D}_r(n)$ satisfies $n/P^s \ge 2^{-R}(\log n)^{A_R}$.

Lemma 2.4 (Prime Exponential Sum Bound on $[n^\delta, n/(\log n)^{12}]$)**.** *Fix* $\delta \in (0, 1/3]$ *and set* $R := \lfloor 1/\delta\rfloor$*,* $A_R := 6R + 16$*, and* $H := \lfloor (\log n)^{R+2}\rfloor$*. Let* $P \in [n^\delta, \frac{n}{(\log n)^{12}}]$*, let* $I \subset (P, 2P]$ *be any subinterval on which the set of active floor indices* $\mathcal{A}(I) \subset \{1, \dots, R\}$ *is fixed with* $\frac{n}{P^r} \ge 2^{-R}(\log n)^{A_R}$ *for all* $r \in \mathcal{A}(I)$*, and let*

$$F(t) \;=\; \sum_{r\in\mathcal{A}(I)} \frac{h_r n}{2t^r}, \qquad \mathbf{h} = (h_r)_{r\in\mathcal{A}(I)} \neq \mathbf{0}, \quad |h_r| \le 2H+1.$$

Then for every smooth weight $w(t)$ on $[P,2P]$ satisfying $|w(t)|\le W_0$ and $\int_P^{2P}|w'(t)|\,dt\le W_0$, we have unconditionally:

(i) *In general, for $T_0:=\frac{|h_{r_0}|n}{P^{r_0}}\ge 2^{-R}(\log n)^{A_R}$ (where $r_0:=\min\{r\in\mathcal{A}(I):h_r\neq 0\}$),*

$$\left|\sum_{p\in I} w(p)(\log p)\,e\big(F(p)\big)\right| \ll_\delta\ W_0P\Big(T_0^{-1/6}+P^{-1/(2^{R+3})}\Big)\log^3 P\ \ll_\delta\ W_0P(\log n)^{-(R+2)}. \tag{2.15}$$

(ii) *Moreover, whenever $T_0\ge n^{\delta/2}$ (which holds in particular on every collar subinterval $J_{r,m_r}\subset\mathcal{C}_r(n)$ where $\mathcal{A}(J_{r,m_r})=\{1,\dots,r-1\}$ and $T_0\ge n^{1-r_0/r}\ge n^{1/r}\ge n^\delta$), there exists $\eta_\delta:=\min\big(\frac{\delta}{20},\frac{\delta}{2^{R+3}}\big)>0$ such that*

$$\left|\sum_{p\in I} w(p)(\log p)\,e\big(F(p)\big)\right| \ll_\delta\ W_0P\,n^{-\eta_\delta}. \tag{2.16}$$

Proof. By partial summation against $w(t)$ and discarding prime powers p^k ($k\ge 2$) at cost $O(P^{1/2})$, it suffices to bound $S(I'):=\sum_{k\in I'}\Lambda(k)\,e(F(k))$ for any subinterval $I'\subset I\subset(P,2P]$. Let $r_0:=\min\{r\in\mathcal{A}(I):h_r\neq 0\}$. Because $t\in[P,2P]$ with $P\ge n^\delta\gg H=\lfloor(\log n)^{R+2}\rfloor$ and $|h_r|\le 2H+1$, for every derivative order $j\ge 1$ the r_0-th term $\frac{h_{r_0}n}{2t^{r_0}}$ strictly dominates all higher terms $r>r_0$:

$$|F^{(j)}(t)|\ =\ \frac{(j+r_0-1)!}{(r_0-1)!}\,\frac{|h_{r_0}|n}{2t^{j+r_0}}\Big(1+O_R\big(HP^{-1}\big)\Big)\ \asymp_{j,R}\ \frac{T_0}{P^j}\qquad(\forall\,t\in[P,2P],\ j\ge 1),$$

where $T_0:=\frac{|h_{r_0}|n}{P^{r_0}}\in\big[2^{-R}(\log n)^{6R+16},\,(2H+1)n^{1-r_0\delta}\big]$. Apply Vaughan's identity [30, Proposition 13.4] with parameters $U=V=P^{1/5}$ to decompose $S(I')=S_{\mathrm{I}}-S'_{\mathrm{I}}+S_{\mathrm{II}}+O(P^{1/5})$ into:

- **Type I sums** (S_{I} with $d\le U=P^{1/5}$ and S'_{I} with $d\le UV=P^{2/5}$): sums of the form $\sum_{d\le D}a_d\sum_{m\in I'/d}e(F(dm))$ (and its smooth $\log m$-weighted analogue), where $|a_d|\le\tau(d)\log P$ and the inner variable $m\asymp M:=P/d\ge P^{3/5}$ ranges over a subinterval of $(P/d,2P/d]$ with smooth phase $g_d(m):=F(dm)$ satisfying $|g_d^{(j)}(m)|=d^j|F^{(j)}(dm)|\asymp_{j,R}T_0/M^j$.
- **Type II bilinear sums** ($U<d\le P/V$, so $P^{1/5}<d\le P^{4/5}$): dyadic bilinear sums of the form $\sum_M\sum_{d\sim P/M}\alpha_d\sum_{m\sim M,\,dm\in I'}\beta_m e(F(dm))$ with $|\alpha_d|,|\beta_m|\le\tau(d)\log P$. Applying Cauchy–Schwarz in $d\sim D:=P/M\in[P^{1/5},P^{4/5}]$ and interchanging summation gives

$$|S_{\mathrm{II}}(D,M)|^2\ \ll\ (D\log^3 P)\left(DM\log^2 P+\sum_{m_1\neq m_2\sim M}|\beta_{m_1}\beta_{m_2}|\left|\sum_{d\in I(m_1,m_2)}e\big(G_{m_1,m_2}(d)\big)\right|\right),$$

where the differenced phase $G_{m_1,m_2}(d):=F(dm_1)-F(dm_2)$ has j-th derivative

$$|G^{(j)}_{m_1,m_2}(d)|\ \asymp_{j,R}\ \frac{|m_1-m_2|}{M}\,\frac{T_0}{D^j}\qquad(\forall\,d\sim D,\ j\ge 1).$$

We distinguish two regimes for T_0:

Regime A ($2^{-R}(\log n)^{6R+16}\le T_0\le P^{3/2}$): In Type I, $M\ge P^{3/5}$. If $T_0\le M$, then $|g_d''(m)|\asymp T_0/M^2$, so van der Corput's second-derivative bound [30, Corollary 8.13] gives $\sum_m e(g_d(m))\ll M(T_0/M^2)^{1/2}+(M^2/T_0)^{1/2}\ll T_0^{1/2}+MT_0^{-1/2}\ll M(P^{-1/10}+T_0^{-1/2})$. If $M<T_0\le P^{3/2}$, then $|g_d^{(3)}(m)|\asymp T_0/M^3$ and the third-derivative van der Corput bound ($j=3$) gives $\ll M\big((T_0/M^3)^{1/6}+(M^3/T_0)^{1/6}\big)\ll M(P^{-1/20}+T_0^{-1/6})$. In Type II, for the terms with $|m_1-m_2|\le MT_0^{-1/3}$ there are $\ll M^2T_0^{-1/3}$ pairs contributing $\ll PT_0^{-1/6}\log^3 P$ after taking the square root; for $|m_1-m_2|>MT_0^{-1/3}$, setting $T_\Delta:=\frac{|m_1-m_2|}{M}T_0\in(T_0^{2/3},T_0]$ and applying the Kusmin–Landau bound [30, Corollary 8.11] when $T_\Delta\le\frac12 D$ and the j-th derivative van der Corput bound ($j\in\{2,3,4\}$) when $T_\Delta>\frac12 D$ bounds the inner d-sum by $\ll D(T_0^{-1/3}+P^{-1/40})$. Summing over d and m_1,m_2 gives $|S(I')|\ll_\delta P(T_0^{-1/6}+P^{-1/40})\log^3 P$. Since $T_0\ge 2^{-R}(\log n)^{6R+16}$, $T_0^{-1/6}\log^3 P\ll_\delta(\log n)^{-(R+8/3)}\ll(\log n)^{-(R+2)}$.

Regime B ($T_0 > P^{3/2}$): Choose the integer derivative order $j := \lfloor \frac{\log(T_0/M)}{\log M} \rfloor + 2 \geq 3$ in Type I (and similarly with D and T_Δ in Type II), where $j \leq 5R+2$ since $T_0 \leq (2H+1)n$ and $M, D \geq P^{1/5} \geq n^{\delta/5}$. Then $M^{j-2} \leq T_0/M < M^{j-1}$, so $\lambda_j := T_0/M^j \in [M^{-2}, M^{-1})$, and the j-th derivative Weyl–van der Corput inequality [30, Theorem 8.25] yields a polynomial power saving $\ll_\delta P\, n^{-\eta_\delta}$ with $\eta_\delta > 0$, proving both (2.15) and (2.16). □

We now state and prove the principal valuation-parity equidistribution theorem.

Theorem 2.5 (Valuation-Parity Equidistribution and Second-Order Discrepancy Law)**.** *For every integer $n \geq 4$, the following statements hold unconditionally:*

(i) ***Second-Order Discrepancy Formula:***

$$\Delta_{\mathrm{parity}}(n) \;=\; \Big(\log 2 - 1 - \Sigma^*_{\mathrm{parity}}(n)\Big)\, n + O\Big(\frac{n}{\log n}\Big). \tag{2.17}$$

(ii) ***Collar-Decomposed Cancellation and Full-Range*** 50–50 ***Weight Split:*** *For every fixed $\delta \in (0, 1/3]$,*

$$\sum_{n^\delta < p \leq \sqrt{n}} (-1)^{v_p(n!)-1} \frac{\log p}{p-1} \;\ll_\delta\; \frac{1}{\log n}, \tag{2.18}$$

and across the full range $3 \leq p \leq n$, $\Sigma^*_{\mathrm{parity}}(n) = o(\log n)$, $\Delta_{\mathrm{parity}}(n) = o(n \log n)$, and

$$S_{\mathrm{odd}}(n) \;=\; \frac{1}{2} n \log n + o(n \log n), \qquad S_{\mathrm{even}}(n) \;=\; \frac{1}{2} n \log n + o(n \log n). \tag{2.19}$$

(iii) ***Multi-Modulus Power-Band Equidistribution:*** *For every integer modulus $h \geq 2$, every residue class $c \bmod h$, and every exponent band $0 \leq a < b \leq 1$,*

$$\sum_{\substack{n^a < p \leq n^b \\ v_p(n!) \equiv c \pmod h}} v_p(n!) \log p \;=\; \Big(\frac{b-a}{h} + o(1)\Big)\, n \log n. \tag{2.20}$$

Proof. (i) Set $R_1 := \lfloor (\log n)^{12} \rfloor$. On $(\frac{n}{R_1}, n]$, Lemma 2.1(i) and (2.5) give

$$\begin{aligned} \sum_{n/R_1 < p \leq n} (-1)^{v_p(n!)} v_p(n!) \log p \;&=\; \sum_{r=1}^{R_1-1} (-1)^r r \left(\vartheta\Big(\frac{n}{r}\Big) - \vartheta\Big(\frac{n}{r+1}\Big) \right) \\ &=\; n \sum_{r=1}^{R_1-1} \frac{(-1)^r}{r+1} + O\Big(\frac{n}{(\log n)^{12}}\Big) \;=\; (\log 2 - 1)\, n + O\Big(\frac{n}{(\log n)^{12}}\Big). \end{aligned}$$

On $(\sqrt{n}, \frac{n}{R_1}]$, writing $v_p(n!) = \lfloor n/p \rfloor = \frac{n}{p} - \{n/p\}$ (where $\{n/p\}$ contributes $\ll \vartheta(n/R_1) \ll \frac{n}{(\log n)^{12}}$) and applying Lemma 2.3 with $H = \lfloor (\log n)^5 \rfloor$ and Lemma 2.4(i) ($\delta = 1/3$, $T_0 \geq R_1 \gg (\log n)^{12}$) with $w(t) = \frac{n}{t}$ ($W_0 = \frac{n}{P}$) across $O(\log n)$ dyadic intervals $(P, 2P]$ bounds the intermediate sum by $O(\frac{n}{\log n})$. Combining this with Lemma 2.2(ii) establishes (2.17).

(ii)–(iii) Fix any $\delta \in (0, 1/3]$ and set $R := \lfloor 1/\delta \rfloor$, $A_R := 6R + 16$, and $H := \lfloor (\log n)^{R+2} \rfloor$. Partition $(n^\delta, \sqrt{n}]$ into the $R-1$ transition collars $\mathcal{C}_r(n) = (\frac{n^{1/r}}{(\log n)^{A_R/r}}, n^{1/r}]$ ($2 \leq r \leq R$) of (2.13) and the complementary inter-collar intervals $\mathcal{D}_r(n) := (n^{1/(r+1)}, \frac{n^{1/r}}{(\log n)^{A_R/r}}]$ ($2 \leq r \leq R-1$, plus $(n^\delta, n^{1/R}]$ if $\delta < 1/R$).

On each collar $\mathcal{C}_r(n)$, partitioning into the $M_r(n) \leq (\log n)^{A_R}$ subintervals J_{r,m_r} of (2.14) fixes $\lfloor n/p^r \rfloor = m_r$, while $\lfloor n/p^s \rfloor = 0$ for $s > r$. Expanding the $r-1$ active floors $(-1)^{\lfloor n/p^s \rfloor}$ ($1 \leq s \leq r-1$) via Lemma 2.3 produces a main product $\prod_{s=1}^{r-1} \widetilde{\Psi}_H(n/p^s)$ and an error product whose zero-frequency constant term is bounded by $\ll_R \frac{1}{H}$ (since $|\widetilde{\Psi}_H(x)| \leq 1 + \widetilde{B}_H(x)$). Summing the zero-frequency error over $p \in \mathcal{C}_r(n)$ contributes $\ll_R \frac{1}{H} \sum_{p \in \mathcal{C}_r(n)} \frac{\log p}{p-1} \ll_R \frac{\log\log n}{(\log n)^{R+2}} \ll (\log n)^{-1}$. For every non-zero frequency tuple $\mathbf{h} = (h_1, \ldots, h_{r-1}) \in \mathbb{Z}^{r-1} \setminus \{\mathbf{0}\}$ ($|h_s| \leq 2H+1$), the active index $r_0 := \min\{1 \leq$

$s \le r-1 : h_s \neq 0\}$ satisfies $r_0 \le r-1$, so for $p \in \mathcal{C}_r(n)$ the phase parameter $T_0 = \frac{|h_{r_0}|n}{P^{r_0}}$ obeys the polynomial lower bound

$$T_0 \;\ge\; \frac{n}{(n^{1/r})^{r_0}} \;=\; n^{1-r_0/r} \;\ge\; n^{1/r} \;\ge\; n^{\delta}.$$

Hence every non-zero frequency tuple on J_{r,m_r} lies in the polynomial power-saving regime of Lemma 2.4(ii) with $w(t) = \frac{1}{t-1}$ ($W_0 \asymp P^{-1}$), contributing $\ll_\delta n^{-\eta_\delta}$ per subinterval and per tuple. Summing over all $M_r(n) \le (\log n)^{A_R}$ subintervals and all $(4H+3)^{r-1} \ll_R (\log n)^{(R-1)(R+2)}$ tuples gives a total oscillatory contribution of $\ll_\delta (\log n)^{A_R+(R-1)(R+2)} n^{-\eta_\delta} \ll_\delta (\log n)^{-1}$.

On each inter-collar interval $\mathcal{D}_r(n)$, covered by $O(\log n)$ dyadic intervals $(P, 2P]$, we expand $\prod_{s=1}^{r}(-1)^{\lfloor n/p^s \rfloor}$ via Lemma 2.3. The zero-frequency constant term of the error product is bounded by $\ll_R \frac{1}{H}$, which summed over $p \in \mathcal{D}_r(n)$ contributes $\ll_R \frac{\log n}{(\log n)^{R+2}} \le (\log n)^{-3}$. For the non-zero frequency tuples $\mathbf{h} \in \mathbb{Z}^r \setminus \{\mathbf{0}\}$, the L^1 norm of the coefficients in $\prod_{s=1}^{r} \widetilde{\Psi}_H(n/p^s)$ (and in the mixed $\widetilde{B}_H$ terms) is $\ll_R (\log H)^r \ll_R (\log\log n)^r$. Applying Lemma 2.4(i) ($W_0 P \asymp 1$) to each non-zero tuple $\mathbf{h}$ and multiplying by the L^1 coefficient norm and the $O(\log n)$ dyadic blocks gives $\ll_\delta (\log\log n)^R (\log n)(\log n)^{-(R+2)} \ll_\delta (\log n)^{-1}$, proving (2.18). Combining (2.18) with Mertens' bound $\left|\sum_{3\le p\le n^\delta} (-1)^{v_p(n!)-1} \frac{\log p}{p-1}\right| \le \delta \log n + O(1)$ and letting $\delta \downarrow 0$ yields $\Sigma^*_{\mathrm{parity}}(n) = o(\log n)$, $\Delta_{\mathrm{parity}}(n) = o(n \log n)$, and (2.19). Replacing $(-1)^{v_p(n!)}$ by the additive characters $e(z v_p(n!)/h)$ $(1 \le z \le h-1)$ via $\mathbf{1}_{v_p(n!)\equiv c \pmod h} = \frac{1}{h}\sum_{z=0}^{h-1} e(-zc/h) e(z v_p(n!)/h)$ on $(n^a, n^b]$ identically establishes (2.20). □

Finally, we record the limiting Rademacher distribution of the r-digit truncated base-p valuation-parity series as n varies, which follows from the Chinese Remainder Theorem applied to the odd-indexed base-p digits of Lemma 2.2(i).

Proposition 2.6 (Jessen–Wintner Rademacher Limit Distribution of Truncated Base-p Parity Series)**.** *For each odd prime $p \ge 3$ and integer $r \ge 1$, let $\xi_p^{(r)}(n) := -\prod_{m=0}^{r-1} (-1)^{d_{2m+1}^{(p)}(n)}$ denote the r-digit truncated base-p parity sign of Lemma 2.2(i), and let $\Sigma_{\mathrm{parity}}^{(r)}(n; B) := \sum_{3\le p\le B} \xi_p^{(r)}(n) \frac{\log p}{p-1}$. As $N \to \infty$ (with $n \in \{1, \dots, N\}$ drawn uniformly), followed by $r \to \infty$ and $B \to \infty$, $\Sigma_{\mathrm{parity}}^{(r)}(n; B)$ converges in distribution and in all moments L^{2m} to the Rademacher random series $\Sigma_\infty := \sum_{p\ge 3} \xi_p \frac{\log p}{p-1}$, where $(\xi_p)_{p\ge 3}$ are independent symmetric Rademacher random variables ($\mathbb{P}(\xi_p = \pm 1) = 1/2$), with characteristic function $\Phi_{\Sigma_\infty}(\tau) = \prod_{p\ge 3} \cos(\frac{\tau \log p}{p-1})$ and a smooth, strictly positive, symmetric probability density on $\mathbb{R}$.*

Proof. For each odd prime p, the digits $d_{2m+1}^{(p)}(n)$ $(0 \le m \le r-1)$ depend only on $n \bmod p^{2r}$, so $\xi_p^{(r)}(n)$ is strictly periodic in n of period p^{2r}. For any fixed $B \ge 3$, $r \ge 1$, and distinct odd primes $3 \le p_1 < \cdots < p_k \le B$, set $M_{k,r} := \prod_{i=1}^{k} p_i^{2r}$. By the Chinese Remainder Theorem, as n runs through a complete period modulo $M_{k,r}$, the residues $n \bmod p_i^{2r}$ vary independently, and within a period modulo p_i^{2r} the r digits $d_{2m+1}^{(p_i)}(n) \in \{0, \dots, p_i - 1\}$ $(0 \le m < r)$ are independent and uniform. Since p_i is odd, $\frac{1}{p_i}\sum_{d=0}^{p_i-1}(-1)^d = \frac{1}{p_i}$, so $\mathbb{E}_{n \bmod p_i^{2r}}[\xi_{p_i}^{(r)}(n)] = -p_i^{-r}$ and $\lim_{N\to\infty} \frac{1}{N}\sum_{n=1}^{N} \prod_{i=1}^{k} \xi_{p_i}^{(r)}(n) = (-1)^k \prod_{i=1}^{k} p_i^{-r}$, which converges to 0 as $r \to \infty$. Since $\sum_{p>B} (\frac{\log p}{p-1})^2 \ll \frac{\log B}{B} \to 0$ as $B \to \infty$, Billingsley's convergence-together theorem yields weak and L^{2m} convergence to Σ_∞, and the Jessen–Wintner law of pure types together with rapid decay of $|\Phi_{\Sigma_\infty}(\tau)|$ as $|\tau| \to \infty$ implies that Σ_∞ has a smooth, strictly positive, symmetric density on $\mathbb{R}$. □

3. Chebyshev–Lucas Valuation Absorption and Proofs of Theorems 1.1 and 1.2

We now apply the valuation-parity equidistribution of Theorem 2.5 to the Pell embedding $m + s_n\sqrt{q_n} = \varepsilon_n^k$ in $K_n = \mathbb{Q}(\sqrt{q_n})$ and prove Theorem 1.1 and Theorem 1.2.

3.1. Basic Invariants of $K_n = \mathbb{Q}(\sqrt{q_n})$ and Chebyshev–Lucas Valuation Laws.

Lemma 3.1 (Fundamental Unit Norm and Growth of q_n, s_n)**.** *For every integer $n \geq 3$ with $n \neq 6$:*

(i) *The square-free integer q_n has at least one prime divisor $p \equiv 3 \pmod 4$. Consequently, the negative Pell equation $x^2 - q_n y^2 = -1$ has no rational or integer solutions, and both the fundamental unit $\varepsilon_n = X_1(q_n) + Y_1(q_n)\sqrt{q_n} > 1$ of $\mathbb{Z}[\sqrt{q_n}]$ and the fundamental unit $\eta_n > 1$ of $\mathcal{O}_{K_n}$ have norm $\mathrm{N}_{K_n/\mathbb{Q}}(\varepsilon_n) = \mathrm{N}_{K_n/\mathbb{Q}}(\eta_n) = +1$, with $[\mathcal{O}_{K_n}^\times : \mathbb{Z}[\sqrt{q_n}]^\times] = d_n \in \{1, 3\}$ ($d_n = 1$ unless $q_n \equiv 5 \pmod 8$). Moreover, for every $n \geq 116$ (and for $8 \leq n \leq 115$ with $n \neq 10$), $\log q_n > 0.4\, n$, while for all $n \geq 8$, $\log \varepsilon_n > 0.2\, n$ and any solution (n, m) to $n! + 1 = m^2$ satisfies the explicit bound $1 \leq k < 2.5 \log n$ (with $2.5 \log n < \frac{n}{5}$ for all $n \geq 116$).*

(ii) *As $n \to \infty$, $\log(s_n\sqrt{q_n}) = \frac{1}{2}\log(n!) = \frac{1}{2}n\log n - \frac{1}{2}n + \frac{1}{4}\log(2\pi n) + O(n^{-1})$ and $\log s_n = \frac{1}{2}n\log n - \frac{1+\log 2}{2}n + O\big(ne^{-c_0\sqrt{\log n}}\big)$.*

Proof. (i) For $n \in \{3, 4, 5\}$, $v_3(n!) = 1$, so $3 \mid q_n$; for $7 \leq n \leq 13$, $v_7(n!) = 1$, so $7 \mid q_n$. For $n \geq 14$, we have $x := n/2 \geq 7$, so Breusch's generalization of Bertrand's postulate to arithmetic progressions modulo 4 [8] guarantees the existence of a prime $p \in (n/2, n]$ with $p \equiv 3 \pmod 4$; for any such prime, $v_p(n!) = 1$, so $p \mid q_n$. Reducing $x^2 - q_n y^2 = -1$ modulo $p \equiv 3 \pmod 4$ ($p \mid q_n$) gives $\left(\frac{-1}{p}\right) = +1$, a contradiction, so $\mathrm{N}(\varepsilon_n) = \mathrm{N}(\eta_n) = +1$. Furthermore, since every prime $p \in (n/2, n]$ divides q_n, Rosser and Schoenfeld's explicit Chebyshev bounds [48, Theorem 10] give $\log q_n \geq \vartheta(n) - \vartheta(n/2) > 0.4\, n$ for all $n \geq 116$ (and direct check verifies $\log q_n > 0.4\, n$ for $8 \leq n \leq 115$ with $n \neq 10$, while at $n = 10$, $q_{10} = 7$ and $\varepsilon_{10} = 8 + 3\sqrt{7}$ satisfies $\log \varepsilon_{10} \approx 2.7686 > 0.2 \times 10 = 2.0$). Thus $\log \varepsilon_n > 0.2\, n$ and $k = \frac{\log(m + s_n\sqrt{q_n})}{\log \varepsilon_n} < \frac{\frac{1}{2}n\log n}{0.2\, n} = 2.5 \log n$ for all $n \geq 8$ (with $2.5 \log n < \frac{n}{5}$ for all $n \geq 116$). Part (ii) follows from Stirling's formula and Lemma 2.1(ii). □

Define the Lucas–Pell sequence $(X_r(q_n), Y_r(q_n))_{r \geq 1}$ by $X_r(q_n) + Y_r(q_n)\sqrt{q_n} := \varepsilon_n^r$.

Lemma 3.2 (Chebyshev–Lucas Factorization and p-Adic Lifting the Exponent)**.** *For every integer $k \geq 1$, we have $X_k(q_n) = T_k(X_1(q_n))$ and $Y_k(q_n) = Y_1(q_n)U_{k-1}(X_1(q_n))$, where $U_{k-1}(x) \in \mathbb{Z}[x]$ is the Chebyshev polynomial of the second kind:*

$$U_{k-1}(X_1(q_n)) \;=\; \sum_{j=0}^{\lfloor (k-1)/2 \rfloor} \binom{k}{2j+1} X_1(q_n)^{k-1-2j}\big(q_n Y_1(q_n)^2\big)^j. \tag{3.1}$$

Moreover:

(i) $\gcd(X_1(q_n)^2 - 1, U_{k-1}(X_1(q_n))^2)$ *divides k^2, and for every prime p dividing $q_n Y_1(q_n)$,*

$$v_p\Big(U_{k-1}(X_1(q_n))\Big) \;=\; v_p(k). \tag{3.2}$$

In particular, for every prime $p \in \mathcal{P}_{\mathrm{odd}}(n)$ in $m + s_n\sqrt{q_n} = \varepsilon_n^k$,

$$v_p\big(q_n Y_1(q_n)^2\big) \;=\; v_p(n!) - 2v_p(k). \tag{3.3}$$

(ii) *For every odd prime $p \nmid q_n$, the projective rank of apparition $\alpha_p(q_n) := \min\{r \geq 1 : p \mid Y_r(q_n)\}$ satisfies $\alpha_p(q_n) \mid (p - \left(\frac{q_n}{p}\right))/2$, and $p \mid Y_k(q_n)$ if and only if $\alpha_p(q_n) \mid k$, in which case $v_p(Y_k(q_n)) = v_p(Y_{\alpha_p(q_n)}(q_n)) + v_p(k/\alpha_p(q_n))$.*

Proof. Expanding $\varepsilon_n^k = (X_1(q_n) + Y_1(q_n)\sqrt{q_n})^k$ by the binomial theorem and substituting $q_n Y_1(q_n)^2 = X_1(q_n)^2 - 1$ gives (3.1). Let p be any prime dividing $M := q_n Y_1(q_n)^2 = X_1(q_n)^2 - 1$,

so $\gcd(p, X_1(q_n)) = 1$. In (3.1), the $j = 0$ term is $kX_1(q_n)^{k-1}$ of p-adic valuation $v_p(k)$. For every $j \geq 1$, the identity $(2j+1)\binom{k}{2j+1} = k\binom{k-1}{2j}$ implies

$$v_p\left(\binom{k}{2j+1}M^j\right) \;\geq\; v_p(k) - v_p(2j+1) + j\,v_p(M) \;\geq\; v_p(k) + (j - v_p(2j+1)).$$

Since $j > v_p(2j+1)$ for all $j \geq 1$ and all primes $p \geq 2$ (as $v_2(2j+1) = 0 < j$ and $p^j \geq 3^j > 2j+1$ for $p \geq 3$), every $j \geq 1$ term has valuation strictly exceeding $v_p(k)$, proving (3.2). For $p \in \mathcal{P}_{\mathrm{odd}}(n)$, $v_p(q_n) = 1$, so $p \mid q_n Y_1(q_n)$ and taking valuations in $n! = q_n Y_1(q_n)^2 U_{k-1}(X_1(q_n))^2$ yields (3.3). Part (ii) follows by applying part (i) to $\varepsilon_n^{\alpha_p(q_n)}$ [14]. □

3.2. **Proof of Theorem 1.1 and Rank-of-Apparition Collapse.**

Proof of Theorem 1.1. For $n \in \{4, 5, 7\}$, direct evaluation gives $q_4 = 6, \varepsilon_4 = 5 + 2\sqrt{6}, k(4) = 1$; $q_5 = 30, \varepsilon_5 = 11 + 2\sqrt{30}, k(5) = 1$; and $q_7 = 35, \varepsilon_7 = 6 + \sqrt{35}, 71 + 12\sqrt{35} = \varepsilon_7^2, k(7) = 2$.

Now let (n, m) be a solution with $n \geq 8$. Since $\varepsilon_n = X_1(q_n) + Y_1(q_n)\sqrt{q_n} > 2Y_1(q_n)\sqrt{q_n}$ and $\varepsilon_n^k = m + s_n\sqrt{q_n} = (2 + O(1/n!))\sqrt{n!}$, taking logarithms gives

$$\log(q_n Y_1(q_n)^2) \;=\; 2\log(Y_1(q_n)\sqrt{q_n}) \;<\; 2\log\varepsilon_n \;=\; \frac{1}{k}\log(n!) + O(1). \tag{3.4}$$

On the other hand, summing (3.3) over all $p \in \mathcal{P}_{\mathrm{odd}}(n)$ gives

$$\log(q_n Y_1(q_n)^2) \;\geq\; S_{\mathrm{odd}}(n) - 2\log k. \tag{3.5}$$

Since $k < 2.5\log n$ by Lemma 3.1(i), Theorem 2.5(ii) ($S_{\mathrm{odd}}(n) = (\frac{1}{2} + o(1))\log(n!)$) combined with (3.4)–(3.5) yields $\frac{1}{2} - o(1) \leq \frac{1}{k}$, forcing $k \in \{1, 2\}$ for all sufficiently large n. Taking logarithms of $\varepsilon_n^k = (2 + o(1))(n!)^{1/2}$ gives (1.7).

Finally, suppose that $q_{n_1} = q_{n_2} = D$ for two solutions $4 \leq n_1 < n_2$. Then every prime $p \in (n_1, n_2]$ must divide $n_2!$ to an even power; since every prime $p \in (n_2/2, n_2]$ has $v_p(n_2!) = 1$, the interval $(n_1, n_2]$ cannot contain any prime in $(n_2/2, n_2]$, so $n_1 \geq n_2/2$, i.e., $n_2 < 2n_1$. By Berndt and Galway [7], the only solutions with $n_1 \leq 10^9$ are $n \in \{4, 5, 7\}$, whose square-free cores $q_4 = 6$, $q_5 = 30$, and $q_7 = 35$ are pairwise distinct. For $n_1 > 10^9$, both $m_{n_1} + s_{n_1}\sqrt{D} = \varepsilon_D^{k_{n_1}}$ and $m_{n_2} + s_{n_2}\sqrt{D} = \varepsilon_D^{k_{n_2}}$ are powers of the same fundamental unit ε_D with $1 \leq k_{n_1} < k_{n_2}$, so $k_{n_2} \geq k_{n_1} + 1$. Since $k_{n_1}\log\varepsilon_D = \frac{1}{2}\log(n_1!) + \log 2 + o(1)$ and $k_{n_1} < 2.5\log n_1$ by Lemma 3.1(i), we have

$$\begin{aligned}(n_2 - n_1)\log(2n_1) \;&>\; \sum_{j=n_1+1}^{n_2} \log j \;=\; \log(n_2!) - \log(n_1!) \;=\; 2(k_{n_2} - k_{n_1})\log\varepsilon_D + O(1/n_1!) \\ &\geq\; \frac{\log(n_1!)}{k_{n_1}} \;>\; \frac{n_1\log n_1 - n_1}{2.5\log n_1} \;>\; 0.38\,n_1,\end{aligned}$$

and hence $n_2 - n_1 > \frac{0.38\,n_1}{\log(2n_1)} > \frac{n_1}{16597}$ for all $n_1 > 10^9$. However, by Schoenfeld's explicit prime gap theorem [50], $\vartheta(x + x/16597) > \vartheta(x)$ for all $x \geq 2010760$, so $(n_1, n_2]$ must contain a prime $p \in (n_1, n_2] \subset (n_2/2, n_2]$, a contradiction. Thus $q_{n_1} \neq q_{n_2}$ for all $4 \leq n_1 < n_2$. □

Remark 3.3 (Dual Even-Valuation Chebyshev–Lucas Elimination of Odd Prime Divisors $\ell \geq 3$ of k). A dual application of Lemma 3.2(i) on the even-valuation prime set $\mathcal{P}_{\mathrm{even}}(n)$ independently eliminates every odd prime divisor $\ell \geq 3$ of k (proving $k = 2^a$). Indeed, if an odd prime $\ell \geq 3$ divides $k = \ell r$, setting $\tilde{\varepsilon} := \varepsilon_n^r = \tilde{X} + \tilde{Y}\sqrt{q_n}$ gives $s_n = \tilde{Y}\,U_{\ell-1}(\tilde{X})$ with $\gcd(q_n\tilde{Y}^2, U_{\ell-1}(\tilde{X})^2) \mid \ell^2$. Thus every prime $p \mid U_{\ell-1}(\tilde{X})$ with $p \neq \ell$ is coprime to q_n and belongs to $\mathcal{P}_{\mathrm{even}}(n)$ with $p \equiv \pm 1 \pmod{\ell}$ [14], while $v_\ell(U_{\ell-1}(\tilde{X})) = 1$ if $\ell \mid q_n\tilde{Y}$ by (3.2), whence

$$\frac{\ell-1}{\ell}\log(n!) - O(1) \;\leq\; \log\left(U_{\ell-1}(\tilde{X})^2\right) \;\leq\; S_{\mathrm{even}}(n) + 2\log\ell + v_2(n!)\log 2 \;=\; \left(\frac{1}{2} + o(1)\right)\log(n!),$$

contradicting $\frac{\ell-1}{\ell} \geq \frac{2}{3} > \frac{1}{2}$ for all sufficiently large n.

As an immediate consequence of Theorem 1.1 and Lemma 3.2(ii), the rank of apparition of ε_n collapses across the entire second harmonic layer $I_2 = (n/3, n/2]$.

Proposition 3.4 (Rank-of-Apparition Collapse on $I_2 = (n/3, n/2]$)**.** *Let (n, m) be a solution to $n! + 1 = m^2$ with $k \in \{1, 2\}$ and $n \geq 8$. For every prime $p \in I_2 = (n/3, n/2]$, we have $p \nmid q_n$, $v_p(s_n) = 1$, and the projective Pell rank of apparition $\alpha_p(q_n)$ satisfies*

$$\alpha_p(q_n) \mid k \in \{1, 2\} \implies \alpha_p(q_n) \in \{1, 2\}, \qquad v_p(Y_{\alpha_p(q_n)}(q_n)) = 1. \tag{3.6}$$

Proof. By Lemma 2.1(i), every prime $p \in I_2 = (n/3, n/2]$ ($p \geq 5$ for $n \geq 9$) satisfies $v_p(n!) = 2$, so $p \nmid q_n$ and $v_p(Y_k(q_n)) = v_p(s_n) = 1$. Since $k \in \{1, 2\}$ by Theorem 1.1, Lemma 3.2(ii) yields $\alpha_p(q_n) \mid k \in \{1, 2\}$ and $v_p(Y_{\alpha_p(q_n)}(q_n)) = 1$. □

3.3. Proof of Theorem 1.2, Fermat Quotients, and Imaginary Quadratic Picard 2-Torsion.

Proof of Theorem 1.2. Let (n, m) be a solution to $n! + 1 = m^2$ with $n \geq 116$. Let $\eta_n > 1$ be the fundamental unit of $\mathcal{O}_{K_n}$ ($\varepsilon_n = \eta_n^{d_n}$ with $d_n \in \{1, 3\}$), and let $\widetilde{R}_p(K_n) := \frac{\log_p(\eta_n)}{\sqrt{D_n}} \in \mathbb{Z}_p$ be the normalized p-adic regulator of $K_n = \mathbb{Q}(\sqrt{q_n})$. By Nagura's theorem [41] (which guarantees a prime in $(x, \frac{6}{5}x)$ for all $x \geq 25$, applied at $x = n/5$ for $n \geq 125$, and verified directly for $116 \leq n \leq 124$), the interval $I_4 = (n/5, n/4]$ is non-empty for every $n \geq 116$, and every prime $p \in I_4$ satisfies $p > n/5 \geq 23.2 > \sqrt{n}$, so $v_p(n!) = \lfloor n/p \rfloor = 4$ and $p \in \mathcal{P}_{\text{even}}(n)$. Moreover, by Lemma 3.1(i), $1 \leq k < 2.5 \log n < \frac{n}{5} < p$, so $p \nmid k$ unconditionally for every $n \geq 116$ and every $p \in I_4$ (and for all sufficiently large n, $k \in \{1, 2\}$ by Theorem 1.1, so $p \nmid k$ for all primes $p \geq 5$).

For any prime $p \in [5, n/2]$ with $p \nmid k$, we have $v_p(2md_nk) = 0$. In $\mathbb{Q}_p(\sqrt{q_n})$,

$$\eta_n^{2d_nk} = \varepsilon_n^{2k} = \left(m + s_n\sqrt{q_n}\right)^2 = (1 + 2q_ns_n^2) + 2ms_n\sqrt{q_n}.$$

Since $p \leq n/2 \implies v_p(n!) \geq 2 \implies v_p(s_n) = \lfloor v_p(n!)/2 \rfloor \geq 1$ and $p \geq 5$, the p-adic logarithm $\log_p(1 + z) = \sum_{j=1}^{\infty} \frac{(-1)^{j-1}}{j} z^j$ is valuation-preserving on $z = 2q_ns_n^2 + 2ms_n\sqrt{q_n} \in p\mathcal{O}_{\mathbb{Q}_p(\sqrt{q_n})}$. Since $v_p(2q_ns_n^2) = v_p(n!) > \lfloor v_p(n!)/2 \rfloor + \frac{1}{2}v_p(q_n) = v_p(2ms_n\sqrt{q_n})$, the ultrametric triangle inequality gives

$$v_p(\widetilde{R}_p(K_n)) = v_p(s_n) = \left\lfloor \frac{v_p(n!)}{2} \right\rfloor \qquad \forall\, p \in [5, n/2],\ \ p \nmid k. \tag{3.7}$$

In particular, every prime $p \in [5, n/4]$ with $p \nmid k$ satisfies $v_p(n!) \geq 4 \implies v_p(\widetilde{R}_p(K_n)) \geq 2 \implies \eta_n^{p^2-1} \equiv 1 \pmod{p^2\mathcal{O}_{K_n}}$ (so p is a Wieferich prime for η_n).

Now let $p \in \mathcal{P}_{\text{even}}(n)$ with $p \geq 5$, $p \nmid k$, and $v_p(n!) \geq 4$ (which includes every prime $p \in I_4 = (n/5, n/4]$ for all $n \geq 116$, and for large n includes all primes in the even harmonic layers $I_{2r} = (\frac{n}{2r+1}, \frac{n}{2r}]$ for $2 \leq r \leq \lfloor(\lfloor\sqrt{n}\rfloor - 1)/2\rfloor$, a set of natural density $\sum_{r=2}^{\infty}(\frac{1}{2r} - \frac{1}{2r+1}) = \frac{5}{6} - \log 2 \approx 0.14019$). Since $2 \mid v_p(n!)$, $p \nmid D_n$ is unramified in K_n, so $\chi_{D_n}(p) \in \{\pm 1\}$ and $v_p(1 - \chi_{D_n}(p)p^{-1}) = -1$. By Leopoldt's p-adic analytic class number formula [61, Theorem 5.24],

$$\left(1 - \frac{\chi_{D_n}(p)}{p}\right)^{-1} L_p(1, \chi_{D_n}) = \frac{2h(q_n)R_p(K_n)}{\sqrt{D_n}} = 2h(q_n)\,\widetilde{R}_p(K_n). \tag{3.8}$$

Taking p-adic valuations in (3.8) via (3.7) yields

$$v_p(L_p(1, \chi_{D_n})) = v_p(h(q_n)) + \frac{1}{2}v_p(n!) - 1 \geq 2 - 1 = 1.$$

By Iwasawa's construction [61, Theorem 7.10], $L_p(1 - s, \chi_{D_n}) = G_p((1 + p)^s - 1, \chi_{D_n})$ for a formal power series $G_p(T, \chi_{D_n}) \in \mathbb{Z}_p[[T]]$. By the Ferrero–Washington theorem [23], $\mu_p(\chi_{D_n}) = 0$, so by the Weierstrass preparation theorem $G_p(T, \chi_{D_n}) = U(T)P_p(T)$ where $U(T) \in \mathbb{Z}_p[[T]]^{\times}$ and $P_p(T) \in \mathbb{Z}_p[T]$ is a distinguished polynomial of degree $\lambda_p(\chi_{D_n}) = \lambda_p(K_n)$ (by the Mazur–Wiles Iwasawa Main Conjecture [40]). Evaluating at $T_1 = (1 + p)^{-1} - 1 \in p\mathbb{Z}_p$ gives $v_p(P_p(T_1)) = v_p(L_p(1, \chi_{D_n})) \geq 1$, which is impossible if $\deg P_p = \lambda_p(K_n) = 0$. Thus $\lambda_p(K_n) \geq 1$, proving (1.8). Consequently, if Greenberg's conjecture $\lambda_p(K_n) = 0$ [27] holds at a single prime $p \in (n/5, n/4]$, no solution to (1.1)

exists with $n \geq 116$; since direct finite verification on $1 \leq n \leq 115$ yields only $(4,5),(5,11)$, and $(7,71)$, these three pairs are the only solutions to (1.1). □

Corollary 3.5 (Rational Fermat Quotient Locks and Sinnott Circular-Unit Congruence)**.** *Let (n,m) be a solution to $n!+1=m^2$ with rational splitting $m-1=d_1x_n^2$ and $m+1=d_2y_n^2$. For every odd prime $p \leq n/2$, the Fermat quotients $q_p(a)=\frac{a^{p-1}-1}{p} \bmod p$ satisfy $q_p(d_1) \equiv q_p(2)-2q_p(x_n) \pmod p$ when $p \mid (m+1)$ and $q_p(-d_2) \equiv q_p(2)-2q_p(y_n) \pmod p$ when $p \mid (m-1)$. Moreover, writing $m \equiv \varepsilon_p \in \{\pm 1\} \pmod{p^{v_p(n!)}}$, in $\mathbb{Q}_p(\sqrt{q_n})$ we have $\frac{d_n k \log_p(\eta_n)}{s_n\sqrt{q_n}} = \varepsilon_p \sum_{j=0}^{\infty} \binom{-1/2}{j} \frac{q_n^j s_n^{2j}}{2j+1}$, and for Sinnott's circular unit $\xi_n = \pm\eta_n^{h(q_n)}$ [54], $h(q_n) \equiv \varepsilon_p d_n k \frac{\log_p(\xi_n)}{s_n\sqrt{q_n}} \pmod{p^{v_p(n!)}}$ for all $p \in [5, n/2]$, while at $p=3$, whenever $n \geq 15$ satisfies $2 \mid v_3(n!)$ (or $n \geq 9$ with $2 \mid v_3(n!)$ and $3 \nmid d_n k$), Kummer's congruence yields $3 \mid (1-\chi_{-3D_n}(3))h(\mathbb{Q}(\sqrt{-3q_n}))$.*

Proof. Reducing $d_1x_n^2 = m-1 \equiv -2 \pmod{p^2}$ (when $p \mid m+1$) and $-d_2y_n^2 \equiv -2 \pmod{p^2}$ (when $p \mid m-1$) via $q_p(-1) \equiv 0 \pmod p$ and $q_p(ab^2) \equiv q_p(a)+2q_p(b) \pmod p$ gives the Fermat-quotient congruences. Taking $\log_p$ of $\varepsilon_p\eta_n^{d_nk} = \exp_p(\varepsilon_p \operatorname{arcsinh}_p(s_n\sqrt{q_n}))$ and multiplying by $h(q_n)$ yields the circular-unit congruence. At $p=3$, when $2 \mid v_3(n!)$ we have $3 \in \mathcal{P}_{\text{even}}(n)$ (so $3 \nmid D_n$ is unramified in K_n and $v_3(1-\chi_{D_n}(3)/3) = -1$) and $v_3(s_n) = \frac{1}{2}v_3(n!)$. For $n \geq 15$ with $2 \mid v_3(n!)$, we have $v_3(n!) \geq 6$ and $v_3(d_nk) \leq 1$, so $v_3(\widetilde{R}_3(K_n)) \geq v_3(s_n) - v_3(d_nk) \geq 2$ and $v_3(L_3(1,\chi_{D_n})) \geq 2-1=1$; Kummer's congruence $(1-\chi_{-3D_n}(3))B_{1,\chi_{-3D_n}} \equiv L_3(1,\chi_{D_n}) \equiv 0 \pmod 3$ [61, Theorem 5.11] then gives the 3-divisibility. □

We conclude this section by showing that the dichotomy $k \in \{1,2\}$ of Theorem 1.1 admits a dual characterization in the Picard group of the imaginary quadratic order $\mathcal{O}_n^- := \mathbb{Z}[\sqrt{-q_n}]$.

Theorem 3.6 (Imaginary Quadratic Picard 2-Torsion Dichotomy)**.** *Let (n,m) be a solution to $n!+1=m^2$ with $n \geq 4$. In the imaginary quadratic order $\mathcal{O}_n^- = \mathbb{Z}[\sqrt{-q_n}]$, the principal ideal $(1+s_n\sqrt{-q_n})$ is the square of a unique invertible integral ideal $\mathfrak{J}_n \subset \mathcal{O}_n^-$ of norm m, defining a 2-torsion class $[\mathfrak{J}_n] \in \operatorname{Pic}(\mathcal{O}_n^-)[2]$ such that*

$$[\mathfrak{J}_n] = 1 \text{ in } \operatorname{Pic}(\mathcal{O}_n^-) \iff k=2, \qquad [\mathfrak{J}_n] = [\mathfrak{d}_2] \neq 1 \text{ in } \operatorname{Pic}(\mathcal{O}_n^-) \iff k=1, \tag{3.9}$$

where $\mathfrak{d}_2 = (\delta_2(n), \sqrt{-q_n})$ is the ambiguous ramified ideal above $\delta_2(n) = \operatorname{core}((m+1)/2)$. Moreover, in the $k=2$ regime, the second-stage ideal $\mathfrak{I}_n \subset \mathcal{O}_{\mathbb{Q}(\sqrt{-q_n})}$ satisfying $(1+Y_1(q_n)\sqrt{-q_n})_{\text{odd}} = \mathfrak{I}_n^2$ has exact order 2 ($[\mathfrak{I}_n] \neq 1$) in $\operatorname{Cl}(\mathbb{Q}(\sqrt{-q_n}))[2]$ unconditionally.

Proof. In $\mathcal{O}_n^- = \mathbb{Z}[\sqrt{-q_n}]$, $(1+s_n\sqrt{-q_n})(1-s_n\sqrt{-q_n}) = 1+q_ns_n^2 = m^2$. Since m is odd and $(1+s_n\sqrt{-q_n})+(1-s_n\sqrt{-q_n}) = 2$, the conjugate ideals $(1+s_n\sqrt{-q_n})$ and $(1-s_n\sqrt{-q_n})$ are coprime and prime to the conductor of $\mathcal{O}_n^-$, so $(1+s_n\sqrt{-q_n}) = \mathfrak{J}_n^2$ for a unique invertible ideal $\mathfrak{J}_n \subset \mathcal{O}_n^-$ of norm m. Since $q_n \geq 6$ is square-free, $\mathcal{O}_n^{-\times} = \{\pm 1\}$, so $\mathfrak{J}_n = (u+v\sqrt{-q_n})$ is principal if and only if $1+s_n\sqrt{-q_n} = \pm((u^2-q_nv^2)+2uv\sqrt{-q_n})$. By Lemma 3.1(i), $u^2-q_nv^2=-1$ is impossible, so the sign is $+$, giving $u^2-q_nv^2=1$, $s_n=2|uv|$, and $m+s_n\sqrt{q_n} = (|u|+|v|\sqrt{q_n})^2 = \varepsilon_n^2$, i.e., $k=2$. In the $k=1$ regime, multiplying $1+s_n\sqrt{-q_n}$ by $\delta_2(n)$ gives $\delta_2(n)(1+s_n\sqrt{-q_n}) = (\delta_2(n)z_2+z_1\sqrt{-q_n})^2$, so $(\mathfrak{d}_2\mathfrak{J}_n)^2 = (\delta_2(n)z_2+z_1\sqrt{-q_n})^2 \implies [\mathfrak{J}_n] = [\mathfrak{d}_2] \neq 1$. Finally, in the $k=2$ regime, if $[\mathfrak{I}_n]=1$, then $X_1(q_n)+Y_1(q_n)\sqrt{q_n}$ would be the square of a smaller unit in $\mathcal{O}_{K_n}^\times$, contradicting the minimality of ε_n. □

4. Arithmetic Structure of the Critical Regime $k = 2$

Theorem 1.1 bifurcates the study of (1.1) into two arithmetic regimes: the critical-capacity regime $k = 2$, where $\log \varepsilon_n = (\frac{1}{4} + o(1))n \log n$ saturates the Chebyshev–Lucas capacity bound (3.5), and the fundamental regime $k = 1$, where $m = X_1(q_n)$ and $s_n = Y_1(q_n)$. In this section we analyze the critical regime $k = 2$, establishing the $(n/2)$-smoothness asymmetry of $m + 1$, the first- and second-stage Euclidean division quartic laws, the stretched-exponential lower bounds on the midpoint cores $\delta_i(n) = \mathrm{core}(X_1(q_n) \mp 1)$, and the Stickelberger–Wilson quadratic reciprocity locks.

4.1. **Three-Consecutive-Integer Decomposition and the Two-Stage Euclidean Division Quartic Law.** When $k = 2$, expanding $m + s_n\sqrt{q_n} = \varepsilon_n^2 = (X_1(q_n) + Y_1(q_n)\sqrt{q_n})^2$ and substituting $X_1(q_n)^2 - q_n Y_1(q_n)^2 = 1$ yields

$$m \;=\; X_1(q_n)^2 + q_n Y_1(q_n)^2 \;=\; 2X_1(q_n)^2 - 1 \;=\; 2q_n Y_1(q_n)^2 + 1, \qquad s_n \;=\; 2X_1(q_n)Y_1(q_n). \tag{4.1}$$

Consequently, the factors $m - 1$ and $m + 1$ in $(m - 1)(m + 1) = n!$ take the form

$$m + 1 \;=\; 2X_1(q_n)^2, \qquad m - 1 \;=\; 2q_n Y_1(q_n)^2 \;=\; 2\big(X_1(q_n) - 1\big)\big(X_1(q_n) + 1\big), \tag{4.2}$$

and Brocard's equation (1.1) becomes the three-consecutive-integer factorial equation

$$n! \;=\; 4\big(X_1(q_n) - 1\big)\, X_1(q_n)^2\, \big(X_1(q_n) + 1\big). \tag{4.3}$$

We begin by recording the prime-support separation and biquadratic splitting laws forced by (4.3).

Proposition 4.1 ($(n/2)$-Smoothness of $m + 1$, Biquadratic Splitting, and Fermat-Quotient Lock for $k = 2$)**.** *Suppose (n, m) is a solution to $n! + 1 = m^2$ in the $k = 2$ regime with $n \geq 8$. Then:*

(i) *Every odd prime $p \leq n$ divides* at most one *of the three consecutive integers $X_1(q_n)-1$, $X_1(q_n)$, $X_1(q_n) + 1$. Specifically, every odd prime $p \in \mathcal{P}_{\mathrm{odd}}(n)$ divides $(X_1(q_n) - 1)(X_1(q_n) + 1) = q_n Y_1(q_n)^2$ to its full factorial power $p^{v_p(n!)}$, whereas every odd prime $p \mid X_1(q_n)$ belongs to $\mathcal{P}_{\mathrm{even}}(n)$ with $p^{v_p(n!)/2} \parallel X_1(q_n)$ and $p \leq n/2$. In particular, $m + 1 = 2X_1(q_n)^2$ is strictly $(n/2)$-smooth, and as $n \to \infty$,*

$$\log \mathrm{rad}(m-1) \;\geq \sum_{p \in \mathcal{P}_{\mathrm{odd}}(n)} \log p \;=\; (\log 2 + o(1))\, n, \qquad \log \mathrm{rad}(m+1) \;\leq\; (1 - \log 2 + o(1))\, n. \tag{4.4}$$

(ii) *Write $X_1(q_n) - 1 = \delta_1(n)u_n^2$ and $X_1(q_n) + 1 = \delta_2(n)v_n^2$ with $\delta_1(n), \delta_2(n)$ square-free. Every odd prime $p \mid X_1(q_n)$ splits completely in the biquadratic field $L_n := \mathbb{Q}(\sqrt{-\delta_1(n)}, \sqrt{\delta_2(n)})$:*

$$\left(\frac{-\delta_1(n)}{p}\right) \;=\; +1, \qquad \left(\frac{\delta_2(n)}{p}\right) \;=\; +1, \qquad \left(\frac{-q_n}{p}\right) \;=\; +1 \qquad \forall\, p \mid X_1(q_n),\ p \geq 3, \tag{4.5}$$

and satisfies the modulo-$p^{v_p(n!)}$ and Fermat-quotient congruences

$$q_n Y_1(q_n)^2 \;\equiv\; -1 \pmod{p^{v_p(n!)}}, \qquad q_p(-q_n) + 2\, q_p(Y_1(q_n)) \;\equiv\; 0 \pmod{p}. \tag{4.6}$$

Proof. (i) Since $\gcd(X_1(q_n), X_1(q_n) \mp 1) = 1$ and $\gcd(X_1(q_n) - 1, X_1(q_n) + 1) \mid 2$, the odd prime factors of $X_1(q_n)-1$, $X_1(q_n)$, and $X_1(q_n)+1$ are pairwise disjoint. By (4.3), any odd prime $p \mid X_1(q_n)$ satisfies $v_p(n!) = 2v_p(X_1(q_n))$, so $p \in \mathcal{P}_{\mathrm{even}}(n)$; by Lemma 2.1(i), every prime $p \in I_1 = (n/2, n]$ has $v_p(n!) = 1$ and thus lies in $\mathcal{P}_{\mathrm{odd}}(n)$, so $X_1(q_n)$ is $(n/2)$-smooth. Summing $\log p$ over $p \in \mathcal{P}_{\mathrm{odd}}(n)$ via Lemma 2.1(ii) gives (4.4).

(ii) Let $p \geq 3$ be a prime dividing $X_1(q_n)$. Since $p \in \mathcal{P}_{\mathrm{even}}(n)$ and $p \leq n/2$, we have $v_p(n!) \geq 2$ and $p^{v_p(n!)/2} \parallel X_1(q_n)$, so $X_1(q_n)^2 \equiv 0 \pmod{p^{v_p(n!)}}$. Reducing $q_n Y_1(q_n)^2 = X_1(q_n)^2 - 1$ modulo $p^{v_p(n!)}$ gives $q_n Y_1(q_n)^2 \equiv -1 \pmod{p^{v_p(n!)}}$, while reducing $\delta_1(n)u_n^2 = X_1(q_n) - 1$ and $\delta_2(n)v_n^2 = X_1(q_n) + 1$ modulo p gives $-\delta_1(n)u_n^2 \equiv +1 \pmod{p}$ and $\delta_2(n)v_n^2 \equiv +1 \pmod{p}$, proving (4.5) and (4.6). □

Proposition 4.1(i) shows that in the $k = 2$ regime, the odd prime factors of $X_1(q_n)$ are drawn exclusively from $\mathcal{P}_{\text{even}}(n)$, while every prime in $\mathcal{P}_{\text{odd}}(n)$ is confined to $q_n Y_1(q_n)^2 = X_1(q_n)^2 - 1$. To turn this prime-support separation into an exact Diophantine equation, define the explicit even-valuation and odd-valuation factorial integers

$$A_n^* := \prod_{\substack{3\le p\le n\\ p\in\mathcal{P}_{\text{even}}(n)}} p^{v_p(n!)}, \qquad B_n^* := \prod_{\substack{3\le p\le n\\ p\in\mathcal{P}_{\text{odd}}(n)}} p^{v_p(n!)}, \qquad Z := \prod_{\substack{3\le p\le n\\ p\in\mathcal{P}_{\text{even}}(n),\, p\mid Y_1(q_n)}} p^{v_p(n!)/2}, \tag{4.7}$$

and set $(A_n, B_n) := (A_n^*, 2^{v_2(n!)-2}B_n^*)$ if $X_1(q_n)$ is odd, and $(A_n, B_n) := (2^{v_2(n!)-2}A_n^*, B_n^*)$ if $X_1(q_n)$ is even (which requires $2 \mid v_2(n!)$ and $q_n \equiv 7 \pmod 8$). The following theorem establishes the resulting two-stage Euclidean division quartic law and stretched-exponential lower bounds on the midpoint cores $\delta_1(n)$ and $\delta_2(n)$.

Theorem 4.2 (Two-Stage Euclidean Division Quartic Law and Stretched-Exponential Midpoint Core Bounds)**.** *Suppose (n, m) is a solution to $n! + 1 = m^2$ in the $k = 2$ regime with $n \ge 8$. Then:*

(i) ***First-Stage Euclidean Division Quartic Law:*** *The integers A_n, B_n, Z satisfy $\gcd(A_n, B_n) = 1$, $X_1(q_n) = \frac{\sqrt{A_n}}{Z}$, $q_n Y_1(q_n)^2 = B_n Z^2$, and the exact quartic Diophantine identity*

$$A_n \;=\; B_n Z^4 + Z^2, \qquad 4\log Z \;=\; (c_2(n) - \Sigma^*_{\text{parity}}(n))n + O\Big(\frac{n}{\log n}\Big) \;=\; o(n\log n), \tag{4.8}$$

where $c_2(n) = -1$ if $X_1(q_n)$ is odd and $c_2(n) = 2\log 2 - 1 \approx 0.38629$ if $X_1(q_n)$ is even. For all sufficiently large n, $0 < Z^2 < B_n$, so Euclidean division $A_n = Q_n B_n + R_n$ $(0 \le R_n < B_n)$ satisfies the simultaneous quotient-remainder lock

$$Q_n \;=\; Z^4, \qquad R_n \;=\; Z^2, \qquad R_n^2 \;=\; Q_n. \tag{4.9}$$

Moreover, splitting $Z = Z_{\le\sqrt{n}} Z_{>\sqrt{n}}$ into its $(\sqrt{n})$-smooth and $(\sqrt{n})$-rough parts flips the sign of every prime $p \le \sqrt{n}$ dividing Z:

$$4\log Z_{>\sqrt{n}} \;=\; \left(c_2(n) + \sum_{3\le p\le\sqrt{n}} \varepsilon_X(p)\frac{\log p}{p-1}\right) n + O\Big(\frac{n}{\log n}\Big), \tag{4.10}$$

where $\varepsilon_X(p) = +1$ if $p \mid X_1(q_n)$ and $\varepsilon_X(p) = -1$ if $p \nmid X_1(q_n)$.

(ii) ***Second-Stage Consecutive Splitting and Dual Quartic Locks:*** *Let $c_0 := \gcd(X_1(q_n) - 1, X_1(q_n) + 1) \in \{1, 2\}$. There exist unique coprime factorizations $B_- B_+ = B_n/c_0^2$ and $Z_- Z_+ = Z$ such that*

$$\frac{X_1(q_n) - 1}{c_0} \;=\; B_- Z_-^2, \qquad \frac{X_1(q_n) + 1}{c_0} \;=\; B_+ Z_+^2, \qquad B_+ Z_+^2 - B_- Z_-^2 \;=\; \frac{2}{c_0}, \tag{4.11}$$

and the first-stage remainder $R_n = Z^2 = Z_-^2 Z_+^2$ satisfies the dual quartic equations

$$B_+ R_n \;=\; B_- Z_-^4 + \frac{2}{c_0} Z_-^2, \qquad B_- R_n \;=\; B_+ Z_+^4 - \frac{2}{c_0} Z_+^2, \tag{4.12}$$

with unconditional Euclidean quotient-remainder locks for all sufficiently large n:

$$\left\lfloor \frac{B_+ R_n}{B_-} \right\rfloor \;=\; Z_-^4, \qquad \left\lceil \frac{B_- R_n}{B_+} \right\rceil \;=\; Z_+^4. \tag{4.13}$$

(iii) ***Stretched-Exponential Lower Bounds on the Midpoint Cores*** $\delta_1(n), \delta_2(n)$***:*** *Let $\delta_1(n) := \text{core}(X_1(q_n) - 1)$ and $\delta_2(n) := \text{core}(X_1(q_n) + 1)$. For both $i \in \{1, 2\}$, the explicit Mertens bound gives $\omega(\delta_i(n)) \ge (8e^\gamma - o(1))\frac{n^{1/4}}{\log n}$ and $\delta_i(n) \ge \exp((2e^\gamma - o(1))n^{1/4})$, and applying the power-band equidistribution (2.20) upgrades these bounds unconditionally for both $i \in \{1, 2\}$ to*

$$P^+(\delta_i(n)) \;\ge\; n^{1/2-o(1)}, \qquad \omega(\delta_i(n)) \;\ge\; n^{1/2-o(1)}, \qquad \delta_i(n) \;\ge\; \exp\!\big(n^{1/2-o(1)}\big). \tag{4.14}$$

Proof. **Part (i):** Since $\mathcal{P}_{\text{odd}}(n)$ and $\mathcal{P}_{\text{even}}(n)$ are disjoint sets of odd primes whose union is the set of all odd primes $3 \le p \le n$, we have $\gcd(A_n^*, B_n^*) = 1$ and $A_n^* B_n^* = \frac{n!}{2^{v_2(n!)}}$. By Proposition 4.1(i), every odd prime $p \in \mathcal{P}_{\text{odd}}(n)$ satisfies $p^{v_p(n!)} \parallel q_n Y_1(q_n)^2$, while every odd prime $p \in \mathcal{P}_{\text{even}}(n)$ divides $s_n = 2X_1(q_n)Y_1(q_n)$ with $\gcd(X_1(q_n), Y_1(q_n)) = 1$, so $p^{v_p(n!)/2} \parallel Y_1(q_n)$ if $p \mid Y_1(q_n)$ (giving $Z = \prod_{p \in \mathcal{P}_{\text{even}}(n), p \mid Y_1} p^{v_p(n!)/2}$) and $p^{v_p(n!)/2} \parallel X_1(q_n)$ if $p \mid X_1(q_n)$ (giving the odd part of $X_1(q_n)$ equal to $\frac{\sqrt{A_n^*}}{Z}$).

For the prime $p = 2$, since $4X_1(q_n)^2(X_1(q_n)^2 - 1) = n!$ and $\gcd(X_1(q_n)^2, X_1(q_n)^2 - 1) = 1$, there are two mutually exclusive cases:

- *Type I ($X_1(q_n)$ odd):* Here $2 \nmid X_1(q_n)^2$, so $2^{v_2(n!)-2} \parallel (X_1(q_n)^2 - 1) = q_n Y_1(q_n)^2$. Setting $A_n := A_n^*$ and $B_n := 2^{v_2(n!)-2} B_n^*$ gives $X_1(q_n) = \frac{\sqrt{A_n}}{Z}$ and $q_n Y_1(q_n)^2 = B_n Z^2$.
- *Type II ($X_1(q_n)$ even):* Here $2 \nmid (X_1(q_n)^2 - 1)$, so $2^{v_2(n!)-2} \parallel X_1(q_n)^2$, which requires $2 \mid v_2(n!)$ and $v_2(X_1(q_n)) = \frac{v_2(n!)-2}{2} \ge 3$ for $n \ge 8$; reducing $q_n Y_1(q_n)^2 = X_1(q_n)^2 - 1 \equiv -1 \pmod 8$ forces $q_n \equiv 7 \pmod 8$ and $2 \nmid Y_1(q_n)$. Setting $A_n := 2^{v_2(n!)-2} A_n^*$ and $B_n := B_n^*$ again gives $X_1(q_n) = \frac{\sqrt{A_n}}{Z}$ and $q_n Y_1(q_n)^2 = B_n Z^2$.

Substituting $X_1(q_n) = \frac{\sqrt{A_n}}{Z}$ and $q_n Y_1(q_n)^2 = B_n Z^2$ into the fundamental Pell equation $X_1(q_n)^2 - q_n Y_1(q_n)^2 = 1$ gives $\frac{A_n}{Z^2} - B_n Z^2 = 1$, and multiplying by Z^2 yields the exact quartic identity

$$A_n \;=\; B_n Z^4 + Z^2.$$

Taking logarithms of $Z^4 = \frac{A_n}{B_n}\big(1 + \frac{1}{B_n Z^2}\big)^{-1}$ and using $\log A_n^* - \log B_n^* = S_{\text{even}}(n) - S_{\text{odd}}(n) = \Delta_{\text{parity}}(n)$ together with $v_2(n!)\log 2 = (\log 2)n + O(\log n)$ and Theorem 2.5(i)–(ii) yields

$$\begin{aligned} 4\log Z \;&=\; \Delta_{\text{parity}}(n) \mp (v_2(n!) - 2)\log 2 - \log\!\left(1 + \frac{1}{B_n Z^2}\right) \\ &=\; (c_2(n) - \Sigma^*_{\text{parity}}(n))n + O\!\left(\frac{n}{\log n}\right) \;=\; o(n \log n), \end{aligned}$$

where $c_2(n) = (\log 2 - 1) - \log 2 = -1$ when $X_1(q_n)$ is odd and $c_2(n) = (\log 2 - 1) + \log 2 = 2\log 2 - 1$ when $X_1(q_n)$ is even. Since $\log B_n \ge S_{\text{odd}}(n) = (\frac{1}{2} - o(1))n \log n$ while $\log(Z^2) = o(n \log n)$, we have $0 < Z^2 < B_n$ for all sufficiently large n. By uniqueness of Euclidean division $A_n = Q_n B_n + R_n$ $(0 \le R_n < B_n)$, we obtain $Q_n = Z^4$, $R_n = Z^2$, and $R_n^2 = Q_n$, proving (4.9). Finally, writing $Z = Z_{\le \sqrt{n}} Z_{> \sqrt{n}}$, every prime $p \le \sqrt{n}$ in $\mathcal{P}_{\text{even}}(n)$ dividing Z contributes $2v_p(n!)\log p = \frac{2\log p}{p-1} n + O(\log n)$ to $4\log Z_{\le \sqrt{n}}$ and $+\frac{\log p}{p-1} n$ to $-\Sigma^*_{\text{parity}}(n) n$, so subtracting $4\log Z_{\le \sqrt{n}}$ from both sides flips the sign of $p \mid Z$ from $+\frac{\log p}{p-1} n$ to $-\frac{\log p}{p-1} n = \varepsilon_X(p)\frac{\log p}{p-1} n$, establishing (4.10).

Part (ii): Factor $X_1(q_n)^2 - 1 = (X_1(q_n) - 1)(X_1(q_n) + 1) = B_n Z^2$. Setting $c_0 := \gcd(X_1(q_n) - 1, X_1(q_n) + 1) \in \{1, 2\}$, the positive integers $\frac{X_1(q_n)-1}{c_0}$ and $\frac{X_1(q_n)+1}{c_0}$ are coprime and have product $\frac{B_n}{c_0^2} Z^2$. Since $\gcd(B_n, Z) = 1$, each prime power dividing $\frac{B_n}{c_0^2}$ or Z^2 divides either $\frac{X_1(q_n)-1}{c_0}$ or $\frac{X_1(q_n)+1}{c_0}$ to its full valuation, yielding unique coprime factorizations $B_- B_+ = B_n / c_0^2$ and $Z_- Z_+ = Z$ satisfying (4.11). Multiplying $B_+ Z_+^2 - B_- Z_-^2 = \frac{2}{c_0}$ by Z_-^2 and by Z_+^2 and substituting $Z_-^2 Z_+^2 = Z^2 = R_n$ gives (4.12). Since $\log(Z_\mp^4) \le 4\log Z = o(n \log n)$ and $\log(B_\mp Z_\mp^2) = \log(\frac{X_1(q_n) \mp 1}{c_0}) = (\frac{1}{4} + o(1))n \log n$, we have

$$\log B_- \;=\; \left(\frac{1}{4} + o(1)\right) n \log n, \qquad \log B_+ \;=\; \left(\frac{1}{4} + o(1)\right) n \log n. \tag{4.15}$$

Hence $0 < \frac{2}{c_0} Z_\mp^2 < B_\mp$ for all sufficiently large n, and Euclidean division in (4.12) yields (4.13).

Part (iii): The odd parts of the square-free cores $\delta_1(n) = \text{core}(X_1(q_n) - 1)$ and $\delta_2(n) = \text{core}(X_1(q_n) + 1)$ are $\delta_1^*(n) = \prod_{p \in \mathcal{P}_{\text{odd}}(n), p \mid B_-} p$ and $\delta_2^*(n) = \prod_{p \in \mathcal{P}_{\text{odd}}(n), p \mid B_+} p$. Fix $i \in \{1, 2\}$ and let $t := \omega(\delta_i^*(n))$. Since $v_2(B_\mp)\log 2 \le v_2(n!)\log 2 = O(n)$, equation (4.15) gives

$$\sum_{p \mid \delta_i^*(n)} v_p(n!)\log p \;=\; \left(\frac{1}{4} + o(1)\right) n \log n. \tag{4.16}$$

Since $v_p(n!) \le \frac{n}{p-1}$ and $p \mapsto \frac{\log p}{p-1}$ is strictly decreasing for $p \ge 3$, the left-hand side of (4.16) is bounded above by the sum over the first t odd primes $3 = p_2 < p_3 < \cdots < p_{t+1}$. By Mertens' first theorem [18, Chapter 7], $\sum_{j=2}^{t+1} \frac{\log p_j}{p_j - 1} = \log p_{t+1} - \gamma - \log 2 + o(1)$, so comparing with (4.16) gives $\log p_{t+1} \ge \frac{1}{4}\log n + \gamma + \log 2 - o(1)$, i.e., $p_{t+1} \ge (2e^{\gamma} - o(1))n^{1/4}$, $\omega(\delta_i(n)) \ge (8e^{\gamma} - o(1))\frac{n^{1/4}}{\log n}$, and $\delta_i(n) \ge \exp(\vartheta(p_{t+1}) - \log 2) \ge \exp((2e^{\gamma} - o(1))n^{1/4})$.

To upgrade the exponent from $1/4$ to $1/2 - o(1)$, fix any $\varepsilon \in (0, 1/6)$. Since every prime $p \mid \delta_i^*(n)$ belongs to $\mathcal{P}_{\mathrm{odd}}(n)$, Theorem 2.5(iii) (equation (2.20) with $h = 2, c = 1, a = 0, b = \frac{1}{2} - \varepsilon$) gives

$$\sum_{\substack{p \mid \delta_i^*(n) \\ p \le n^{1/2-\varepsilon}}} v_p(n!)\log p \;\le \sum_{\substack{3 \le p \le n^{1/2-\varepsilon} \\ p \in \mathcal{P}_{\mathrm{odd}}(n)}} v_p(n!)\log p \;=\; \left(\frac{1}{4} - \frac{\varepsilon}{2} + o(1)\right) n \log n.$$

Subtracting this from (4.16) shows that the prime divisors of $\delta_i^*(n)$ exceeding $n^{1/2-\varepsilon}$ carry factorial weight

$$\sum_{\substack{p \mid \delta_i^*(n) \\ p > n^{1/2-\varepsilon}}} v_p(n!)\log p \;\ge\; \left(\frac{\varepsilon}{2} - o(1)\right) n \log n.$$

Since every prime $p > n^{1/2-\varepsilon}$ satisfies $v_p(n!) \le \frac{n}{p-1} \le 2n^{1/2+\varepsilon}$ and $\log p \le \log n$, dividing by $2n^{1/2+\varepsilon}\log n$ yields $P^+(\delta_i(n)) \ge n^{1/2-\varepsilon}$, $\omega(\delta_i(n)) \ge (\frac{\varepsilon}{4} - o(1))n^{1/2-\varepsilon}$, and $\log \delta_i(n) \ge (\frac{\varepsilon}{4} - o(1))n^{1/2-\varepsilon}\log n$. Since $\varepsilon > 0$ is arbitrary, this establishes (4.14). □

4.2. **Stickelberger Base-p Reciprocity and the Conductor-Free Quadratic Residue Lock.** We next show that Stickelberger's p-adic factorial congruence determines the quadratic symbol $\left(\frac{-q_n}{p}\right)$ at every unramified prime $p \in \mathcal{P}_{\mathrm{even}}(n)$ directly in terms of the base-p digits of n, and forces a conductor-free quadratic residue lock on $\mathrm{core}(Z)$.

Lemma 4.3 (Stickelberger's p-Adic Factorial Congruence [56])**.** *Let p be an odd prime, and let $n = \sum_{j=0}^{K} d_j^{(p)}(n)p^j$ $(0 \le d_j^{(p)}(n) < p)$ be the base-p expansion of a positive integer n. Then*

$$\frac{n!}{p^{v_p(n!)}} \;\equiv\; (-1)^{v_p(n!)} \prod_{j=0}^{K} (d_j^{(p)}(n)!) \pmod{p}. \tag{4.17}$$

Proof. We proceed by induction on $n \ge 1$. For $1 \le n < p$, $v_p(n!) = 0$ and $n = d_0^{(p)}(n)$, so both sides equal $n! \pmod p$. For $n \ge p$, write $n = qp + d_0$ with $q = \lfloor n/p \rfloor \ge 1$ and $0 \le d_0 < p$. Separating the q multiples of p in $\{1, \dots, n\}$ (whose product is $p^q q!$) and grouping the remaining $qp + d_0 - q = q(p-1) + d_0$ coprime integers into q complete residue blocks modulo p plus the trailing block $\{qp+1, \dots, qp+d_0\}$ gives

$$\frac{n!}{p^{v_p(n!)}} \;=\; \frac{q!}{p^{v_p(q!)}} \cdot \left(\prod_{i=0}^{q-1}\prod_{r=1}^{p-1}(ip+r)\right) \cdot \prod_{r=1}^{d_0}(qp+r) \;\equiv\; \frac{q!}{p^{v_p(q!)}} \cdot ((p-1)!)^q \cdot d_0! \pmod{p}.$$

By Wilson's theorem, $(p-1)! \equiv -1 \pmod p$, and since $v_p(n!) = q + v_p(q!)$ and $d_j^{(p)}(q) = d_{j+1}^{(p)}(n)$ for $j \ge 0$, the induction hypothesis applied to q yields (4.17). □

Theorem 4.4 (Stickelberger Base-p Reciprocity, Inert Prime Absorption, and Conductor-Free Residue Lock)**.** *Let $n \ge 8$ be an integer.*

(i) *For every prime $p \in \mathcal{P}_{\mathrm{even}}(n) \setminus \{2\}$, writing $n = \sum_{j=0}^{K} d_j^{(p)}(n)p^j$ in base p, we have*

$$\begin{aligned} \left(\frac{-q_n}{p}\right) &= \left(\frac{-1}{p}\right) \prod_{j=0}^{K} \left(\frac{d_j^{(p)}(n)!}{p}\right), \\ \left(\frac{-q_n}{p}\right) &= \left(\frac{-(2r)!\,(n-2rp)!}{p}\right) \qquad \text{on } I_{2r} = \left(\frac{n}{2r+1}, \frac{n}{2r}\right], \end{aligned} \tag{4.18}$$

which at the Gauss–Wilson midpoint $n - 2rp = \frac{p-1}{2} - m$ *(*$p \equiv 1 \pmod 4$*) reduces to* $\left(\frac{-q_n}{p}\right) = \left(\frac{(2r)!}{p}\right)\left(\frac{2}{p}\right)^{m+1}\prod_{j=0}^{m-1}\left(\frac{2j+1}{p}\right)$.

(ii) *Suppose* (n, m) *is a solution to* $n! + 1 = m^2$ *in the* $k = 2$ *regime. Then every inert prime* $p \in \mathcal{P}^{-}_{\mathrm{even}}(n) := \{p \in \mathcal{P}_{\mathrm{even}}(n) \setminus \{2\} : \left(\frac{-q_n}{p}\right) = -1\}$ *is absorbed to its full factorial half-power* $p^{v_p(n!)/2} \parallel Z$*, and for every ramified prime* $p \in \mathcal{P}_{\mathrm{odd}}(n)$ *with* $p \equiv 1 \pmod 4$*,*

$$\left(\frac{\mathrm{core}(Z)}{p}\right) = \left(\frac{2^{v_2(X_1(q_n))}\, c_{\mathrm{even}}(n)}{p}\right), \qquad c_{\mathrm{even}}(n) := \prod_{\substack{3\le q\le n \\ v_q(n!)\equiv 2 \pmod 4}} q. \tag{4.19}$$

Proof. (i) For any odd prime $p \in \mathcal{P}_{\mathrm{even}}(n)$, $2 \mid v_p(n!)$, so $(-1)^{v_p(n!)} = +1$ and $\frac{n!}{p^{v_p(n!)}} = q_n\left(\frac{s_n}{p^{v_p(n!)/2}}\right)^2$ differs from q_n by a non-zero square in $\mathbb{F}_p^\times$. Taking Legendre symbols modulo p in (4.17) gives the first formula in (4.18). On the even harmonic layer $I_{2r} = \left(\frac{n}{2r+1}, \frac{n}{2r}\right]$ $(2 \le 2r < \sqrt{n})$, we have $p > \sqrt{n}$, so $K = 1$, $d_1^{(p)}(n) = \lfloor n/p \rfloor = 2r$, and $d_0^{(p)}(n) = n - 2rp$, giving the second formula in (4.18). When $p \equiv 1 \pmod 4$, pairing j with $p - j \equiv -j \pmod p$ for $1 \le j \le \frac{p-1}{2}$ in $(p-1)! \equiv -1 \pmod p$ gives $\left(((p-1)/2)!\right)^2 \equiv (-1)^{(p+1)/2} = -1 \pmod p$, so Euler's criterion gives $\left(\frac{((p-1)/2)!}{p}\right) = (-1)^{(p-1)/4} = \left(\frac{2}{p}\right)$. Writing $d_0 = n - 2rp = \frac{p-1}{2} - m$ $(m \ge 0)$ and using $\frac{p-1}{2} - j \equiv -\frac{2j+1}{2} \pmod p$ yields $\left(\frac{-q_n}{p}\right) = \left(\frac{(2r)!}{p}\right)\left(\frac{2}{p}\right)^{m+1}\prod_{j=0}^{m-1}\left(\frac{2j+1}{p}\right)$.

(ii) By Proposition 4.1(ii), every odd prime $p \mid X_1(q_n)$ satisfies $\left(\frac{-q_n}{p}\right) = +1$; consequently, no prime $p \in \mathcal{P}^{-}_{\mathrm{even}}(n)$ can divide $X_1(q_n)$, forcing $p^{v_p(n!)/2} \parallel Z$ for all $p \in \mathcal{P}^{-}_{\mathrm{even}}(n)$. Dually, for every ramified prime $p \in \mathcal{P}_{\mathrm{odd}}(n)$ with $p \equiv 1 \pmod 4$, reducing $X_1(q_n)^2 - q_n Y_1(q_n)^2 = 1$ modulo $p \mid q_n$ gives $X_1(q_n) = \frac{\sqrt{A_n}}{Z} \equiv \pm 1 \pmod p$. Since $\left(\frac{-1}{p}\right) = +1$, taking Legendre symbols modulo p gives $\left(\frac{Z}{p}\right) = \left(\frac{\sqrt{A_n}}{p}\right)$. In $\sqrt{A_n^*} = \prod_{q \in \mathcal{P}_{\mathrm{even}}(n)\setminus\{2\}} q^{v_q(n!)/2}$, the exponent $v_q(n!)/2$ is odd if and only if $v_q(n!) \equiv 2 \pmod 4$, so $\mathrm{core}(\sqrt{A_n^*}) = c_{\mathrm{even}}(n)$, establishing (4.19). □

5. The Fundamental Regime $k = 1$ and Rédei–Reichardt Cohomology

We now turn to the fundamental regime $k = 1$ (where $m = X_1(q_n)$ and $s_n = Y_1(q_n)$), together with the intrinsic continued-fraction midpoint structure of $\sqrt{q_n}$ valid for all $n \geq 8$. In this section we establish the continued-fraction midpoint bias, the reciprocal cross-symmetry and effective Landau–Siegel sub-exponential core bounds, the exact homogeneous $\mathbb{F}_2$-Laplacian equivalence, the Galois cohomology / Rédei–Reichardt 4-rank exact sequence, and the reduced dual quartic laws.

5.1. Midpoint Core Bipartition, Cross-Symmetry, and Effective Sub-Exponential Core Bounds.

We first establish the properties of the continued-fraction midpoint of $\sqrt{q_n}$ and the one-sided character bias on $I_1 = (n/2, n]$.

Proposition 5.1 (Continued-Fraction Midpoint Denominator and Square-Root Conductor Bias)**.** *Let $n \geq 8$ be an integer, let $\varepsilon_n = X_1(q_n) + Y_1(q_n)\sqrt{q_n}$ be the fundamental unit of $\mathbb{Z}[\sqrt{q_n}]$, and write*

$$X_1(q_n) - 1 \;=\; \delta_1^{\mathrm{mid}}(n)\, u_n^2, \qquad X_1(q_n) + 1 \;=\; \delta_2^{\mathrm{mid}}(n)\, v_n^2, \qquad \delta_2^{\mathrm{mid}}(n)\, v_n^2 - \delta_1^{\mathrm{mid}}(n)\, u_n^2 \;=\; 2, \tag{5.1}$$

where $\delta_i^{\mathrm{mid}}(n) := \mathrm{core}(X_1(q_n) \mp 1)$ have odd parts $\delta_i^{\mathrm{mid}*}(n) = \delta_i^{\mathrm{mid}}(n)/\gcd(\delta_i^{\mathrm{mid}}(n), 2)$. Then:

(i) *$\delta_1^{\mathrm{mid}*}(n)\,\delta_2^{\mathrm{mid}*}(n) = q_n^*$ is a coprime factorization of the odd part $q_n^* > 1$ of the square-free integer q_n into two distinct factors, immediately forcing*

$$\delta_{\min}(n) \;:=\; \min\Big(\delta_1^{\mathrm{mid}*}(n),\, \delta_2^{\mathrm{mid}*}(n)\Big) \;<\; \sqrt{q_n^*} \;\leq\; \sqrt{q_n}. \tag{5.2}$$

Moreover, the period length $L(q_n)$ of the regular continued fraction of $\sqrt{q_n}$ is even, and at the half-period $M = L(q_n)/2$, the complete quotient denominator $Q_M^* < 2\sqrt{q_n}$ satisfies $Q_M^* = 2^a \delta_{i_n}^{\mathrm{mid}}(n)$ for some $a \in \{-1, 0, 1\}$ and $i_n \in \{1, 2\}$.

(ii) *Let $\psi_1(p) := \left(\frac{-2\delta_1^{\mathrm{mid}}(n)}{p}\right)$ and $\psi_2(p) := \left(\frac{2\delta_2^{\mathrm{mid}}(n)}{p}\right)$, and let $\psi_{\min} \in \{\psi_1, \psi_2\}$ be the character corresponding to $\delta_{\min}(n)$, of conductor $Q_{\min} \leq 8\delta_{\min}(n) < 8\sqrt{q_n}$. On $I_1 = (n/2, n]$, $\psi_{\min}$ satisfies the unconditional one-sided bias*

$$\sum_{p \in (n/2, n]} \psi_{\min}(p) \log p \;=\; \vartheta(n) - \vartheta(n/2) - \sum_{\substack{p \in (n/2, n] \\ p \mid \delta_{\min}(n)}} \log p \;\geq\; \left(\frac{1 - \log 2}{2} - o(1)\right) n \;\approx\; 0.153426\, n. \tag{5.3}$$

Proof. (i) Since $\delta_1^{\mathrm{mid}}(n)\delta_2^{\mathrm{mid}}(n)(u_n v_n)^2 = X_1(q_n)^2 - 1 = q_n Y_1(q_n)^2$ and $\gcd(X_1(q_n) - 1, X_1(q_n) + 1) \mid 2$, taking p-adic valuations modulo 2 at every odd prime p shows that $\delta_1^{\mathrm{mid}*}(n)\delta_2^{\mathrm{mid}*}(n) = q_n^*$ with $\gcd(\delta_1^{\mathrm{mid}*}(n), \delta_2^{\mathrm{mid}*}(n)) = 1$. Because $q_n^* > 1$ is square-free, $\delta_1^{\mathrm{mid}*}(n) \neq \delta_2^{\mathrm{mid}*}(n)$, so $\min(\delta_1^{\mathrm{mid}*}(n), \delta_2^{\mathrm{mid}*}(n)) < \sqrt{q_n^*} \leq \sqrt{q_n}$, proving (5.2). Moreover, by Lemma 3.1(i), $x^2 - q_n y^2 = -1$ has no solutions, so $L(q_n)$ is even, and by Lagrange's midpoint symmetry theorem for regular continued fractions [43, §25], the complete quotient denominator $Q_{L(q_n)/2}^*$ at the half-period divides $2q_n$, is square-free up to a factor of 2, satisfies $0 < Q_{L(q_n)/2}^* < 2\sqrt{q_n}$, and equals $2^a \delta_{i_n}^{\mathrm{mid}}(n)$ ($a \in \{-1, 0, 1\}, i_n \in \{1, 2\}$).

(ii) Every prime $p \in I_1 = (n/2, n]$ has $v_p(n!) = 1$ and therefore divides $q_n^* = \delta_1^{\mathrm{mid}*}(n)\delta_2^{\mathrm{mid}*}(n)$. Reducing $\delta_2^{\mathrm{mid}}(n)v_n^2 - \delta_1^{\mathrm{mid}}(n)u_n^2 = 2$ modulo $p \in I_1$ shows that if $p \mid \delta_1^{\mathrm{mid}}(n)$, then $\psi_1(p) = 0$ and $\psi_2(p) = \left(\frac{2\delta_2^{\mathrm{mid}}(n)}{p}\right) = \left(\frac{(2\delta_2^{\mathrm{mid}}(n)v_n)^2}{p}\right) = +1$; symmetrically, if $p \mid \delta_2^{\mathrm{mid}}(n)$, then $\psi_2(p) = 0$ and $\psi_1(p) = +1$. Thus $\psi_{\min}(p) = +1$ for all $p \in I_1$ with $p \nmid \delta_{\min}(n)$ and $\psi_{\min}(p) = 0$ for $p \in I_1$ with $p \mid \delta_{\min}(n)$. Since $\sum_{p \in I_1, p \mid \delta_{\min}(n)} \log p \leq \log \delta_{\min}(n) < \frac{1}{2} \log q_n = (\frac{\log 2}{2} + o(1))n$ and $\vartheta(n) - \vartheta(n/2) = (\frac{1}{2} - o(1))n$, subtracting yields (5.3). □

Combining Proposition 5.1 with Theorem 2.5 and the explicit formula for $\psi(x, \psi_i)$ yields the reciprocal cross-symmetry laws, radical lower bounds, and effective Landau–Siegel sub-exponential lower bounds on $\delta_1^{\mathrm{mid}}(n)$ and $\delta_2^{\mathrm{mid}}(n)$.

Theorem 5.2 (Midpoint Core Bipartition, Cross-Symmetry, Radical Bounds, and Effective Landau–Siegel Lower Bounds)**.** *Let $n \geq 4$ be an integer, and for any solution (n, m) to $n! + 1 = m^2$, write $u_1 := \frac{m-1}{2} = \delta_1(n) z_1^2$ and $u_2 := \frac{m+1}{2} = \delta_2(n) z_2^2$ with $\delta_1(n), \delta_2(n) \geq 1$ square-free. Then:*

(i) ***Midpoint Core Bipartition, Cross-Symmetry, and Radical Bounds:*** *We have $\delta_1(n)\delta_2(n) = q_n$, $\gcd(\delta_1(n), \delta_2(n)) = 1$, $z_1 z_2 = s_n/2$, $\delta_2(n) z_2^2 - \delta_1(n) z_1^2 = 1$, and $\delta_1(n) > 1$ for all $n \geq 3$ (with $\delta_2(n) = 1 \iff k = 2$ and $\delta_2(n) > 1 \iff k = 1$). Decomposing the odd parts $u_i^* = B^{(i)}(Z_i^*)^2$ into $B^{(i)} := \prod_{p \in \mathcal{P}_{\mathrm{odd}}(n), p \mid u_i^*} p^{v_p(n!)}$ and $Z_i^* := \prod_{q \in \mathcal{P}_{\mathrm{even}}(n), q \mid u_i^*} q^{v_q(n!)/2}$ yields the exact reciprocal cross-symmetry laws*

$$\begin{aligned} \log B^{(2)} - \log((Z_1^*)^2) &= \frac{1}{2}\Big((2w_2^{(1)} - v_2(n!)) \log 2 - \Delta_{\mathrm{parity}}(n)\Big) + O\Big(e^{-\frac{1}{3} n \log n}\Big) = O(n), \\ \log B^{(1)} - \log((Z_2^*)^2) &= \frac{1}{2}\Big((2w_2^{(2)} - v_2(n!)) \log 2 - \Delta_{\mathrm{parity}}(n)\Big) + O\Big(e^{-\frac{1}{3} n \log n}\Big) = O(n), \end{aligned} \tag{5.4}$$

where $w_2^{(i)} = v_2(u_i)$, together with the unconditional radical bounds for both $m-1$ and $m+1$:

$$P^+(m \mp 1) \geq n^{1/2 - o(1)}, \qquad \omega(m \mp 1) \geq n^{1/2 - o(1)}, \qquad \mathrm{rad}(m \mp 1) \geq \exp\big(n^{1/2 - o(1)}\big). \tag{5.5}$$

(ii) ***Effective Sub-Exponential Midpoint Core Bounds and Non-Solubility of $x^2 - q_n y^2 = \pm 2$:*** *For every integer $n \geq 8$ (regardless of whether $n! + 1 = m^2$ is soluble), there exists an absolute effective constant $c_0 > 0$ such that for all sufficiently large n,*

$$\delta_1^{\mathrm{mid}}(n) > \exp\big(c_0 \sqrt{\log n}\big), \qquad \delta_2^{\mathrm{mid}}(n) > \exp\big(c_0 \sqrt{\log n}\big). \tag{5.6}$$

Consequently, the Pell-type equations $x^2 - q_n y^2 = 2$ and $x^2 - q_n y^2 = -2$ have no integer solutions for all sufficiently large n.

Proof. (i) For any solution (n, m) to $n! + 1 = m^2$ with $n \geq 4$, $u_1 = \frac{m-1}{2}$ and $u_2 = \frac{m+1}{2}$ are coprime positive integers with $u_2 - u_1 = 1$ and $u_1 u_2 = \frac{n!}{4} = q_n (s_n/2)^2$. Writing $u_1 = \delta_1(n) z_1^2$ and $u_2 = \delta_2(n) z_2^2$ with $\delta_1(n), \delta_2(n) \geq 1$ square-free gives $\delta_1(n)\delta_2(n) = q_n$, $\gcd(\delta_1(n), \delta_2(n)) = 1$, $z_1 z_2 = s_n/2$, and $\delta_2(n) z_2^2 - \delta_1(n) z_1^2 = 1$. If $\delta_1(n) = 1$, then $z_1^2 - q_n z_2^2 = -1$, which is impossible by Lemma 3.1(i). Since $(z_2\sqrt{\delta_2(n)} + z_1\sqrt{\delta_1(n)})^2 = m + s_n\sqrt{q_n} = \varepsilon_n^k$ with $k \in \{1, 2\}$ by Theorem 1.1, we have $\delta_2(n) = 1 \iff k = 2$ and $\delta_2(n) > 1 \iff k = 1$.

For (5.4), note that $\log B^{(1)} + \log((Z_1^*)^2) = \log u_1^* = \frac{1}{2}\log(n!) - w_2^{(1)} \log 2 + O((n!)^{-1/2})$, while $\log B^{(1)} + \log B^{(2)} = S_{\mathrm{odd}}(n) = \frac{1}{2}\log(n!) - \frac{1}{2}\Delta_{\mathrm{parity}}(n) - \frac{1}{2} v_2(n!) \log 2$. Subtracting these two identities eliminates $\log B^{(1)}$ and $\frac{1}{2}\log(n!)$ identically, giving (5.4). For (5.5), let $S_{\mp}$ be the set of odd prime divisors of $m \mp 1$. Since $\sum_{p \in S_{\mp}} v_p(n!) \log p = (\frac{1}{2} + o(1)) n \log n$ and Theorem 2.5(iii) bounds $\sum_{3 \leq p \leq n^{1/2-\varepsilon}} v_p(n!) \log p = (\frac{1}{2} - \varepsilon + o(1)) n \log n$ for any $\varepsilon \in (0, 1/4)$, the primes $p \in S_{\mp}$ with $p > n^{1/2-\varepsilon}$ carry weight $\geq (\varepsilon - o(1)) n \log n$ with $v_p(n!) \leq 2n^{1/2+\varepsilon}$, forcing $P^+(m \mp 1) \geq n^{1/2-\varepsilon}$, $\omega(m \mp 1) \geq (\frac{\varepsilon}{2} - o(1)) n^{1/2-\varepsilon}$, and $\log \mathrm{rad}(m \mp 1) \geq (\frac{\varepsilon}{2} - o(1)) n^{1/2-\varepsilon} \log n$.

(ii) By Proposition 5.1(i)–(ii), for every $n \geq 7$ and $i \in \{1, 2\}$, the character $\psi_i(p)$ of conductor $Q_i \leq 8\delta_i^{\mathrm{mid}}(n)$ is non-principal: $\psi_1(p) = \left(\frac{-2\delta_1^{\mathrm{mid}}(n)}{p}\right)$ is imaginary quadratic ($-2\delta_1^{\mathrm{mid}}(n) < 0$) and hence never principal, while $\psi_2(p) = \left(\frac{2\delta_2^{\mathrm{mid}}(n)}{p}\right)$ is principal if and only if $\delta_2^{\mathrm{mid}}(n) = 2$, which would give $X_1(q_n) + 1 = 2v_n^2$ and $X_1(q_n) - 1 = 2q_n^*(u_n')^2$ and hence $v_n^2 - q_n(u_n')^2 = 1$ with $0 < v_n < X_1(q_n)$, contradicting the minimality of the fundamental unit ε_n. Moreover, ψ_i satisfies

$$\sum_{n/2 < p \leq n} \psi_i(p) \log p = \vartheta(n) - \vartheta(n/2) - \sum_{\substack{p \in (n/2, n] \\ p \mid \delta_i^{\mathrm{mid}}(n)}} \log p \geq \frac{1}{2} n - O\Big(n e^{-c_1\sqrt{\log n}}\Big) - \log \delta_i^{\mathrm{mid}}(n). \tag{5.7}$$

On the other hand, by the explicit formula for $\psi(x, \psi_i)$ and the de la Vallée Poussin zero-free region [18, Chapter 19], if $Q_i \leq \exp(2c_0\sqrt{\log n})$ for a small absolute constant $c_0 > 0$, then $L(s, \psi_i)$

has at most one simple real Landau–Siegel zero $\beta_1 \in (1 - \frac{c_2}{\log Q_i}, 1)$, and

$$\sum_{n/2<p\le n} \psi_i(p)\log p \;=\; -\frac{n^{\beta_1} - (n/2)^{\beta_1}}{\beta_1} + O\Big(n\exp\big(-c_3\sqrt{\log n}\big)\Big).$$

Because $\beta_1 > 0 \implies n^{\beta_1} > (n/2)^{\beta_1}$, the Landau–Siegel zero contribution $-\frac{n^{\beta_1}-(n/2)^{\beta_1}}{\beta_1} < 0$ is *strictly negative*. Thus $\sum_{n/2<p\le n}\psi_i(p)\log p \le O(n\exp(-c_3\sqrt{\log n})) = o(n)$ effectively, which contradicts (5.7) unless $\delta_i^{\mathrm{mid}}(n) > \exp(c_0\sqrt{\log n})$. In particular, if $x^2 - q_n y^2 = \pm 2$ had an integer solution (x, y), then $(x^2 \mp 1)^2 - q_n(xy)^2 = 1$ would be a unit in $\mathbb{Z}[\sqrt{q_n}]$ of the form ε_n^{2r+1}, forcing $\min(\delta_1^{\mathrm{mid}}(n), \delta_2^{\mathrm{mid}}(n)) \in \{1, 2\}$, contradicting (5.6). □

5.2. **Exact Homogeneous $\mathbb{F}_2$-Laplacian Equivalence, Galois Cohomology, and Reduced Dual Quartics.** We now show that the coprime prime-power bipartition $B(y) - A(y) = 1$ ($A(y)B(y) = n!/4$) decouples into a hierarchy of homogeneous linear systems over $\mathbb{F}_2$ across the 2-adic valuation layers

$$\mathcal{L}_r(n) \;:=\; \{3 \le p \le n : v_2(v_p(n!)) = r\} \qquad (r \ge 0), \tag{5.8}$$

where $\mathcal{L}_0(n) = \mathcal{P}_{\mathrm{odd}}(n) \setminus \{2\}$ and $\bigsqcup_{r\ge1}\mathcal{L}_r(n) = \mathcal{P}_{\mathrm{even}}(n) \setminus \{2\}$.

Lemma 5.3 (Exact Homogeneous Laplacian Equivalence and 2-Adic Layer Cascade)**.** *Let $n \ge 6$ be an integer, and for each binary vector $y = (y_p)_{p\le n} \in \{0,1\}^{\pi(n)}$, define the coprime prime-power products $A(y) := \prod_{p\le n} p^{w'_p(n)(1-y_p)}$ and $B(y) := \prod_{p\le n} p^{w'_p(n)y_p}$ ($w'_2(n) = v_2(n!) - 2$, $w'_p(n) = v_p(n!)$ for odd $p \le n$), so that $A(y)B(y) = n!/4$.*

(i) *A positive integer m solves $n! + 1 = m^2$ if and only if $y \in \{0,1\}^{\pi(n)} \setminus \{\mathbf{0}, \mathbf{1}\}$ satisfies $B(y) - A(y) = 1$ (with $A(y) = \frac{m-1}{2}$ and $B(y) = \frac{m+1}{2}$), which in turn implies that y satisfies the homogeneous 2-adic cascade:*

- Level 0 ($\mathcal{L}_0(n)$): *The restriction $y_{\mathcal{L}_0(n)} \in \mathbb{F}_2^{\mathcal{L}_0(n)}$ solves the homogeneous linear system $M_n^{(0)} y_{\mathcal{L}_0(n)} = \mathbf{0}$ over $\mathbb{F}_2$ formed by the ramified rows $p \in \mathcal{L}_0(n)$, the $p = 2$ modulo-8 rows, and the unramified split-even rows $q \in \mathcal{P}^+_{\mathrm{even}}(n)$, while $y_{\mathcal{P}^-_{\mathrm{even}}(n)}$ is uniquely locked by $y_q = \frac{1}{2}(1 - \left(\frac{c_2(y_{\mathcal{L}_0(n)})}{q}\right))$.*
- Level 1 ($\mathcal{L}_1(n) = \{p \le n : v_p(n!) \equiv 2 \pmod 4\}$): *For each $y_{\mathcal{L}_0(n)} \in \ker_{\mathbb{F}_2}(M_n^{(0)})$, the restriction $y_{\mathcal{L}_1^+(n)} \in \mathbb{F}_2^{\mathcal{L}_1^+(n)}$ ($\mathcal{L}_1^+(n) = \mathcal{L}_1(n) \cap \mathcal{P}^+_{\mathrm{even}}(n)$) solves the linear system $M_n^{(1)} y_{\mathcal{L}_1^+(n)} = \mathbf{b}_n^{(1)}(y_{\mathcal{L}_0(n)})$ over $\mathbb{F}_2$ induced by the quadratic characters of the reduced square roots on $\mathcal{L}_0(n) \cup \mathcal{P}^+_{\mathrm{even}}(n)$.*

(ii) *For every odd prime $p \le n/2$, let $N_p := \frac{n!}{4p^{v_p(n!)}}$. Then y satisfies the homogeneous Fermat-quotient equation over $\mathbb{F}_p$:*

$$\sum_{\ell\le n,\, \ell\ne p} w'_\ell(n)\, q_p(\ell)\, y_\ell \;+\; q_p(N_p)\, y_p \;\equiv\; 0 \pmod p, \tag{5.9}$$

and for every prime $p \in I_2 = (n/3, n/2]$ ($p \ge 5$), Wolstenholme's theorem $(p-1)!\sum_{j=1}^{p-1}\frac{1}{j} \equiv 0 \pmod{p^2}$ yields the closed-form evaluation

$$4N_p \;=\; \frac{n!}{p^2} \;\equiv\; 2\cdot(n-2p)!\,\Big(1 + 2p(H_{n-2p} - W_p)\Big) \pmod{p^2}, \tag{5.10}$$

where $W_p := \frac{(p-1)!+1}{p} \bmod p$ is the Wilson quotient and $H_r := \sum_{j=1}^{r}\frac{1}{j} \bmod p$.

Proof. (i) Writing $A(y) = c_1(y)x^2$ and $B(y) = c_2(y)z^2$ with $c_1(y)c_2(y) = q_n$, reducing $B(y) - A(y) = 1$ modulo each $p \in \mathcal{L}_0(n)$, modulo 8, and modulo each $q \in \mathcal{P}^+_{\mathrm{even}}(n)$ (where $\left(\frac{-c_1(y)c_2(y)}{q}\right) = \left(\frac{-q_n}{q}\right) = +1$, so $\left(\frac{c_2(y)}{q}\right) = \left(\frac{-c_1(y)}{q}\right)$ regardless of whether $y_q = 0$ or $y_q = 1$) gives a homogeneous linear system $M_n^{(0)} y_{\mathcal{L}_0(n)} = \mathbf{0}$ depending *only* on $y_{\mathcal{L}_0(n)}$, while on $q \in \mathcal{P}^-_{\mathrm{even}}(n)$ reducing $B(y) - A(y) = 1 \pmod q$

locks y_q deterministically. Once $y_{\mathcal{L}_0(n)}$ and $y_{\mathcal{P}^-_{\text{even}}(n)}$ are fixed, the odd primes entering the square roots x, z to an odd power are precisely $\mathcal{L}_1(n)$, whose quadratic residue constraints at $\mathcal{L}_0(n)\cup\mathcal{P}^+_{\text{even}}(n)$ form the linear system $M_n^{(1)} y_{\mathcal{L}_1^+(n)} = \mathbf{b}_n^{(1)}(y_{\mathcal{L}_0(n)})$.

(ii) For any odd prime $p \le n/2$, $v_p(n!) \ge 2$. If $y_p = 0$, then $p^2 \mid A(y)$, so $B(y) = A(y) + 1 \equiv +1 \pmod{p^2}$ and $q_p(B(y)) = \sum_{\ell\neq p} w'_\ell(n) q_p(\ell) y_\ell \equiv 0 \pmod p$. If $y_p = 1$, then $p^2 \mid B(y)$, so $A(y) = B(y) - 1 \equiv -1 \pmod{p^2}$; since $A(y)\prod_{\ell\neq p} \ell^{w'_\ell(n) y_\ell} = N_p$ and $q_p(-1) \equiv 0 \pmod p$, applying q_p gives $\sum_{\ell\neq p} w'_\ell(n) q_p(\ell) y_\ell + q_p(N_p) \equiv 0 \pmod p$, proving (5.9). Finally, for $p \in (n/3, n/2]$ ($p \ge 5$), we have $v_p(n!) = 2$ and

$$4N_p \;=\; \frac{n!}{p^2} \;=\; (p-1)!\cdot 1 \cdot \left(\prod_{j=1}^{p-1}(p+j)\right)\cdot 2 \cdot \left(\prod_{j=1}^{n-2p}(2p+j)\right).$$

By Wolstenholme's theorem [29], $\prod_{j=1}^{p-1}(p+j) \equiv (p-1)!(1 + p\sum_{j=1}^{p-1}\frac{1}{j}) \equiv (p-1)! \pmod{p^2}$. Substituting $(p-1)! = -1 + pW_p \implies ((p-1)!)^2 \equiv 1 - 2pW_p \pmod{p^2}$ and $\prod_{j=1}^{n-2p}(2p+j) \equiv (n-2p)!(1 + 2pH_{n-2p}) \pmod{p^2}$ yields (5.10). □

Proposition 5.4 (Explicit $\mathbb{F}_2$ Block Structure of $M_n^{(0)}$ and All-Ones Quadratic Reciprocity Duality)**.** *Let $n \ge 8$, let $\mathcal{P}_{\text{odd}}(n) \setminus \{2\} = \{p_1 < p_2 < \cdots < p_t\}$, and for any odd prime p and integer a with $p \nmid a$, define the additive Legendre index $\ell_p(a) \in \mathbb{F}_2$ by $(-1)^{\ell_p(a)} := \left(\frac{a}{p}\right)$. For any solution (n, m) to $n! + 1 = m^2$, let $\mathbf{x} = (x_1, \ldots, x_t)^{\mathsf{T}} \in \mathbb{F}_2^t$ encode the odd part of $\delta_2(n)$ via $\delta_2^*(n) = \prod_{j=1}^t p_j^{x_j}$.*

(i) *In Branch A ($2^{v_2(n!)-2} \mid u_1$, so u_2 is odd and $\delta_2(n) = \delta_2^*(n) \equiv 1 \pmod 8$), $\mathbf{x}$ satisfies the homogeneous $\mathbb{F}_2$-linear system $\mathcal{M}_n^{(1)}\mathbf{x} = \mathbf{0} \pmod 2$, where $\mathcal{M}_n^{(1)} \in \mathbb{F}_2^{(t+|\mathcal{P}^+_{\text{even}}(n)|+2)\times t}$ consists of:*

- Ramified rows ($1 \le i \le t$): $\sum_{j\neq i} \ell_{p_i}(p_j)x_j + \ell_{p_i}(-q_n^*/p_i)x_i \equiv 0 \pmod 2$;
- Unramified split-even rows ($q \in \mathcal{P}^+_{\text{even}}(n)$): $\sum_{j=1}^t \ell_q(p_j)x_j \equiv 0 \pmod 2$;
- Modulo-8 rows: $\sum_{j=1}^t \frac{p_j-1}{2}x_j \equiv 0 \pmod 2$ *and* $\sum_{j=1}^t \frac{p_j^2-1}{8}x_j \equiv 0 \pmod 2$.

(ii) *In Branch B ($2^{v_2(n!)-2} \mid u_2$, so u_1 is odd and $\delta_1(n) = \delta_1^*(n) \equiv 7 \pmod 8$), the complementary vector $\tilde{\mathbf{x}} = \mathbf{1} - \mathbf{x} \in \mathbb{F}_2^t$ (encoding $\delta_1^*(n) = \prod_{j=1}^t p_j^{\tilde{x}_j}$) satisfies the inhomogeneous $\mathbb{F}_2$-linear system $\mathcal{M}_n^{(1)}\tilde{\mathbf{x}} = \mathbf{b}_n^{(-1)} \pmod 2$, where $(\mathbf{b}_n^{(-1)})_{p_i} = \ell_{p_i}(-1)$, $(\mathbf{b}_n^{(-1)})_q = \ell_q(-1)$, and $(\mathbf{b}_n^{(-1)})_{4,8} = (1,0)^{\mathsf{T}}$.*

(iii) All-Ones Quadratic Reciprocity Duality: *Whenever $2 \mid v_2(n!)$ and $q_n \equiv 7 \pmod 8$, the all-ones vector $\mathbf{1} = (1, 1, \ldots, 1)^{\mathsf{T}} \in \mathbb{F}_2^t$ satisfies $\mathcal{M}_n^{(1)}\mathbf{1} = \mathbf{b}_n^{(-1)} \pmod 2$ identically, corresponding to the $k = 2$ specialization $\delta_1(n) = q_n, \delta_2(n) = 1$, and reducing Branch B to the homogeneous coset $\tilde{\mathbf{x}} \in \mathbf{1} + \ker_{\mathbb{F}_2}(\mathcal{M}_n^{(1)})$.*

Proof. In Branch A, reducing $\delta_2^*(n)z_2^2 - 2^{v_2(n!)-2}\delta_1^*(n)z_1^2 = 1$ modulo 8 forces $\delta_2^*(n) \equiv 1 \pmod 8$ (giving the two modulo-8 rows). Reducing modulo $p_i \in \mathcal{P}_{\text{odd}}(n) \setminus \{2\}$ gives $\left(\frac{\delta_2^*(n)}{p_i}\right) = +1$ if $x_i = 0$ ($p_i \mid \delta_1^*(n)$) and $\left(\frac{-2^{v_2(n!)-2}\delta_1^*(n)}{p_i}\right) = +1$ if $x_i = 1$ ($p_i \mid \delta_2^*(n)$); since $\delta_1^*(n) = (q_n^*/p_i)/(\delta_2^*(n)/p_i)$ when $x_i = 1$ and $2^{v_2(n!)-2}\delta_1^*(n) = q_n/\delta_2^*(n)$, both cases combine into the single linear equation $\sum_{j\neq i}\ell_{p_i}(p_j)x_j + \ell_{p_i}(-q_n/p_i)x_i = 0$ over $\mathbb{F}_2$. Similarly, for $q \in \mathcal{P}^+_{\text{even}}(n)$, $\left(\frac{-q_n}{q}\right) = +1$ implies $\left(\frac{\delta_2(n)}{q}\right) = \left(\frac{-\delta_1(n)}{q}\right) = +1$, giving $\sum_{j=1}^t \ell_q(p_j)x_j = 0$. In Branch B, replacing (u_1, u_2) by $(-u_2, -u_1)$ shifts every Legendre symbol by $\left(\frac{-1}{\cdot}\right)$, giving $\mathcal{M}_n^{(1)}\tilde{\mathbf{x}} = \mathbf{b}_n^{(-1)}$. Finally, when $2 \mid v_2(n!)$ and $q_n \equiv 7 \pmod 8$, we have $q_n = q_n^* = \prod_{j=1}^t p_j$, so $(\mathcal{M}_n^{(1)}\mathbf{1})_{p_i} = \ell_{p_i}(q_n/p_i) + \ell_{p_i}(-q_n/p_i) = \ell_{p_i}(-1)$, $(\mathcal{M}_n^{(1)}\mathbf{1})_q = \ell_q(q_n) = \ell_q(-1)$ (since $\left(\frac{-q_n}{q}\right) = +1$), and $q_n \equiv 7 \pmod 8$ gives $(1,0)^{\mathsf{T}}$ on the modulo-8 rows, proving $\mathcal{M}_n^{(1)}\mathbf{1} = \mathbf{b}_n^{(-1)}$. □

Theorem 5.5 (Rédei–Reichardt 4-Rank Cohomology and Reduced Dual Quartic Laws)**.** *Let $n \geq 8$ be an integer.*

(i) ***Galois Cohomology Exact Sequence and Rédei–Reichardt 4-Rank Identity:*** *Let R_n^{ram} denote the ramified block of the Level-0 homogeneous matrix $M_n^{(0)}$ over $\mathbb{F}_2$. Hilbert's Theorem 90 yields the exact sequence of $\mathbb{F}_2$-vector spaces*

$$0 \longrightarrow H^1(\mathrm{Gal}(K_n/\mathbb{Q}),\, \mathcal{O}_{K_n}^{\times,+}) \longrightarrow \ker_{\mathbb{F}_2}(R_n^{\mathrm{ram}}) \xrightarrow{\Psi_+} \mathrm{Cl}^+(q_n)[2] \cap 2\,\mathrm{Cl}^+(q_n) \longrightarrow 0, \tag{5.11}$$

where $H^1(\mathrm{Gal}(K_n/\mathbb{Q}), \mathcal{O}_{K_n}^{\times,+}) = \{\overline{1}, \overline{\eta_n}\} \simeq \mathbb{F}_2$ is spanned by $y = \mathbf{0}$ (the $k = 2$ branch) and the continued-fraction midpoint vector $y^{\mathrm{mid}}(n)$ (the $k = 1$ branch), so that

$$\dim_{\mathbb{F}_2} \ker_{\mathbb{F}_2}(R_n^{\mathrm{ram}}) \;=\; 1 + \mathrm{rank}_4\, \mathrm{Cl}^+(q_n). \tag{5.12}$$

(ii) ***Deterministic $\mathcal{P}^-_{\mathrm{even}}(n)$-Absorption and Reduced Dual Quartic Laws:*** *For any solution (n, m) to $n! + 1 = m^2$ with $\frac{m-1}{2} = \delta_1(n) z_1^2$ and $\frac{m+1}{2} = \delta_2(n) z_2^2$, every inert prime $q \in \mathcal{P}^-_{\mathrm{even}}(n)$ divides z_1 if and only if $\left(\frac{\delta_2(n)}{q}\right) = +1$ and divides z_2 if and only if $\left(\frac{\delta_2(n)}{q}\right) = -1$. Absorbing these deterministic prime powers into $\widetilde{B}_1, \widetilde{B}_2$ leaves only the split-even factor $A^*_{n,+} := \prod_{q \in \mathcal{P}^+_{\mathrm{even}}(n)} q^{v_q(n!)} = (W_1 W_2)^2$, which satisfies the dual quartic equations*

$$\widetilde{B}_2 A^*_{n,+} \;=\; \widetilde{B}_1 W_1^4 + W_1^2, \qquad \widetilde{B}_1 A^*_{n,+} \;=\; \widetilde{B}_2 W_2^4 - W_2^2. \tag{5.13}$$

Proof. (i) Let $\mathcal{A}_n := \{\mathfrak{a}(y) = \prod_{p \in \mathcal{L}_0(n)} \mathfrak{p}_p^{y_p} : y \in \mathbb{F}_2^{\mathcal{L}_0(n)}\} \simeq \mathbb{F}_2^{\mathcal{L}_0(n)}$ be the elementary abelian 2-group of square-free products of ramified prime ideals of $\mathcal{O}_{K_n}$ (augmented at $p = 2$ when $2 \mid D_n$). For each $p \in \mathcal{L}_0(n)$, the row of R_n^{ram} at p asserts that $\left(\frac{c_2(y)}{p}\right) = +1$ when $y_p = 0$ and $\left(\frac{-c_1(y)}{p}\right) = +1$ when $y_p = 1$, which is equivalent to the local Hilbert symbol condition $\left(\frac{c_2(y),\, q_n}{v}\right) = +1$ at every place v of $\mathbb{Q}$. By Hasse's norm theorem and Gauss's genus theory [24], $R_n^{\mathrm{ram}} y = \mathbf{0}$ if and only if $c_2(y) \in \mathrm{N}_{K_n/\mathbb{Q}}(K_n^{\times,+})$, i.e., $[\mathfrak{a}(y)]_+ \in \mathrm{Cl}^+(q_n)[2] \cap 2\,\mathrm{Cl}^+(q_n)$. Moreover, $\mathfrak{a}(y) \in \ker(\Psi_+)$ if and only if $\mathfrak{a}(y) = (\gamma)\mathcal{O}_{K_n}$ for some totally positive $\gamma \in K_n^{\times,+}$, in which case $u := \sigma(\gamma)/\gamma \in \mathcal{O}_{K_n}^{\times,+}$ defines a 1-cocycle class $[u] \in H^1(\mathrm{Gal}(K_n/\mathbb{Q}), \mathcal{O}_{K_n}^{\times,+}) = \mathcal{O}_{K_n}^{\times,+}/(\mathcal{O}_{K_n}^{\times,+})^2 = \{\overline{1}, \overline{\eta_n}\} \simeq \mathbb{F}_2$ (since $\mathrm{N}(\eta_n) = +1$ by Lemma 3.1(i)). The trivial class $[u] = \overline{1}$ gives $y = \mathbf{0}$ ($k = 2$), while the non-trivial class $[u] = \overline{\eta_n}$ gives $\gamma^2 = c_2(y)\varepsilon_n$, which is the unique continued-fraction midpoint vector $y = y^{\mathrm{mid}}(n)$ ($k = 1$). By the Rédei–Reichardt theorem, $\dim_{\mathbb{F}_2}(\mathrm{Cl}^+(q_n)[2] \cap 2\,\mathrm{Cl}^+(q_n)) = \mathrm{rank}_4\, \mathrm{Cl}^+(q_n)$, proving (5.11) and (5.12).

(ii) For each inert prime $q \in \mathcal{P}^-_{\mathrm{even}}(n)$, $\left(\frac{-\delta_1(n)}{q}\right)\left(\frac{\delta_2(n)}{q}\right) = \left(\frac{-q_n}{q}\right) = -1$. In $\delta_2(n) z_2^2 - \delta_1(n) z_1^2 = 1$, $q \mid z_1 z_2$ with $\gcd(z_1, z_2) = 1$: if $q \mid z_1$, reducing modulo q gives $\left(\frac{\delta_2(n)}{q}\right) = +1$; if $q \mid z_2$, reducing modulo q gives $\left(\frac{-\delta_1(n)}{q}\right) = +1 \iff \left(\frac{\delta_2(n)}{q}\right) = -1$. Absorbing these deterministically partitioned prime powers into $\widetilde{B}_1, \widetilde{B}_2$ gives $u_1 = \widetilde{B}_1 W_1^2$ and $u_2 = \widetilde{B}_2 W_2^2$ with $(W_1 W_2)^2 = A^*_{n,+} = \prod_{q \in \mathcal{P}^+_{\mathrm{even}}(n)} q^{v_q(n!)}$ and $\widetilde{B}_2 W_2^2 - \widetilde{B}_1 W_1^2 = 1$. Multiplying by W_1^2 and by W_2^2 yields (5.13). □

6. Pairwise $\mathrm{SL}_2(\mathbb{Z})$ Duality, the Capacity Diamond, and Proof of Theorem 1.3(i)

Having established the single-solution arithmetic structure of the $k = 2$ and $k = 1$ regimes, we now investigate the simultaneous Diophantine structure of pairs of solutions $4 \le n_1 < n_2$ to (1.1). By combining a universal $\mathrm{SL}_2(\mathbb{Z})$ four-GCD factorization system with the cross-symmetry law of Theorem 5.2(i) and odd-harmonic Vaaler diagonal non-vanishing, we prove the consecutive and square-shift exclusion theorems, the Universal Capacity Diamond, and Theorem 1.3(i).

6.1. The Universal $\mathrm{SL}_2(\mathbb{Z})$ Four-GCD System and Twin Norm Identities in $\mathbb{Z}[\sqrt{P}]$.

Let (n_1, m_1) and (n_2, m_2) be any two solutions to $n! + 1 = m^2$ with $4 \le n_1 < n_2$, and set $P := \frac{n_2!}{n_1!} = \prod_{j=n_1+1}^{n_2} j$. Define the six positive integers

$$A_i := \frac{m_i - 1}{2}, \qquad B_i := \frac{m_i + 1}{2} \quad (i = 1, 2), \qquad U := \frac{m_2 - m_1}{2}, \qquad V := \frac{m_2 + m_1}{2}. \tag{6.1}$$

Notice that $\gcd(A_1, B_1) = \gcd(A_2, B_2) = 1$, $A_1 B_1 = \frac{n_1!}{4}$, $A_2 B_2 = P\,\frac{n_1!}{4}$, $UV = (P-1)\,\frac{n_1!}{4}$, and

$$U \;=\; A_2 - A_1 \;=\; B_2 - B_1, \qquad V \;=\; A_2 + B_1 \;=\; B_2 + A_1. \tag{6.2}$$

Theorem 6.1 (Universal $\mathrm{SL}_2(\mathbb{Z})$ Four-GCD System and Twin Norm Identities in $\mathbb{Z}[\sqrt{P}]$). *Let (n_1, m_1) and (n_2, m_2) be any two solutions to $n! + 1 = m^2$ with $4 \le n_1 < n_2$, and set $P = \frac{n_2!}{n_1!}$. Then the four greatest common divisors $X := \gcd(A_1, A_2)$, $Y := \gcd(B_1, B_2)$, $Z := \gcd(A_1, B_2)$, and $W := \gcd(B_1, A_2)$ are pairwise coprime positive integers satisfying $A_1 = XZ$, $B_1 = YW$, and $XYZW = \frac{n_1!}{4}$. Moreover:*

(i) *There exist unique positive integers a, b, c, d with $ab = P - 1$ and $cd = P$ such that $A_2 = cXW$, $B_2 = dYZ$, $U = aXY$, $V = bZW$,*

$$\begin{pmatrix} X \\ Z \end{pmatrix} \;=\; \begin{pmatrix} b & -d \\ c & -a \end{pmatrix} \begin{pmatrix} W \\ Y \end{pmatrix}, \qquad dY^2 - cX^2 \;=\; b, \qquad cW^2 - dZ^2 \;=\; a, \tag{6.3}$$

and the explicit four-squares identities hold:

$$\begin{aligned} 2ac\,X^2 &= Pm_1 - m_2 - (P-1), & 2ad\,Y^2 &= Pm_1 - m_2 + (P-1), \\ 2bd\,Z^2 &= Pm_1 + m_2 - (P-1), & 2bc\,W^2 &= Pm_1 + m_2 + (P-1). \end{aligned} \tag{6.4}$$

(ii) *In the quadratic ring $\mathbb{Z}[\sqrt{P}]$, the elements $\alpha_1 := dY + X\sqrt{P}$ (of norm bd) and $\alpha_2 := cW + Z\sqrt{P}$ (of norm ac) satisfy the exact algebraic identities*

$$\alpha_1\,\alpha_2 \;=\; Pm_1 + m_2\sqrt{P}, \qquad \alpha_1\,\overline{\alpha_2} \;=\; P - \sqrt{P}, \qquad \frac{\alpha_1}{\alpha_2} \;=\; \frac{P - \sqrt{P}}{ac}. \tag{6.5}$$

(iii) *At every odd prime $p \le n_1$ with $p \nmid P(P-1)$, the four characters $\chi_X(p) = \left(\frac{bd}{p}\right)$, $\chi_Y(p) = \left(\frac{-bc}{p}\right)$, $\chi_Z(p) = \left(\frac{ac}{p}\right)$, $\chi_W(p) = \left(\frac{-ad}{p}\right)$ of conductor $\le 4P(P-1)$ satisfy $\chi_T(p) = +1$ whenever $p \mid T$ ($T \in \{X, Y, Z, W\}$), and obey the Quadrant IV lock*

$$\left(1 + \left(\frac{-P}{p}\right)\right)\left(1 + \left(\frac{-(P-1)}{p}\right)\right)\left(1 - \left(\frac{bd}{p}\right)\right) \;=\; 0. \tag{6.6}$$

Proof. Since $\gcd(A_1, B_1) = 1$ and $\gcd(A_2, B_2) = 1$, the four greatest common divisors $X = \gcd(A_1, A_2)$, $Y = \gcd(B_1, B_2)$, $Z = \gcd(A_1, B_2)$, and $W = \gcd(B_1, A_2)$ are pairwise coprime. Every prime power $p^e \parallel A_1 B_1 = n_1!/4$ divides either A_1 or B_1, and since $A_1 B_1 \mid A_2 B_2$, it also divides either A_2 or B_2. Thus $A_1 = \gcd(A_1, A_2 B_2) = XZ$, $B_1 = \gcd(B_1, A_2 B_2) = YW$, and $XYZW = A_1 B_1 = n_1!/4$.

(i) By definition, $XW \mid A_2$ and $YZ \mid B_2$ with $\gcd(XW, YZ) = 1$, while from (6.2), $X = \gcd(A_1, A_2) \mid U$ and $Y = \gcd(B_1, B_2) \mid U$ with $\gcd(X, Y) = 1$, so $XY \mid U$; similarly, $ZW \mid V$. Writing $A_2 = cXW$, $B_2 = dYZ$, $U = aXY$, and $V = bZW$ gives $cd = \frac{A_2 B_2}{XYZW} = P$ and $ab = \frac{UV}{XYZW} = P - 1$.

Substituting these into the four linear relations of (6.2) and dividing by X, Y, W, Z respectively yields

$$Z \;=\; cW - aY, \qquad W \;=\; dZ - aX, \qquad Y \;=\; bZ - cX, \qquad X \;=\; bW - dY,$$

which is the $\mathrm{SL}_2(\mathbb{Z})$ matrix system in (6.3) of determinant $cd - ab = P - (P-1) = 1$. Substituting $W = \frac{X+dY}{b}$ and $Z = \frac{Y+cX}{b}$ into $YW - XZ = B_1 - A_1 = 1$ gives $dY^2 - cX^2 = b$, and substituting $Y = \frac{cW-Z}{a}$ and $X = \frac{dZ-W}{a}$ into $YW - XZ = 1$ gives $cW^2 - dZ^2 = a$.

Multiplying $dY^2 - cX^2 = b$ by a gives $adY^2 - acX^2 = ab = P - 1$. Squaring both sides and using $cd = P$, $2aXY = 2U = m_2 - m_1$, and $m_2^2 = Pm_1^2 - (P-1)$ yields

$$a^2\big(dY^2 + cX^2\big)^2 \;=\; (P-1)^2 + P(m_2 - m_1)^2 \;=\; P^2m_1^2 - 2Pm_1m_2 + m_2^2 \;=\; \big(Pm_1 - m_2\big)^2.$$

Since $Pm_1 > m_2$, taking positive square roots gives $adY^2 + acX^2 = Pm_1 - m_2$; adding and subtracting $adY^2 - acX^2 = P - 1$ gives $2acX^2 = Pm_1 - m_2 - (P-1)$ and $2adY^2 = Pm_1 - m_2 + (P-1)$, and the identical calculation on $cW^2 - dZ^2 = a$ with $2bZW = m_2 + m_1$ gives $2bdZ^2 = Pm_1 + m_2 - (P-1)$ and $2bcW^2 = Pm_1 + m_2 + (P-1)$.

(ii)–(iii) In $\mathbb{Z}[\sqrt{P}]$, expanding $\alpha_1\alpha_2 = (dY + X\sqrt{P})(cW + Z\sqrt{P}) = cdYW + PXZ + (cXW + dYZ)\sqrt{P} = P(YW + XZ) + (A_2 + B_2)\sqrt{P} = Pm_1 + m_2\sqrt{P}$ and $\alpha_1\overline{\alpha_2} = (dY + X\sqrt{P})(cW - Z\sqrt{P}) = P(YW - XZ) - (dYZ - cXW)\sqrt{P} = P - \sqrt{P}$ establishes (6.5). Finally, for any odd prime $p \le n_1$ with $p \nmid P(P-1)$, p divides $XYZW = n_1!/4$. Reducing $dY^2 - cX^2 = b$ and $cW^2 - dZ^2 = a$ modulo p gives $\chi_X(p) = \left(\frac{bd}{p}\right) = +1$ if $p \mid X$, $\chi_Y(p) = \left(\frac{-bc}{p}\right) = +1$ if $p \mid Y$, $\chi_Z(p) = \left(\frac{ac}{p}\right) = +1$ if $p \mid Z$, and $\chi_W(p) = \left(\frac{-ad}{p}\right) = +1$ if $p \mid W$. Since $\chi_Y(p) = \left(\frac{-P}{p}\right)\chi_X(p)$, $\chi_Z(p) = \left(\frac{P(P-1)}{p}\right)\chi_X(p)$, and $\chi_W(p) = \left(\frac{-(P-1)}{p}\right)\chi_X(p)$, all four characters coincide with $\left(\frac{bd}{p}\right)$ whenever $\left(\frac{-P}{p}\right) = \left(\frac{-(P-1)}{p}\right) = +1$, proving (6.6). □

Proposition 6.2 (Quadratic-Radical Identities, Quartic Core-Split Thue System, Square-Parameter X-Splitting, and Three-Solution Pfaffian System)**.** *Under the hypotheses and notation of Theorem 6.1:*

(i) ***Exact Quadratic-Radical Identities:*** *The components (X, Y, Z, W) satisfy*

$$\big(\sqrt{d}\,Y - \sqrt{c}\,X\big)^2 \;=\; \frac{(\sqrt{P}-1)^2}{a(\sqrt{P}\,m_1 + m_2)}, \qquad \big(\sqrt{c}\,W - \sqrt{d}\,Z\big)^2 \;=\; \frac{(\sqrt{P}-1)^2}{b(m_2 - \sqrt{P}\,m_1)}, \tag{6.7}$$

and in $\mathbb{Q}(\sqrt{P})$, $(dY + X\sqrt{P})^2 = \frac{d}{a}(\sqrt{P}-1)(m_2 + m_1\sqrt{P})$ and $(cW + Z\sqrt{P})^2 = \frac{c}{b}(\sqrt{P}+1)(m_2 + m_1\sqrt{P})$.

(ii) ***Quartic Core-Split Thue System:*** *Decomposing each $T \in \{X, Y, Z, W\}$ into its square-free core $\delta_T := \mathrm{core}(T)$ and square part $t_0^2 := T/\delta_T$ transforms the dual Pell equations into the simultaneous degree-4 Thue system*

$$d\,\delta_Y^2 y_0^4 - c\,\delta_X^2 x_0^4 \;=\; b, \qquad c\,\delta_W^2 w_0^4 - d\,\delta_Z^2 z_0^4 \;=\; a, \tag{6.8}$$

where $\delta_X\delta_Y\delta_Z\delta_W = \mathrm{core}(n_1!/4)$ divides $2q_{n_1}$.

(iii) ***Consecutive Square-Parameter Secondary Splitting of X:*** *For consecutive solutions $(n, n+1)$ ($ab = n, cd = n+1$), the character $\chi_X(p) = \left(\frac{bd}{p}\right)$ in (6.6) belongs to the subgroup $\langle\left(\frac{-n}{\cdot}\right), \left(\frac{-(n+1)}{\cdot}\right)\rangle$ if and only if either $b = b_0^2, d = d_0^2$ are both squares (as at $n = 4 = 2^2$, where $b = 4 = 2^2, d = 1 = 1^2$) or $a = a_0^2, c = c_0^2$ are both squares. When $b = b_0^2$ and $d = d_0^2$, the equation $d_0^2Y^2 - cX^2 = b_0^2$ factors over $\mathbb{Z}$ as*

$$(d_0Y - b_0)(d_0Y + b_0) \;=\; cX^2, \qquad \gcd(d_0Y - b_0, d_0Y + b_0) \mid 2, \tag{6.9}$$

inducing a secondary coprime factorization $X = X_1X_2$ ($c = c_1c_2$ up to 2-powers) satisfying $c_2X_2^2 - c_1X_1^2 = 2b_0 \le 2\sqrt{n}$.

(iv) ***Three-Solution Octant-Pair Pfaffian Linear System:*** *For any three solutions $4 \le n_1 < n_2 < n_3$, partitioning the primes $p \le n_1$ into the 8 sign octants $G_{\sigma_1\sigma_2\sigma_3}$ ($\sigma_i \in \{\pm 1\}$)*

and setting $\alpha = G_{+++}G_{---}$, $\beta = G_{++-}G_{--+}$, $\gamma = G_{+-+}G_{-+-}$, $\delta = G_{+--}G_{-++}$ *yields the coupled linear system*

$$a_{13}\gamma - a_{12}\beta \;=\; \widetilde{a}_{23}\delta, \qquad b_{13}\beta - b_{12}\gamma \;=\; \widetilde{a}_{23}\alpha, \qquad (a_{13}b_{13} - a_{12}b_{12})\beta\gamma \;=\; \widetilde{a}_{23}\gamma(a_{13}\alpha + b_{12}\delta). \tag{6.10}$$

Proof. (i) From (6.4), $dY^2 + cX^2 = \frac{Pm_1 - m_2}{a}$, while $2\sqrt{cd}\,XY = 2\sqrt{P}\,XY = \frac{\sqrt{P}(m_2 - m_1)}{a}$. Subtracting gives

$$\left(\sqrt{d}\,Y - \sqrt{c}\,X\right)^2 \;=\; \frac{Pm_1 - m_2 - \sqrt{P}(m_2 - m_1)}{a} \;=\; \frac{(\sqrt{P}-1)(\sqrt{P}\,m_1 - m_2)}{a} \;=\; \frac{(\sqrt{P}-1)^2}{a(\sqrt{P}\,m_1 + m_2)},$$

since $(\sqrt{P}\,m_1 - m_2)(\sqrt{P}\,m_1 + m_2) = Pm_1^2 - m_2^2 = P - 1$. The identity for $(\sqrt{c}\,W - \sqrt{d}\,Z)^2$ follows identically from $cW^2 + dZ^2 = \frac{Pm_1 + m_2}{b}$ and $2\sqrt{P}\,ZW = \frac{\sqrt{P}(m_2 + m_1)}{b}$. Expanding $(dY + X\sqrt{P})^2 = d(dY^2 + cX^2 + 2XY\sqrt{P}) = \frac{d}{a}(\sqrt{P}-1)(m_2 + m_1\sqrt{P})$ and $(cW + Z\sqrt{P})^2 = c(cW^2 + dZ^2 + 2ZW\sqrt{P}) = \frac{c}{b}(\sqrt{P}+1)(m_2 + m_1\sqrt{P})$ establishes the remaining identities in $\mathbb{Q}(\sqrt{P})$.

(ii) Writing each pairwise coprime positive integer $T \in \{X, Y, Z, W\}$ in its canonical square-free / square decomposition $T = \delta_T t_0^2$ with $\delta_T = \mathrm{core}(T)$ square-free gives $T^2 = \delta_T^2 t_0^4$; substituting these into the dual Pell equations $dY^2 - cX^2 = b$ and $cW^2 - dZ^2 = a$ of (6.3) yields (6.8), where $\delta_X\delta_Y\delta_Z\delta_W = \mathrm{core}(XYZW) = \mathrm{core}(n_1!/4) \mid 2q_{n_1}$.

(iii) For consecutive solutions $(n, n+1)$, we have $ab = P - 1 = n$ and $cd = P = n + 1$. Since $bd > 0$ is positive whereas $-n < 0$ and $-(n+1) < 0$ are negative, the real character $\chi_X(p) = \left(\frac{bd}{p}\right)$ belongs to the four-element group $\left\langle \left(\frac{-n}{\cdot}\right), \left(\frac{-(n+1)}{\cdot}\right)\right\rangle = \left\{1, \left(\frac{-n}{\cdot}\right), \left(\frac{-(n+1)}{\cdot}\right), \left(\frac{n(n+1)}{\cdot}\right)\right\}$ if and only if either bd is a square or $ac = \frac{n(n+1)}{bd}$ is a square. Because $\gcd(b, d) \mid \gcd(n, n+1) = 1$, bd is a square if and only if $b = b_0^2$ and $d = d_0^2$ are both squares. In that case, $d_0^2Y^2 - b_0^2 = cX^2$ factors as $(d_0Y - b_0)(d_0Y + b_0) = cX^2$; since $\gcd(d_0Y, b_0) \mid \gcd(cX^2, b_0^2) = 1$, we have $\gcd(d_0Y - b_0, d_0Y + b_0) \mid 2$, inducing the coprime splitting $X = X_1X_2$ with $c_2X_2^2 - c_1X_1^2 = 2b_0 \le 2\sqrt{n}$.

(iv) For three solutions $4 \le n_1 < n_2 < n_3$, partitioning the prime powers of $n_1!/4$ into the 8 sign octants $G_{\sigma_1\sigma_2\sigma_3}$ ($\sigma_i \in \{\pm 1\}$) and forming the four octant pairs $\alpha = G_{+++}G_{---}$, $\beta = G_{++-}G_{--+}$, $\gamma = G_{+-+}G_{-+-}$, $\delta = G_{+--}G_{-++}$ gives $X_{12}Y_{12} = \alpha\beta$, $Z_{12}W_{12} = \gamma\delta$, $X_{13}Y_{13} = \alpha\gamma$, $Z_{13}W_{13} = \beta\delta$, and $\gcd(A_2, A_3)\gcd(B_2, B_3)$ divisible by $\alpha\delta$. Substituting $U_{12} = a_{12}\alpha\beta$, $U_{13} = a_{13}\alpha\gamma$, $V_{12} = b_{12}\gamma\delta$, $V_{13} = b_{13}\beta\delta$, and $U_{23} = \widetilde{a}_{23}\alpha\delta$ into $U_{13} - U_{12} = U_{23} = V_{13} - V_{12}$ and dividing by α and δ yields the two linear equations in (6.10), and eliminating δ between them gives the third equation in (6.10). □

6.2. **Consecutive-Index Rigidity and Unconditional Square-Shift Exclusions.** Specializing Theorem 6.1 to consecutive indices $(n, n+1)$ and to square-ratio shifts $P = t^2$ yields the following unconditional exclusion laws.

Theorem 6.3 (Consecutive-Index Rigidity and Unconditional Square-Shift Exclusions)**.** *Let* (n_1, m_1) *and* (n_2, m_2) *be two solutions to* $n! + 1 = m^2$ *with* $4 \le n_1 < n_2$.

(i) ***Uniqueness of the*** $n = 4$ ***CRT Torus Boundary Collapse:*** *For consecutive indices* $(n, n+1)$*, the CRT torus boundary collapse* $m_{n+1} = \frac{n!}{2} - 1$ *(*$\varepsilon_p(n+1) = -1$ *for all* $p \le n$*, i.e.,* $X = W = 1$*) occurs if and only if* $n! = 4(n+2)$*, whose* ***unique*** *positive integer solution is* $n = 4$*, yielding the consecutive pair* $(4, 5)$ *and* $(5, 11)$.

(ii) ***Quarter-Factorial Equipartition and Exclusion of*** $(k_n, k_{n+1}) = (2, 2)$***:*** *For consecutive solutions* $(n, n+1)$ *with* $n \ge 6$*, all four GCD components satisfy* $X, Y, Z, W > 1$ *and* $\log X, \log Y, \log Z, \log W = \frac{1}{4}\log(n!) + O(\log n)$*, and* $(k_n, k_{n+1}) \ne (2, 2)$ *for all sufficiently large* n.

(iii) ***Unconditional Square-Shift Exclusions:*** *No two solutions* $n_1 < n_2$ *with* $P < 4(n_1! + 1)$ *can have* $P = \frac{n_2!}{n_1!} = t^2$ *a perfect square; in particular, for every integer* $t \ge 2$*,* $(t^2 - 1)! + 1$ *and* $(t^2)! + 1$ *can* ***never*** *both be perfect squares. Likewise, for consecutive squares* $n_1 = j^2$ *and* $n_2 = (j+1)^2$*,* $\log X, \log Y, \log Z, \log W = \frac{1}{4}\log((j^2)!) + O(j \log j)$ *unconditionally excludes* $(k_{j^2}, k_{(j+1)^2}) = (2, 2)$ *for all sufficiently large* j.

Proof. (i) For $n \geq 4$, $M_n := \frac{n!}{2}$ satisfies $M_n > m_{n+1} > m_n > 1$. The CRT torus boundary collapse $\varepsilon_p(n+1) = -1$ for all $p \leq n$ is equivalent to $m_{n+1} \equiv -1 \pmod{M_n}$, i.e., $m_{n+1} = M_n - 1$ (so $B_{n+1} = dYZ = M_n/2$ and $X = W = 1$). Squaring gives

$$(n+1)n! + 1 \;=\; m_{n+1}^2 \;=\; (M_n - 1)^2 \;=\; \frac{(n!)^2}{4} - n! + 1 \quad\Longleftrightarrow\quad n! \;=\; 4(n+2).$$

Since $n! < 4(n+2)$ for $1 \leq n \leq 3$, $4! = 24 = 4(6)$ at $n = 4$, and $n! > 4(n+2)$ strictly for all $n \geq 5$ by induction, $n = 4$ is the unique positive integer solution.

(ii) For $n \geq 6$, the four-squares identity $2acX^2 = (n+1)m_n - m_{n+1} - n \geq (n+1-\sqrt{n+1})\sqrt{n!} - n > 2n(n+1) \geq 2ac$ gives $X, Y, Z, W > 1$ and $\log X, \log Y, \log Z, \log W = \frac{1}{4}\log(n!) + O(\log n)$. If $k_n = k_{n+1} = 2$, then $m_n + 1 = 2YW = 2X_1(q_n)^2$ and $m_{n+1} + 1 = 2dYZ = 2X_1(q_{n+1})^2$, so no prime $p \in \mathcal{P}_{\mathrm{odd}}(n)$ with $p \nmid (n+1)$ can divide Y, W, or Z. Thus every prime $p \in \mathcal{P}_{\mathrm{odd}}(n)$ with $p \nmid (n+1)$ divides X to its full factorial power $p^{v_p(n!)}$, forcing $\log X \geq S_{\mathrm{odd}}(n) - O(n \log\log n) = (\frac{1}{2} - o(1))n \log n$, which contradicts $\log X = (\frac{1}{4} + o(1))n \log n$.

(iii) Suppose two solutions $4 \leq n_1 < n_2$ satisfy $P = \frac{n_2!}{n_1!} = t^2$ for some integer $t \geq 2$ with $P < 4(n_1! + 1)$. Then $m_2^2 - 1 = P(m_1^2 - 1) = t^2(m_1^2 - 1)$ rearranges to

$$t^2 m_1^2 - m_2^2 \;=\; t^2 - 1 \quad\Longleftrightarrow\quad (tm_1 - m_2)(tm_1 + m_2) \;=\; t^2 - 1.$$

Since $t^2 m_1^2 - m_2^2 = t^2 - 1 > 0$, we have $tm_1 > m_2$, so $tm_1 - m_2 \geq 1$ is a positive integer, forcing $tm_1 + m_2 \leq t^2 - 1$. However, $tm_1 + m_2 > tm_1 = t\sqrt{n_1! + 1} > t^2 - 1$ whenever $t < 2\sqrt{n_1! + 1}$. In particular, for consecutive integers $n_1 = t^2 - 1$ and $n_2 = t^2$ ($t \geq 2$), we have $P = t^2 < 4((t^2-1)! + 1)$, so $(t^2-1)! + 1$ and $(t^2)! + 1$ can **never** both be perfect squares. Similarly, for consecutive squares $n_1 = j^2$ and $n_2 = (j+1)^2$, we have $\log P = (2j+1)\log(j^2) + O(j) = O(j \log j) = o(\log((j^2)!))$, so (6.4) again forces $\log X, \log Y, \log Z, \log W = \frac{1}{4}\log((j^2)!) + O(j \log j)$, which unconditionally rules out $(k_{j^2}, k_{(j+1)^2}) = (2, 2)$ for all large j by the same $\mathcal{P}_{\mathrm{odd}}(j^2)$ weight saturation $S_{\mathrm{odd}}(j^2) = (\frac{1}{2} - o(1))\log((j^2)!) > (\frac{1}{4} + o(1))\log((j^2)!)$. □

6.3. **Odd-Harmonic Vaaler Diagonal Non-Vanishing, the Universal Capacity Diamond, and Proof of Theorem 1.3(i).** We now prove the Universal Capacity Diamond and complete the proof of Theorem 1.3(i). The crucial analytic observation is that when we correlate the valuation parities $(-1)^{v_p(n_1!)}$ and $(-1)^{v_p(n_2!)}$ of two solutions $n_1 < n_2$ with gap $h := n_2 - n_1 = n_1^\alpha$ ($\alpha \in [0, 1)$), the Vaaler trigonometric polynomial $\widetilde{\Psi}_H(x)$ for $(-1)^{\lfloor x \rfloor}$ is supported *exclusively on odd harmonics* $2m + 1 \neq 0$, so on the diagonal $2m_1 + 1 = -(2m_2 + 1)$ the difference phase $(2m_2 + 1)\frac{h}{2p}$ never vanishes.

Lemma 6.4 (Odd-Harmonic Vaaler Diagonal Non-Vanishing for Parity-Flip Primes)**.** *Let $4 \leq n_1 < n_2$ with $h := n_2 - n_1 = n_1^\alpha$ ($\alpha \in [0, 1)$), and set $P = n_2!/n_1!$. Then the logarithmic factorial weight of the parity-flip primes $\{p \leq n_1 : 2 \nmid v_p(P)\} = \{p \leq n_1 : v_p(n_1!) \not\equiv v_p(n_2!) \pmod 2\}$ satisfies unconditionally as $n_1 \to \infty$:*

$$\sum_{\substack{3 \leq p \leq n_1 \\ 2 \nmid v_p(P)}} v_p(n_1!) \log p \;=\; \left(\frac{\alpha}{2} + o(1)\right) n_1 \log n_1. \tag{6.11}$$

Proof. Set $y_2 := h(\log n_1)^2 + 2$ and $y_1 := \max(3, \frac{h}{(\log n_1)^{12}})$.

(1) *Upper Range $p \geq y_2$:* Any prime $p \geq y_2$ with $2 \nmid v_p(P)$ satisfies $v_p(P) \geq 1$, so $p \mid P = \prod_{j=n_1+1}^{n_1+h} j$. Since $v_p(n_1!) \leq \frac{n_1}{p-1} \leq \frac{n_1}{y_2 - 1}$ for all $p \geq y_2$, we have

$$\sum_{\substack{y_2 \leq p \leq n_1 \\ 2 \nmid v_p(P)}} v_p(n_1!) \log p \;\leq\; \frac{n_1}{y_2 - 1} \sum_{p \mid P} \log p \;\leq\; \frac{n_1 \log P}{y_2 - 1} \;\leq\; \frac{n_1 h \log(2n_1)}{h(\log n_1)^2} \;=\; O\!\left(\frac{n_1}{\log n_1}\right).$$

(2) *Transition Collar $y_1 < p < y_2$ and Tiny Range $p \le n_1^\delta$ ($\delta \in (0, \alpha/4)$):* By Mertens' theorem, $\sum_{y_1<p<y_2} v_p(n_1!)\log p \le n_1 \sum_{y_1<p<y_2} \frac{\log p}{p-1} \ll n_1 \log\log n_1 = o(n_1 \log n_1)$, and $\sum_{3\le p\le n_1^\delta} v_p(n_1!)\log p = (\delta + o(1))n_1 \log n_1$.

(3) *Oscillatory Range $n_1^\delta < p \le y_1$:* On this range, write $\mathbf{1}_{2\nmid v_p(P)} = \frac{1}{2} - \frac{1}{2}(-1)^{v_p(n_1!)}(-1)^{v_p(n_2!)}$. The main term $\frac{1}{2}$ contributes $\frac{1}{2}\sum_{n_1^\delta<p\le y_1} v_p(n_1!)\log p = (\frac{\alpha-\delta}{2} + o(1))n_1 \log n_1$. For the correlation sum $\sum_{n_1^\delta<p\le y_1}(-1)^{v_p(n_1!)+v_p(n_2!)} v_p(n_1!)\log p$, expanding $(-1)^{\lfloor n_1/p\rfloor}$ and $(-1)^{\lfloor n_2/p\rfloor}$ via the Vaaler polynomial $\widetilde{\Psi}_H$ of Lemma 2.3 ($H = \lfloor(\log n_1)^3\rfloor$), note that in $\widetilde{\Psi}_H(x) = 2\Psi_H(x) - 4\Psi_{2H+1}(x/2)$, every even harmonic cancels identically, leaving only odd frequencies $h_1 = 2m_1 + 1$ and $h_2 = 2m_2 + 1$. Thus for every pair of odd frequencies (h_1, h_2), the p^{-1} phase is

$$\frac{h_1 n_1 + h_2 n_2}{2p} = \frac{(h_1 + h_2)n_1 + h_2 h}{2p}.$$

If $h_1 + h_2 \neq 0$, then $|(h_1 + h_2)n_1 + h_2 h| \ge 2n_1 - 2Hh \gg n_1$, giving derivative parameter $T \gg n_1/P$. If $h_1 + h_2 = 0$ (the diagonal $h_1 = -h_2 = -(2m_2+1)$), then because $h_2 = 2m_2 + 1 \neq 0$ is odd, the phase is $\frac{(2m_2+1)h}{2p} \neq 0$ with derivative parameter $T = \frac{|2m_2+1|h}{2P} \ge \frac{h}{2y_1} \ge \frac{1}{2}(\log n_1)^{12}$. Applying Lemma 2.4 on dyadic intervals in $(n_1^\delta, y_1]$ cancels the correlation sum to $o(n_1 \log n_1)$. Letting $\delta \downarrow 0$ yields (6.11).

□

Theorem 6.5 (Universal Capacity Diamond and Proof of Theorem 1.3(i))**.** *For any solution (n, m) to $n! + 1 = m^2$, define its odd-core polarization weight $\theta(n) := \frac{\log B^{(2)}(n)}{S_{\mathrm{odd}}(n)} \in [0, 1]$ and, for any $\varepsilon \in (0, 1/4)$, its upper-band polarization weight*

$$\theta_\varepsilon(n) := \frac{\displaystyle\sum_{n^{1-\varepsilon/2}<p\le n,\, p\mid B^{(2)}(n)} v_p(n!)\log p}{\displaystyle\sum_{n^{1-\varepsilon/2}<p\le n,\, p\in\mathcal{P}_{\mathrm{odd}}(n)} v_p(n!)\log p} \in [0, 1]$$

(where $\theta(n) = \theta_\varepsilon(n) = 0$ if $k_n = 2$). For any two solutions $4 \le n_1 < n_2$ with $n_2 - n_1 = n_1^\alpha$ ($\alpha \in [0, 1)$), the pair $(\theta(n_1), \theta(n_2))$ lies inside the ***Capacity Diamond***

$$\frac{1-\alpha}{2} - o(1) \le \theta(n_1) + \theta(n_2) \le \frac{3+\alpha}{2} + o(1), \qquad |\theta(n_1) - \theta(n_2)| \le \frac{1+\alpha}{2} + o(1). \tag{6.12}$$

Consequently, any two solutions $n_1 < n_2$ in the $k = 2$ regime ($\theta(n_1) = \theta(n_2) = 0$) are separated by $n_2 - n_1 > n_1^{1-\varepsilon}$ for every $\varepsilon > 0$, proving Theorem 1.3(i), and for every $\eta \in (0, 1/2]$, $\#\{n \le N : |\theta_\varepsilon(n) - 1/2| \ge \eta\} \le (\lfloor(2\eta^2\varepsilon)^{-1}\rfloor + 1)N^\varepsilon$.

Proof. Let $4 \le n_1 < n_2$ be any two solutions with $h = n_2 - n_1 = n_1^\alpha$ ($\alpha \in [0, 1)$). From (6.4), since $\log P \le h \log n_2 = O(n_1^\alpha \log n_1) = o(n_1 \log n_1)$, both ZW and XY satisfy

$$\log(ZW) = \frac{1}{2}\log(n_1!) + O(\log P), \qquad \log(XY) = \frac{1}{2}\log(n_1!) + O(\log P). \tag{6.13}$$

Decompose $A_i^* = B^{(1)}(n_i)Z_1^*(n_i)^2$ and $B_i^* = B^{(2)}(n_i)Z_2^*(n_i)^2$ ($i = 1, 2$) as in Theorem 5.2(i). Consider any odd prime $p \le n_1$ with $2 \mid v_p(P)$ (so $v_p(n_1!) \equiv v_p(n_2!) \pmod 2$):

- If $p \mid Z = \gcd(A_1, B_2)$, then either $p \in \mathcal{P}_{\mathrm{odd}}(n_1) \cap \mathcal{P}_{\mathrm{odd}}(n_2)$ (so $p^{v_p(n_1!)} \mid B^{(2)}(n_2)$) or $p \in \mathcal{P}_{\mathrm{even}}(n_1) \cap \mathcal{P}_{\mathrm{even}}(n_2)$ (so $p^{v_p(n_1!)} \mid Z_1^*(n_1)^2$).
- If $p \mid W = \gcd(B_1, A_2)$, then either $p \in \mathcal{P}_{\mathrm{odd}}(n_1) \cap \mathcal{P}_{\mathrm{odd}}(n_2)$ (so $p^{v_p(n_1!)} \mid B^{(2)}(n_1)$) or $p \in \mathcal{P}_{\mathrm{even}}(n_1) \cap \mathcal{P}_{\mathrm{even}}(n_2)$ (so $p^{v_p(n_1!)} \mid Z_1^*(n_2)^2$).

Consequently, ZW divides $2^{v_2(n_1!)} B^{(2)}(n_1)B^{(2)}(n_2)Z_1^*(n_1)^2 Z_1^*(n_2)^2 \prod_{3\le p\le n_1, 2\nmid v_p(P)} p^{v_p(n_1!)}$. Taking logarithms and applying the cross-symmetry law (5.4) ($\log(B^{(2)}(n_i)Z_1^*(n_i)^2) = (\theta(n_i) + o(1))n_1 \log n_1$) together with Lemma 6.4 and (6.13) gives

$$\frac{1}{2} n_1 \log n_1 \le (\theta(n_1) + \theta(n_2) + \frac{\alpha}{2} + o(1))n_1 \log n_1 \quad \Longrightarrow \quad \theta(n_1) + \theta(n_2) \ge \frac{1-\alpha}{2} - o(1).$$

Repeating the identical divisibility argument with $(B^{(1)}, Z_2^*)$ (where $\log(B^{(1)}(n_i)Z_2^*(n_i)^2) = (1 - \theta(n_i) + o(1))n_1 \log n_1$) gives $\theta(n_1) + \theta(n_2) \le \frac{3+\alpha}{2} + o(1)$, and applying both arguments to $XY = \gcd(A_1, A_2)\gcd(B_1, B_2)$ gives $|\theta(n_1) - \theta(n_2)| \le \frac{1+\alpha}{2} + o(1)$, establishing (6.12).

When both n_1 and n_2 lie in the $k = 2$ regime, $\theta(n_1) = \theta(n_2) = 0$, so (6.12) forces $\frac{1-\alpha}{2} \le o(1) \implies \alpha \ge 1 - o(1)$, proving $n_2 - n_1 > n_1^{1-\varepsilon}$ and $\#\mathcal{B}_{k=2}(N) \ll_\varepsilon N^\varepsilon$ (Theorem 1.3(i)). Finally, in the weighted Hilbert space $\ell^2(w)$ on primes $p \le N'$ with weights $w_p = \frac{v_p(N'!)\log p}{\log(N'!)}$, each solution $n \in I = [N', N' + (N')^{1-\varepsilon}]$ defines a unit sign vector $\mathbf{u}_n$ ($u_n(p) = \varepsilon_p(n)$) satisfying $|\langle \mathbf{u}_{n_1}, \mathbf{u}_{n_2}\rangle_w| = |\frac{\log(XY)-\log(ZW)}{\log(N'!)}| + O((N')^{-\varepsilon}) \le 3(N')^{-\varepsilon}$ by (6.13), while any η-polarized solution correlates by $\ge \eta\sqrt{2\varepsilon} - o(1)$ with the fixed parity reference vector $\mathbf{v}_I$ on $((N')^{1-\varepsilon/2}, N']$. Applying Rankin's spherical cap Cauchy–Schwarz inequality $t^2\eta_0^2 \le t + 3t(t-1)(N')^{-\varepsilon}$ bounds the number of η-polarized solutions in I by $\lfloor (2\eta^2\varepsilon)^{-1} \rfloor + 1$. □

7. Support-Reduced Padé–Wronskian Approximations and Proof of Theorem 1.3(ii)

In this section we prove Theorem 1.3(ii) and establish unconditional finiteness for the generalized Brocard–Ramanujan equation $n! + A = m^k$ outside $n! + c^2 = m^2$. Let $P(x) \in \mathbb{Z}[x]$ be a polynomial of degree $r \geq 2$ and let $s \in \mathbb{Z} \setminus \{0\}$. By the classical depression substitution $x \mapsto r a_r x + a_{r-1}$ and scaling of s by $(r a_r)^{2r}$ (Berend and Osgood [6, Lemma 1]; Bui, Pratt, and Zaharescu [12, Section 3]), we may assume without loss of generality that

$$P(x) \;=\; a_r x^r + a_{r-\nu} x^{r-\nu} + \cdots + a_0 \;\in\; \mathbb{Z}[x], \qquad a_r > 0, \quad s > 0, \tag{7.1}$$

where $a_{r-1} = 0$ and $\nu := r - \deg(P(x) - a_r x^r) \in \{2, \ldots, r\}$ denotes the degree gap between the leading term $a_r x^r$ and the next non-zero term of $P(x)$ (for Brocard's equation $n! + 1 = m^2$, $P(x) = x^2 - 1$ has $r = 2$ and $\nu = 2$).

7.1. The $(m+1)$-Variable Binomial-Product Padé System and Siegel's Lemma.

Fix an integer $m \geq 2$ and a real parameter $\varepsilon_0 \in (0, m-1)$. Let $M := \lfloor N^\theta \rfloor$ with $0 < \theta < \frac{1}{m+2}$. Let $r \leq \beta_1 < \beta_2 < \cdots < \beta_m \leq M$ be positive integer multiples of r, and consider the $m+1$ algebraic functions

$$\omega_0(x) \;:=\; 1, \qquad \omega_i(x) \;:=\; \prod_{j=1}^{\beta_i - 1} (1 - jx)^{-1/r} \;=\; \sum_{\ell=0}^{\infty} b_{i,\ell}\, x^\ell \qquad (1 \leq i \leq m) \tag{7.2}$$

analytic in the disk $|x| < \frac{1}{\beta_m - 1}$. By the binomial expansion $(1-jx)^{-1/r} = \sum_{u=0}^{\infty} \binom{-1/r}{u} (-jx)^u$ and Cauchy's integral estimates [12, Lemma 4.2], each Taylor coefficient $b_{i,\ell} \in \mathbb{Q}$ satisfies $r^{2\ell} b_{i,\ell} \in \mathbb{Z}$ and $|b_{i,\ell}| \leq 2^{\beta_i} (2\beta_i)^\ell \leq 2^M (2M)^\ell$.

Lemma 7.1 (Initial $(m+1)$-Variable Padé Polynomials)**.** *Let $D \geq 10m$ be an integer, let $\varepsilon_0 \in (0, m-1)$, and set $O := \lfloor (m + 1 - \varepsilon_0) D \rfloor$. There exist polynomials $P_0(x), P_1(x), \ldots, P_m(x) \in \mathbb{Z}[x]$ of degree at most D, not all identically zero, such that the Padé remainder series*

$$R(x) \;:=\; \sum_{i=0}^{m} P_i(x)\, \omega_i(x) \;=\; \sum_{v=0}^{\infty} r_v\, x^v \tag{7.3}$$

satisfies $r_v = 0$ for all $0 \leq v \leq O$ (so $\operatorname{ord}_{x=0} R(x) \geq O + 1$), and the coefficients $p_{i,k}$ of $P_i(x) = \sum_{k=0}^{D} p_{i,k} x^k$ $(0 \leq i \leq m)$ and Taylor coefficients r_v $(v > O)$ satisfy the upper bounds

$$|p_{i,k}|,\; |r_v| \;\leq\; 4^{m \varepsilon_0^{-1} M} \left(2 m r^4 M\right)^{\frac{(m-\varepsilon_0)(m+1-\varepsilon_0)}{\varepsilon_0} D} (2M)^v. \tag{7.4}$$

Proof. Write $P_i(x) = \sum_{k=0}^{D} p_{i,k} x^k$. Expanding Cauchy products in (7.3) shows that the condition $r_v = 0$ for $0 \leq v \leq O$ is equivalent to the homogeneous linear system of $O+1$ equations

$$r^{2O}\, p_{0,v} \cdot \mathbf{1}_{v \leq D} \;+\; \sum_{i=1}^{m} \sum_{k=0}^{\min(v, D)} p_{i,k} \left(r^{2O} b_{i, v-k}\right) \;=\; 0 \qquad (0 \leq v \leq O), \tag{7.5}$$

where every coefficient $r^{2O} b_{i,v-k}$ is an integer of absolute value at most $r^{2O} 2^M (2M)^O$. Consider first the subsystem of (7.5) indexed by $D+1 \leq v \leq O$: since $p_{0,v} \cdot \mathbf{1}_{v \leq D} = 0$ for $v \geq D+1$, this subsystem consists of

$$M_{\mathrm{eq}} \;:=\; O - D \;\leq\; (m - \varepsilon_0)\, D$$

homogeneous linear equations with integer coefficients in the

$$N_{\mathrm{var}} \;:=\; m(D+1) \;>\; mD$$

unknowns $\{p_{i,k} : 1 \leq i \leq m,\, 0 \leq k \leq D\}$. Since $\varepsilon_0 \in (0, m-1)$, we have $N_{\mathrm{var}} - M_{\mathrm{eq}} > \varepsilon_0 D > 0$, so the Dirichlet box-principle exponent in Siegel's lemma [12, Lemma 4.3] is bounded above by

$$\frac{M_{\mathrm{eq}}}{N_{\mathrm{var}} - M_{\mathrm{eq}}} \;\leq\; \frac{(m - \varepsilon_0) D}{\varepsilon_0 D} \;=\; \frac{m - \varepsilon_0}{\varepsilon_0}.$$

Applying Siegel's lemma yields a non-zero integer solution $\{\tilde{p}_{i,k} : 1 \le i \le m,\ 0 \le k \le D\}$ bounded in absolute value by $\left(m(D+1)r^{2O}2^M(2M)^O\right)^{\frac{m-\varepsilon_0}{\varepsilon_0}}$. Multiplying each $\tilde{p}_{i,k}$ by r^{2O} to define $p_{i,k} := r^{2O}\tilde{p}_{i,k}$ $(1 \le i \le m)$ ensures that $r^{2O} \mid p_{i,k}$, so the remaining $D+1$ equations of (7.5) for $0 \le v \le D$ uniquely define integers

$$p_{0,v} \;:=\; -\sum_{i=1}^{m}\sum_{k=0}^{v}\frac{p_{i,k}}{r^{2O}}\left(r^{2O}b_{i,v-k}\right) \;\in\; \mathbb{Z} \qquad (0 \le v \le D).$$

Using $O \le (m+1-\varepsilon_0)D$ and $|b_{i,v-k}| \le 2^M(2M)^v$ yields (7.4). □

7.2. **Puiseux Linear Independence, the Support-Reduced Wronskian, and Gauss's Lemma Multiplicity Lock.** Define the logarithmic derivatives $A_0(x) := 0$ and

$$A_i(x) \;:=\; \frac{\omega_i'(x)}{\omega_i(x)} \;=\; \sum_{j=1}^{\beta_i-1}\frac{j}{r(1-jx)} \quad (1 \le i \le m), \qquad T(x) \;:=\; r\prod_{j=1}^{M}(1-jx) \;\in\; \mathbb{Z}[x], \tag{7.6}$$

so that $T(x)A_i(x) \in \mathbb{Z}[x]$ and $\deg(T) = M$. For every integer $k \ge 0$ and $0 \le i \le m$, define the differential iterates

$$\begin{aligned} P_i^{[k]}(x) \;&:=\; \left(T(x)\frac{d}{dx} + T(x)A_i(x)\right)^k P_i(x) \;\in\; \mathbb{Z}[x], \\ R^{[k]}(x) \;&:=\; \left(T(x)\frac{d}{dx}\right)^k R(x) \;=\; \sum_{i=0}^{m} P_i^{[k]}(x)\,\omega_i(x). \end{aligned} \tag{7.7}$$

Lemma 7.2 (Linear Independence of $\omega_0, \dots, \omega_m$ and Monomial Products over $\mathbb{C}(x)$)**.** *For any integer $d \ge 1$, any shifts $r \le \beta_1 < \beta_2 < \cdots < \beta_d$ $(r \mid \beta_i)$, and any degree bound $1 \le E \le r-1$, the $\binom{d+E}{d}$ monomial-product functions*

$$\Omega_{\mathbf{e}}(x) \;:=\; \prod_{i=1}^{d}\omega_i(x)^{e_i} \;=\; \prod_{i=1}^{d}\prod_{j=1}^{\beta_i-1}(1-jx)^{-e_i/r}, \qquad \mathbf{e} = (e_1,\dots,e_d) \in \mathbb{Z}_{\ge 0}^d,\ \ |\mathbf{e}| := \sum_{i=1}^{d} e_i \le E, \tag{7.8}$$

are linearly independent over $\mathbb{C}(x)$. In particular, taking $E = 1$ and $d = m$, the functions $\omega_0(x), \omega_1(x), \dots, \omega_m(x)$ are linearly independent over $\mathbb{C}(x)$.

Proof. We proceed by induction on $d \ge 1$. Suppose there exists a relation $\sum_{|\mathbf{e}|\le E} U_{\mathbf{e}}(x)\Omega_{\mathbf{e}}(x) = 0$ with $U_{\mathbf{e}}(x) \in \mathbb{C}[x]$. Grouping terms according to the last exponent $e_d \in \{0, 1, \dots, E\}$ gives

$$\sum_{e_d=0}^{E}\Phi_{e_d}(x)\,\omega_d(x)^{e_d} \;=\; 0, \qquad \Phi_{e_d}(x) \;:=\; \sum_{\substack{(e_1,\dots,e_{d-1}) \\ e_1+\cdots+e_{d-1} \le E-e_d}} U_{(e_1,\dots,e_d)}(x)\prod_{i=1}^{d-1}\omega_i(x)^{e_i}. \tag{7.9}$$

Consider the local Puiseux expansion in a punctured neighborhood of the branch point $x_* := \frac{1}{\beta_d - 1}$. Because $\beta_i - 1 < \beta_d - 1$ for all $1 \le i \le d-1$, each function $\omega_i(x)$ $(1 \le i \le d-1)$ is holomorphic and non-vanishing at $x = x_*$. Consequently, each coefficient $\Phi_{e_d}(x)$ is holomorphic at $x = x_*$, so if $\Phi_{e_d}(x) \not\equiv 0$, its order of vanishing at $x = x_*$ is a non-negative integer $k_{e_d} \in \mathbb{Z}_{\ge 0}$. On the other hand, $\omega_d(x) = (1-(\beta_d-1)x)^{-1/r}\prod_{j=1}^{\beta_d-2}(1-jx)^{-1/r}$ has fractional Puiseux valuation $-\frac{1}{r}$ at $x = x_*$, so the e_d-th term $\Phi_{e_d}(x)\omega_d(x)^{e_d}$ has Puiseux valuation

$$v_{x_*}\!\left(\Phi_{e_d}(x)\omega_d(x)^{e_d}\right) \;=\; k_{e_d} - \frac{e_d}{r} \;\in\; -\frac{e_d}{r} + \mathbb{Z}.$$

Since $0 \le e_d \le E \le r-1$, the cosets $-\frac{e_d}{r} + \mathbb{Z}$ $(0 \le e_d \le E)$ in $\mathbb{Q}/\mathbb{Z}$ are pairwise distinct. Therefore no two terms with different e_d can cancel in the Puiseux expansion at $x = x_*$, forcing $\Phi_{e_d}(x) \equiv 0$ for every $0 \le e_d \le E$, which by the induction hypothesis on $d-1$ forces $U_{\mathbf{e}}(x) \equiv 0$ for all $|\mathbf{e}| \le E$. □

When $\varepsilon_0 \ge 1$ in Lemma 7.1, some of the polynomials $P_i(x)$ may vanish identically, causing the full $(m+1)\times(m+1)$ Wronskian determinant $\det(P_i^{[k]}(x))_{0\le k,i\le m}$ to have a column of zeros. We overcome this obstacle by passing to the non-zero support $S \subset \{0, 1, \dots, m\}$.

Lemma 7.3 (Support-Reduced Wronskian Non-Vanishing and Gauss's Lemma Multiplicity Bound)**.** *Let $(P_0, \dots, P_m)$ be the Padé polynomials constructed in Lemma 7.1 for $\varepsilon_0 \in (0, m-1)$, and let*

$$S \;:=\; \{0 \le i \le m : P_i(x) \not\equiv 0\} \;=\; \{i_1 < i_2 < \cdots < i_s\} \tag{7.10}$$

be the non-zero support of $(P_0, \dots, P_m)$, of cardinality $s := |S|$. Assume $D > (m+1)^2 M$. Then:

(i) *The $s \times s$ support-reduced Wronskian determinant*

$$\Delta_S(x) \;:=\; \det\Big(P_{i_a}^{[k]}(x)\Big)_{0 \le k \le s-1,\, 1 \le a \le s} \;\in\; \mathbb{Z}[x] \tag{7.11}$$

*is **not identically zero** ($\Delta_S(x) \not\equiv 0$), has degree $\deg(\Delta_S) \le sD + \binom{s}{2}M$, and vanishes at $x = 0$ to order $\operatorname{ord}_{x=0} \Delta_S(x) \ge O - (s-1)$. Consequently, the support size satisfies $\lceil m + 1 - \varepsilon_0 \rceil \le s \le m+1$ (in particular, $s \ge 2$), and the sum of the multiplicities of all non-zero roots of $\Delta_S(x)$ in $\mathbb{C}^\times$ satisfies*

$$\sum_{\alpha \in \mathbb{C}^\times} \operatorname{ord}_{x=\alpha} \Delta_S(x) \;\le\; \deg(\Delta_S) - \operatorname{ord}_{x=0} \Delta_S(x) \;\le\; (s - (m+1-\varepsilon_0))D + \binom{s}{2} M + s. \tag{7.12}$$

(ii) *For any $\alpha = 1/n_0 \in \mathbb{Q} \setminus \{0\}$ with $n_0 > M$, let $a := \operatorname{ord}_{x=\alpha} \Delta_S(x) \ge 0$, and define the normalized operators $P_i^k(x) := \frac{T(x)^k}{k!}\big(\frac{d}{dx} + A_i(x)\big)^k P_i(x) \in \mathbb{Q}[x]$. Then the $(a+s) \times s$ rational matrix $\big(P_{i_a}^k(\alpha)\big)_{0 \le k \le a+s-1,\, 1 \le a \le s}$ has full column rank s.*

(iii) *For any distinct integers $N \le n_1 < n_2 < \cdots < n_t < 2N$, let $a_r := \operatorname{ord}_{x=1/n_r} \Delta_S(x)$. Then $\prod_{r=1}^t (n_r x - 1)^{a_r}$ divides $\Delta_S(x)$ in $\mathbb{Z}[x]$, so $\prod_{r=1}^t n_r^{a_r}$ divides the non-zero integer leading coefficient $c_{\mathrm{lead}}(\Delta_S) \in \mathbb{Z} \setminus \{0\}$, yielding the Gauss's Lemma multiplicity bound*

$$\sum_{r=1}^{t} a_r \;\le\; s\,\frac{(m-\varepsilon_0)(m+1-\varepsilon_0)}{\varepsilon_0}\,\frac{\log M}{\log N}\,D + O_m\!\left(\frac{D + M\log N}{\log N}\right). \tag{7.13}$$

Proof. (i) Since $(P_0, \dots, P_m) \neq (0, \dots, 0)$, we have $s = |S| \ge 1$. Because $P_{i_a}(x) \not\equiv 0$ for every $1 \le a \le s$ by definition of S and $\omega_{i_1}(x), \dots, \omega_{i_s}(x)$ are linearly independent over $\mathbb{C}(x)$ by Lemma 7.2, Siegel's differential Wronskian lemma [53, Chapter 2, Lemma 4] (cf. [12, Lemma 5.4]) applies directly: if $\Delta_S(x) \equiv 0$, then the rows of $(P_{i_a}^{[k]}(x))_{0 \le k \le s-1, 1 \le a \le s}$ satisfy a non-trivial relation $\sum_{k=0}^{\mu} B_k(x) P_{i_a}^{[k]}(x) = 0$ over $\mathbb{C}[x]$ of minimal order $\mu \le s-1$, which implies that the s non-zero functions $f_a(x) := P_{i_a}(x)\omega_{i_a}(x)$ $(1 \le a \le s)$ all satisfy the homogeneous linear differential equation $\sum_{k=0}^{\mu} B_k(x)(T\frac{d}{dx})^k f = 0$ of order $\mu \le s-1$. Since the solution space of an order-μ linear ODE has dimension $\mu \le s-1$ over $\mathbb{C}$, the functions $f_1, \dots, f_s$ would be $\mathbb{C}$-linearly dependent, yielding a non-trivial relation $\sum_{a=1}^{s} c_a P_{i_a}(x)\omega_{i_a}(x) = 0$ with $c_a \in \mathbb{C}$ and $P_{i_a}(x) \not\equiv 0$, contradicting Lemma 7.2. Thus $\Delta_S(x) \not\equiv 0$.

Since $P_i(x) \equiv 0$ for all $i \notin S$, we have $R^{[k]}(x) = \sum_{a=1}^{s} P_{i_a}^{[k]}(x)\omega_{i_a}(x)$ for every $k \ge 0$. Multiplying the first column of $\Delta_S(x)$ by $\omega_{i_1}(x)$ and adding $\omega_{i_a}(x)$ times the a-th column for $2 \le a \le s$ replaces the first column by $(R^{[k]}(x))_{0 \le k \le s-1}$. Expanding by cofactors $\Delta_{1,k}(x) \in \mathbb{Z}[x]$ along the first column gives

$$\Delta_S(x)\,\omega_{i_1}(x) \;=\; \sum_{k=0}^{s-1} \Delta_{1,k}(x)\,R^{[k]}(x). \tag{7.14}$$

Since $\operatorname{ord}_{x=0} R(x) \ge O + 1$ and each application of $T(x)\frac{d}{dx}$ decreases the order of vanishing at $x = 0$ by at most 1, we have $\operatorname{ord}_{x=0} R^{[k]}(x) \ge O + 1 - k \ge O - (s-1)$ for all $0 \le k \le s-1$. Since $\omega_{i_1}(0) = 1 \neq 0$, (7.14) gives $\operatorname{ord}_{x=0} \Delta_S(x) \ge O - (s-1)$. Moreover, $\deg(P_{i_a}^{[k]}) \le D + kM$, so $\deg(\Delta_S) \le sD + \binom{s}{2}M$. Comparing $\deg(\Delta_S) \ge \operatorname{ord}_{x=0} \Delta_S(x)$ gives $sD + \binom{s}{2}M \ge (m+1-\varepsilon_0)D - s$, which for $D > (m+1)^2 M$ forces $s \ge \lceil m + 1 - \varepsilon_0 \rceil \ge 2$ and proves (7.12).

(ii) Since $T(\alpha) = r\prod_{j=1}^{M}(1 - j/n_0) \neq 0$ for $n_0 > M$, the operators $(T\frac{d}{dx})^k$ and $\frac{T(x)^k}{k!}\frac{d^k}{dx^k}$ span the same $\mathbb{Q}$-vector space of differential operators at $x = \alpha$; differentiating (7.14) up to order $a = \operatorname{ord}_{x=\alpha} \Delta_S(x)$ at $x = \alpha$ via Leibniz's rule shows that $(P_{i_a}^k(\alpha))_{0 \le k \le a+s-1, 1 \le a \le s}$ has rank s [12, Lemmas 5.6 and 6.3].

(iii) Since $n_r \geq N \geq 2$, each linear polynomial $n_r x - 1 \in \mathbb{Z}[x]$ is primitive and irreducible in the unique factorization domain $\mathbb{Z}[x]$, and distinct integers $n_1 < \cdots < n_t$ give pairwise coprime polynomials $n_r x - 1$ in $\mathbb{Q}[x]$ and hence non-associate prime elements in $\mathbb{Z}[x]$. Since $(n_r x - 1)^{a_r}$ divides $\Delta_S(x)$ in $\mathbb{Q}[x]$ and $n_r x - 1$ is primitive, Gauss's Lemma implies that $\prod_{r=1}^{t}(n_r x - 1)^{a_r}$ divides $\Delta_S(x)$ in $\mathbb{Z}[x]$:

$$\Delta_S(x) \;=\; Q(x)\prod_{r=1}^{t}(n_r x - 1)^{a_r}, \qquad Q(x) \in \mathbb{Z}[x] \setminus \{0\}.$$

Comparing the leading coefficients on both sides gives $\prod_{r=1}^{t} n_r^{a_r} \mid c_{\mathrm{lead}}(\Delta_S) \in \mathbb{Z} \setminus \{0\}$. Since $n_r \geq N$, we have $N^{\sum_{r=1}^{t} a_r} \leq \prod_{r=1}^{t} n_r^{a_r} \leq |c_{\mathrm{lead}}(\Delta_S)|$. Bounding $|c_{\mathrm{lead}}(\Delta_S)|$ by Hadamard's inequality and (7.4) gives $|c_{\mathrm{lead}}(\Delta_S)| \leq (2mr^4M)^{s\frac{(m-\varepsilon_0)(m+1-\varepsilon_0)}{\varepsilon_0}D + O_m(D/\log M + M)}$, and taking logarithms and dividing by $\log N$ yields (7.13). □

Before completing the proof of Theorem 1.3(ii), we record the explicit Leibniz derivative and factorial-root approximation estimates at $x = 1/n_0$.

Lemma 7.4 (Leibniz Derivative Estimates and Factorial-Root Approximation at $x = 1/n_0$)**.** *Under the hypotheses of Lemma 7.1 and Lemma 7.3, let $N \leq n_0 < 2N$ with $n_0 > 2M$, and set $P_i^k(x) := \frac{T(x)^k}{k!}\big(\frac{d}{dx} + A_i(x)\big)^k P_i(x) \in \mathbb{Q}[x]$ and $R^k(x) := \frac{T(x)^k}{k!}\frac{d^k}{dx^k}R(x) = \sum_{i=0}^{m} P_i^k(x)\omega_i(x)$.*

(i) *For every $0 \leq k \leq a + s - 1$, the rational numbers $P_i^k(1/n_0)$ $(0 \leq i \leq m)$ have a common denominator dividing $Z := r^{a+s} n_0^{D+(a+s)M}$, and satisfy the upper bounds*

$$|P_i^k(1/n_0)| \;\leq\; u\,U^D, \qquad |R^k(1/n_0)| \;\leq\; w\,W^{-D}, \tag{7.15}$$

where $U = (2mr^4M)^{\frac{(m-\varepsilon_0)(m+1-\varepsilon_0)}{\varepsilon_0}}$, $W = N^{m+1-\varepsilon_0}(8mr^4M)^{-\frac{m(m+1-\varepsilon_0)}{\varepsilon_0}}$, and the sub-exponential factors are $u = (4rN)^{M^{m+2}} 10^{m\varepsilon_0^{-1}M}$ and $w = (2rN)^{2M^{m+2}} 4^{m\varepsilon_0^{-1}M}$.

(ii) *Suppose $sn_0! = P(x_0)$ and $s(n_0 - \beta_i)! = P(x_i)$ $(1 \leq i \leq d)$, where $P(x) = a_r x^r + a_{r-\nu}x^{r-\nu} + \cdots + a_0$ has degree $r \geq 2$ and degree gap $\nu \in \{2, \ldots, r\}$. Then for every multi-index $\mathbf{e} = (e_1, \ldots, e_d) \in \mathbb{Z}_{\geq 0}^d$ with $|\mathbf{e}| \leq E \leq \min(r-1, \nu-1)$, setting $q_{\mathbf{e}} := x_0^{E-|\mathbf{e}|}\prod_{i=1}^{d}(n_0^{\beta_i/r}x_i)^{e_i} \in \mathbb{N}$, we have*

$$\left|\Omega_{\mathbf{e}}\Big(\frac{1}{n_0}\Big) - \frac{q_{\mathbf{e}}}{x_0^E}\right| \;\ll_{P,E}\; M^E\, x_0^{-\nu} \;\leq\; x_0^{-\nu+o(1)}. \tag{7.16}$$

Proof. (i) Since $T(x) = r\prod_{j=1}^{M}(1 - jx)$ and $T(x)A_i(x) \in \mathbb{Z}[x]$ have degree $\leq M$, induction on k via Leibniz's rule shows that $r^k k!\, P_i^k(x) = r^k P_i^{[k]}(x) + \sum_{j=0}^{k-1} c_{k,j}(x) P_i^{[j]}(x)$ is a polynomial in $\mathbb{Z}[x]$ of degree $\leq D + kM$. Evaluating at $x = 1/n_0$ shows that $r^{a+s} n_0^{D+(a+s)M} P_i^k(1/n_0) \in \mathbb{Z}$ for all $0 \leq k \leq a + s - 1$. Cauchy's integral formula on $|z - 1/n_0| = \frac{1}{2n_0}$ combined with (7.4) bounds $|P_i^k(1/n_0)| \leq uU^D$, while differentiating $R(x) = \sum_{v=O+1}^{\infty} r_v x^v$ term-by-term at $x = 1/n_0 \leq 1/N$ gives

$$|R^k(1/n_0)| \;\leq\; \frac{|T(1/n_0)|^k}{k!}\sum_{v=O+1}^{\infty} v(v-1)\cdots(v-k+1)\,|r_v|\,n_0^{-(v-k)} \;\leq\; w\,W^{-D},$$

since $O + 1 \geq (m + 1 - \varepsilon_0)D$ and $\sum_{v>O}\binom{v}{k}(2M/n_0)^v \leq 2\binom{O+1}{k}(4M/N)^{O+1}$ for $N > 8M$.

(ii) Dividing $sn_0! = a_r x_0^r(1 + O(x_0^{-\nu}))$ by $s(n_0 - \beta_i)! = a_r x_i^r(1 + O(x_i^{-\nu}))$ gives

$$\frac{n_0!}{(n_0 - \beta_i)!} \;=\; n_0^{\beta_i}\prod_{j=1}^{\beta_i - 1}\left(1 - \frac{j}{n_0}\right) \;=\; \left(\frac{x_i}{x_0}\right)^r (1 + O(x_0^{-\nu})).$$

Raising both sides to the power $-1/r$ and multiplying by $n_0^{\beta_i/r}$ (which is an integer since $r \mid \beta_i$) yields $\omega_i(1/n_0) = \prod_{j=1}^{\beta_i - 1}(1 - j/n_0)^{-1/r} = \frac{n_0^{\beta_i/r}x_i}{x_0}(1 + O(x_0^{-\nu}))$. Taking products over $1 \leq i \leq d$ with exponents e_i $(\sum e_i \leq E)$ gives $\Omega_{\mathbf{e}}(1/n_0) = \frac{q_{\mathbf{e}}}{x_0^E}(1 + O(Ex_0^{-\nu})) = \frac{q_{\mathbf{e}}}{x_0^E} + O(M^E x_0^{-\nu})$, since $\omega_i(1/n_0) = 1 + O(\beta_i^2/n_0) \ll M$. □

7.3. **Proof of Theorem 1.3(ii) and Generalized Finiteness for $n!+A=m^k$.**

Proof of Theorem 1.3(ii). **Step 1 (The $(m+1)$-Variable Support-Reduced Padé–Wronskian Bound and the $6/7+\varepsilon$ Theorem):** Fix $m\ge 2$, $\varepsilon_0\in(0,m-1)$, and $0<\theta<\min(\frac{\varepsilon_0(m-1-\varepsilon_0)}{(m+1-\varepsilon_0)(2m-\varepsilon_0)},\frac{1}{m+2})$, and set $M:=\lfloor N^\theta\rfloor$. Let $N\le n_1<n_2<\cdots<n_R<2N$ be the solutions in $[N,2N)$ to $s\cdot n!=P(x)$. Grouping these solutions into disjoint $(mr+1)$-tuples $(n_q,n_{q+1},\dots,n_{q+mr})$, at most $(mr+1)\frac{N}{M}$ solutions belong to tuples of diameter $n_{q+mr}-n_q>M$. In every tuple of diameter $\le M$, the Pigeonhole Principle modulo r produces $m+1$ solutions $n-\beta_m<\cdots<n-\beta_1<n$ in the same residue class modulo r, so $r\le\beta_1<\cdots<\beta_m\le M$ and $r\mid\beta_i$. Covering $[N,2N)$ by $\le\log N+1$ short intervals $I_{N'}=[N',N'+\frac{N}{\log N})$ and summing over the $\le M^m$ choices of shifts $(\beta_1,\dots,\beta_m)$ gives

$$R\ \ll_{P,m}\ \frac{N}{M}\ +\ M^m(\log N)\max_{\substack{r\le\beta_1<\cdots<\beta_m\le M\\ r\mid\beta_i}}\ \max_{N\le N'<2N} t(\beta_1,\dots,\beta_m;N'),\tag{7.17}$$

where $t=t(\beta_1,\dots,\beta_m;N')$ is the number of integers $n\in I_{N'}$ satisfying $s(n-\beta_i)!=P(x_i)$ for $0\le i\le m$ ($\beta_0=0$).

To establish $R\ll_{P,s,m}\frac{N}{M}=N^{1-\theta}$, it suffices to prove that $t\le\frac{N}{M^{m+1}\log N}$. Suppose for contradiction that $t>\frac{N}{M^{m+1}\log N}$. By Lemma 7.3(i), there exists at least one $n_0\in I_{N'}$ among these t solutions such that $a:=\operatorname{ord}_{x=1/n_0}\Delta_S(x)\le\frac{\deg(\Delta_S)}{t}\ll_m\frac{DM^{m+1}\log N}{N}$. Set

$$U\ :=\ (2mr^4M)^{\frac{(m-\varepsilon_0)(m+1-\varepsilon_0)}{\varepsilon_0}},\qquad W\ :=\ \frac{N^{m+1-\varepsilon_0}}{(8mr^4M)^{\frac{m(m+1-\varepsilon_0)}{\varepsilon_0}}},\qquad C\ :=\ 2N,$$

and choose $D:=1+\lfloor\frac{\log(bcw((N+N/\log N)!)^{1/r})}{\log(W/C)}\rfloor\asymp_{P,s,m}N$ as in [12, Proposition 3.4], where $c=(2rN)^{M^{m+3}}$, $u=(4rN)^{M^{m+2}}10^{m\varepsilon_0^{-1}M}$, and $w=(2rN)^{2M^{m+2}}4^{m\varepsilon_0^{-1}M}$. Since $\theta<\frac{1}{m+2}$, we have $aM\ll_m M^{m+2}\log N=o(N)$, so $c,u,w=(N!)^{o(1)}$. By Lemma 7.3(ii), there exist $0\le k_1<k_2<\cdots<k_s\le a+s-1$ such that the $s\times s$ matrix $p_{i_a,j}:=P_{i_a}^{k_j}(1/n_0)$ $(1\le a,j\le s)$ is invertible, with common denominator $Z:=r^{a+s}n_0^{D+(a+s)M}\le cC^D$, $|p_{i_a,j}|\le uU^D$, and

$$\left|\sum_{a=1}^{s}p_{i_a,j}\,\omega_{i_a}(1/n_0)\right|\ =\ |R^{k_j}(1/n_0)|\ \le\ wW^{-D}\qquad(1\le j\le s).$$

Let $x,x_1,\dots,x_m\in\mathbb{N}$ satisfy $sn_0!=P(x)$ and $s(n_0-\beta_i)!=P(x_i)$, and set $p_0:=x>0$ and $p_i:=n_0^{\beta_i/r}x_i>0$ $(1\le i\le m)$. Since $P(x)$ is depressed ($\nu\ge 2$), $\rho:=\max_{1\le i\le m}|\omega_i(1/n_0)-\frac{p_i}{x}|\ll_P x^{-2+o(1)}=(N!)^{\frac{1}{r}(-2+o(1))}$. Because $p_i>0$ for every $0\le i\le m$, the s-dimensional subvector $(p_{i_1},\dots,p_{i_s})$ is non-zero, and since $\det(p_{i_a,j})\ne 0$, there exists $j\in\{1,\dots,s\}$ such that $\sum_{a=1}^s p_{i_a,j}p_{i_a}\ne 0$. Clearing the denominator $Z\le cC^D$ and applying the triangle inequality to $p_{i,j}p_i=-xp_{i,j}(\omega_i(1/n_0)-\frac{p_i}{x})+xp_{i,j}\omega_i(1/n_0)$ gives

$$\frac{1}{cC^D}\ \le\ \frac{1}{Z}\ \le\ \left|\sum_{a=1}^{s}p_{i_a,j}p_{i_a}\right|\ \le\ sxuU^D\rho\ +\ xwW^{-D}.$$

By our choice of D, $xwW^{-D}\le\frac{1}{2cC^D}$, so $\rho\gg_{P,s,m}x^{-1}(CU)^{-D}\gg(N!)^{\frac{1}{r}(-1-\chi-o(1))}$, where

$$\chi\ =\ \frac{\log(CU)}{\log(W/C)}\ =\ \frac{1+\theta\,\frac{(m-\varepsilon_0)(m+1-\varepsilon_0)}{\varepsilon_0}}{m-\varepsilon_0-\theta\,\frac{m(m+1-\varepsilon_0)}{\varepsilon_0}}+O\Big(\frac{1}{\log N}\Big).$$

This contradicts $\rho\ll(N!)^{\frac{1}{r}(-2+o(1))}$ whenever $\chi<1-\delta$, which rearranges to

$$\theta\ <\ \theta_m(\varepsilon_0)\ :=\ \frac{\varepsilon_0(m-1-\varepsilon_0)}{(m+1-\varepsilon_0)(2m-\varepsilon_0)}.\tag{7.18}$$

Evaluating at $m = 5$ and $\varepsilon_0 = 5 - \sqrt{5} \in (2, 3)$ gives $m - 1 - \varepsilon_0 = \sqrt{5} - 1$, $m + 1 - \varepsilon_0 = \sqrt{5} + 1$, and $2m - \varepsilon_0 = 5 + \sqrt{5} = \sqrt{5}(\sqrt{5} + 1)$, so

$$\begin{aligned}\frac{\varepsilon_0(4-\varepsilon_0)}{(6-\varepsilon_0)(10-\varepsilon_0)} &= \frac{(5-\sqrt{5})(\sqrt{5}-1)}{(\sqrt{5}+1)\cdot\sqrt{5}(\sqrt{5}+1)} = \left(\frac{\sqrt{5}-1}{\sqrt{5}+1}\right)^2 \\ &= \frac{7-3\sqrt{5}}{2} \approx 0.145898 > \frac{1}{7} \approx 0.142857,\end{aligned}$$

so $\theta_5(5 - \sqrt{5}) = \frac{7-3\sqrt{5}}{2} > \frac{1}{7}$. Thus every $\theta < \frac{1}{m+2} = \frac{1}{7}$ is admissible at $m = 5$ (see Table 1), and summing over dyadic intervals $[N/2^j, N/2^{j-1})$ establishes (1.10).

TABLE 1. Progression of unconditional power-saving exponents θ and upper bounds $O(N^{1-\theta})$ for $\mathcal{N}_{P,s}(N)$ and $\mathcal{N}_{\text{Brocard}}(N)$ under the Support-Reduced Padé–Wronskian method.

Shifts m	Wronskian Size	Optimal $\varepsilon_0 \in (0, m-1)$	$\theta_m(\varepsilon_0)$	Pigeonhole Cap $\frac{1}{m+2}$	Unconditional Counting Bound
2 [12]	3×3	$2 - \sqrt{2}$	$17 - 12\sqrt{2} \approx 0.02944$	1/4	$O(N^{33/34})$
3 (Support-Reduced)	$s \times s$ $(s \le 4)$	$3 - \sqrt{3}$	$7 - 4\sqrt{3} \approx 0.07180$	1/5	$O(N^{0.92821})$
4 (Support-Reduced)	$s \times s$ $(s \le 5)$	2	$1/9 \approx 0.11111$	1/6	$O_\varepsilon(N^{8/9+\varepsilon})$
5 (Support-Reduced)	$\mathbf{s \times s}$ $\mathbf{(4 \le s \le 6)}$	$\mathbf{5 - \sqrt{5}}$	$\mathbf{\frac{7-3\sqrt{5}}{2} \approx 0.14590 > \frac{1}{7}}$	**1/7**	$\mathbf{O_\varepsilon(N^{6/7+\varepsilon})}$

Step 2 (Decoupled Monomial-Product Padé–Wronskian Exponents for $\nu \ge 3$): Suppose $r \ge 3$ and $\nu \ge 3$. Fix $d \ge 1$ and $1 \le E \le \min(r-1, \nu-1)$ with $m := \binom{d+E}{d} - 1 \ge 2$. By Lemma 7.2, the $m + 1 = \binom{d+E}{d}$ monomial-product functions $\Omega_{\mathbf{e}}(x) = \prod_{i=1}^{d} \omega_i(x)^{e_i}$ $(|\mathbf{e}| \le E)$ are linearly independent over $\mathbb{C}(x)$. Because all $m + 1$ basis functions are generated from only d shifts $\beta_1 < \cdots < \beta_d \le M$, the Pigeonhole Principle over $[1, M]^d$ replaces M^m in (7.17) by M^d, so $t > \frac{N}{M^{d+1}\log N}$ yields root multiplicity $a \ll \frac{DM^{d+1}\log N}{N}$ and $(a + s)M = o(D)$ holds for all $\theta < \frac{1}{d+2}$. At $x = 1/n_0$, Lemma 7.4(ii) gives $\Omega_{\mathbf{e}}(1/n_0) = \frac{q_{\mathbf{e}}}{x_0^E} + O(x_0^{-\nu+o(1)})$ with common denominator x_0^E. Clearing x_0^E in the non-zero support Wronskian row $\sum_{a=1}^{s} p_{i_a,j} q_{\mathbf{e}_a} \ne 0$ and choosing D so that $x_0^E w W^{-D} \le \frac{1}{2cC^D}$ yields $\rho := \max_{|\mathbf{e}| \le E} |\Omega_{\mathbf{e}}(1/n_0) - q_{\mathbf{e}}/x_0^E| \gg x_0^{-E}(CU)^{-D} \gg x_0^{-E(1+\chi+o(1))}$, which contradicts $\rho \ll x_0^{-\nu+o(1)}$ whenever $E(1 + \chi) < \nu \iff \chi < \frac{\nu-E}{E}$, i.e.,

$$\begin{aligned}\theta < \theta_{d,E,\nu}(\varepsilon_0) &:= \frac{\varepsilon_0((\nu-E)(m-\varepsilon_0)-E)}{(m+1-\varepsilon_0)(\nu m - E\varepsilon_0)}, \\ \theta^*_{d,E,\nu} &:= \min\left(\max_{\varepsilon_0 \in (0,m-1)} \theta_{d,E,\nu}(\varepsilon_0),\ \frac{1}{d+2}\right).\end{aligned} \tag{7.19}$$

Evaluating (7.19) for $\nu \ge 3$ (Table 2) completes the proof of Theorem 1.3(ii). □

TABLE 2. Unconditional counting bounds $\mathcal{N}_{P,s}(N) \ll_{P,s,\varepsilon} N^{1-\theta^*_{d,E,\nu}+\varepsilon}$ obtained via the Decoupled Monomial-Product Padé–Wronskian method for polynomial-factorial equations $s \cdot n! = P(x)$ with degree gap $\nu \ge 2$.

Degree Gap ν	Shifts d	Product Degree E	Basis Functions $m+1 = \binom{d+E}{d}$	Optimal ε_0	$\theta^*_{d,E,\nu}$	Unconditional Counting Bound
$\nu = 2$ (e.g. $n! + 1 = m^2$)	5	1	6 $(m = 5)$	$5 - \sqrt{5} \approx 2.7639$	$1/7 \approx 0.14286$	$O_\varepsilon(N^{6/7+\varepsilon})$
$\nu = 3$ (e.g. $s \cdot n! + A = x^3$)	4	1	5 $(m = 4)$	$\frac{40-2\sqrt{85}}{9} \approx 2.3957$	$1/6 \approx 0.16667$	$O_\varepsilon(N^{5/6+\varepsilon})$
$\nu = 4$ (e.g. $s \cdot n! + A = x^4$)	3	1	4 $(m = 3)$	$\frac{18-2\sqrt{21}}{5} \approx 1.7670$	$1/5 = 0.20000$	$O_\varepsilon(N^{4/5+\varepsilon})$
$\nu = 5$ (e.g. $s \cdot n! + A = x^5$)	2	2	6 $(m = 5)$	$\frac{90-7\sqrt{30}}{17} \approx 3.0388$	0.21062	$O(N^{0.78939})$
$\nu \ge 6$ (with $d = 2, E = 2$)	2	2	6 $(m = 5)$	$\frac{60-3\sqrt{70}}{11} \approx 3.1728$	$1/4 = 0.25000$	$O_\varepsilon(N^{3/4+\varepsilon})$
$\nu \to \infty$ (single shift $d = 1$)	1	$E \ge 6$	$E + 1$ $(m = E)$	4	$1/3 \approx 0.33333$	$O_\varepsilon(N^{2/3+\varepsilon})$

7.4. **Unconditional Finiteness for $n!+A=m^k$ Outside the Quadratic Square-Shift Family.** The support-reduced Wronskian and monomial-product Padé theorems above yield unconditional power-saving upper bounds $\mathcal{N}_{P,s}(N) \ll_{P,s,\varepsilon} N^{1-\theta^*_{d,E,\nu}+\varepsilon}$ for every polynomial $P(x) \in \mathbb{Z}[x]$ of degree $r \geq 2$, including the critical split quadratic family $P(x) = x^2 - c^2 = (x-c)(x+c)$ to which Brocard's equation $n!+1 = m^2$ ($c=1$) belongs. To delineate the exact arithmetic boundary between polynomial-factorial equations that admit unconditional finiteness proofs via algebraic number theory and those that require Diophantine approximation bounds, we conclude this section by proving that for the shifted power family $n! + A = m^k$ ($A \in \mathbb{Z} \setminus \{0\}, k \geq 2$), the square-shift quadratic family ($k=2, A=c^2$) is the *unique* locally unobstructed family: whenever $k \geq 3$ or $A \neq c^2$, a combination of the Chebotarev density theorem and balanced p-adic valuation cascades forces unconditional finiteness.

Theorem 7.5 (Unconditional Finiteness for $n! + A = m^k$ Outside $n! + c^2 = m^2$)**.** *For any fixed non-zero integer $A \neq 0$ and exponent $k \geq 2$, the generalized Brocard–Ramanujan equation $n!+A = m^k$ has at most finitely many positive integer solutions (n,m) whenever $k \geq 3$ (for all $A \neq 0$, including $A = c^k$) or whenever $k = 2$ and A is not a positive integer square c^2.*

Proof. If A is not of the form c^k for $c \in \mathbb{Z}$ (and if $8 \mid k$, not of the form $2^{k/2}c^k$), then by the Grunwald–Wang theorem [1, Chapter 10], the polynomial $f(x) = x^k - A \in \mathbb{Z}[x]$ has an irreducible factor over $\mathbb{Q}$ of degree $d \geq 2$. In $G = \mathrm{Gal}(\mathbb{Q}(\zeta_k, A^{1/k})/\mathbb{Q})$, Jordan's theorem on derangements in transitive permutation groups implies that the subset $C \subset G$ fixing no root of $f(x)$ satisfies $|C|/|G| \geq \frac{1}{d} \geq \frac{1}{k}$. By the Chebotarev density theorem [59], the primes $p \nmid kA$ for which $x^k \equiv A \pmod{p}$ has no solution have positive natural density $\geq \frac{1}{k}$. Letting $p_0(A,k)$ be the least such prime, any solution (n,m) must satisfy $n < p_0(A,k)$ (since $n \geq p_0 \implies p_0 \mid n! \implies m^k \equiv A \pmod{p_0}$).

If $A = c^k$ ($c \neq 0$) and $k \geq 3$ has an odd prime divisor $p_0 \mid k$, setting $M = m^{k/p_0}$ and $c_0 = c^{k/p_0}$ gives $n! = M^{p_0} - c_0^{p_0} = (M - c_0)\Phi_{p_0}(M, c_0)$, where $\Phi_{p_0}(M,c_0) = \frac{M^{p_0}-c_0^{p_0}}{M-c_0}$ satisfies $\gcd(M - c_0, \Phi_{p_0}(M,c_0)) \mid p_0 c_0^{p_0-1}$. Every prime $q \leq n$ with $q \nmid p_0 c_0$ dividing $\Phi_{p_0}(M,c_0)$ satisfies $q \equiv 1 \pmod{p_0}$, so every prime $q \leq n$ with $q \nmid p_0 c_0$ and $q \not\equiv 1 \pmod{p_0}$ divides $M - c_0$ to its full factorial power $q^{v_q(n!)}$. By the Prime Number Theorem for arithmetic progressions, these primes contribute $\log|M - c_0| \geq (\frac{p_0-2}{p_0-1} - o(1))n\log n \geq (\frac{1}{2} - o(1))n\log n$, contradicting $\log|M - c_0| \leq (\frac{1}{p_0} + o(1))n\log n \leq (\frac{1}{3} + o(1))n\log n$. If $k \geq 3$ is a power of 2, then $4 \mid k$, reducing to $n! = M^4 - c_0^4 = (M-c_0)(M+c_0)(M^2+c_0^2)$; since $M^4 - c_0^4$ has the irreducible quadratic factor $M^2 + c_0^2$ over $\mathbb{Q}$, the polynomial-factorial finiteness theorem of Berend and Harmse [5, Theorem 1.1] (and Pollack and Shapiro [44] for $c_0 = 1$) implies that there are only finitely many solutions. Finally, for $k = 2$ and $A \equiv 2, 3 \pmod 4$, reducing $n! + A = m^2 \pmod 4$ rules out all $n \geq 4$. □

8. Harmonic Analysis on the CRT Torus and p-Adic Binomial Lifting

In this section we develop the Chinese Remainder Theorem (CRT) torus formulation of (1.1) on $\mathbb{T} = \mathbb{R}/\mathbb{Z}$, establishing the CRT Partition of Unity, the Universal Torus Projection Law, the Zero-Dimensional Locked-Torus Collapse Theorem, the exact resonance-free L^{2k} Parseval identities for the Riesz cosine product, and simultaneous p-adic binomial series lifting.

8.1. **CRT Basis Phases, Partition of Unity, Universal Projection, and Locked-Torus Collapse.** For each integer $n \geq 4$, let $M_n := \frac{n!}{2}$, and set $w_2(n) := v_2(n!) - 1 \geq 2$ and $w_p(n) := v_p(n!) \geq 1$ for odd primes $3 \leq p \leq n$, so that $M_n = \prod_{p \leq n} q_p(n)$ is the product of the $\pi(n)$ pairwise coprime prime powers $q_p(n) := p^{w_p(n)}$. For each prime $p \leq n$, let $a_p(n) \in \{1, \dots, q_p(n) - 1\}$ be the modular inverse of $M_n/q_p(n)$ mod $q_p(n)$, and define the CRT torus basis phase

$$\theta_p(n) \;:=\; \frac{a_p(n)}{q_p(n)} \;=\; \frac{a_p(n)}{p^{w_p(n)}} \;\in\; \mathbb{T} \;=\; \mathbb{R}/\mathbb{Z}. \tag{8.1}$$

For any $x \in \mathbb{T}$, let $\|x\|_{\mathbb{T}} := \min_{z \in \mathbb{Z}} |x - z| \in [0, 1/2]$.

Theorem 8.1 (CRT Partition of Unity, Punctured Torus Discrepancy, Universal Projection, and Zero-Dimensional Locked-Torus Collapse)**.** *Let $n \geq 4$ be an integer, and for each sign vector $\boldsymbol{\epsilon} = (\epsilon_p)_{p \leq n} \in \{-1, 1\}^{\pi(n)}$, define $\Sigma_n(\boldsymbol{\epsilon}) := \sum_{p \leq n} \epsilon_p \theta_p(n) \in \mathbb{T}$. Then:*

(i) ***CRT Partition of Unity and Punctured Discrepancy Equivalence:*** *The map $\boldsymbol{\epsilon} \mapsto M_n \Sigma_n(\boldsymbol{\epsilon}) \bmod M_n$ is a bijection from $\{-1, 1\}^{\pi(n)}$ onto $\{m \in \mathbb{Z}/M_n\mathbb{Z} : m^2 \equiv 1 \pmod{n!}\}$. Evaluating at $\boldsymbol{\epsilon} = \pm\mathbf{1}$ gives the **CRT Partition of Unity***

$$\sum_{p \leq n} \theta_p(n) \;\equiv\; \frac{1}{M_n} \;=\; \frac{2}{n!} \pmod 1, \qquad \big\|\Sigma_n(\pm\mathbf{1})\big\|_{\mathbb{T}} \;=\; \frac{2}{n!}, \tag{8.2}$$

while every non-trivial sign vector $\boldsymbol{\epsilon} \in \{-1, 1\}^{\pi(n)} \setminus \{\pm\mathbf{1}\}$ satisfies $\|\Sigma_n(\boldsymbol{\epsilon})\|_{\mathbb{T}} = \frac{2\sqrt{k_{\boldsymbol{\epsilon}} n! + 1}}{n!}$ for a unique integer $k_{\boldsymbol{\epsilon}} \geq 1$. Consequently, the punctured torus discrepancy $\mathrm{disc}^{\circ}_{\mathbb{T}}(n) := \min_{\boldsymbol{\epsilon} \neq \pm\mathbf{1}} \|\Sigma_n(\boldsymbol{\epsilon})\|_{\mathbb{T}}$ satisfies $\mathrm{disc}^{\circ}_{\mathbb{T}}(n) \geq \delta_n := \frac{2\sqrt{n!+1}}{n!}$, with equality if and only if $n! + 1 = m^2$ is soluble.

(ii) ***Universal Torus Projection Law Across Factorial Levels:*** *For every integer $n \geq 4$ and every prime $p \leq n + 1$, the CRT basis phases satisfy the exact scaling relation in $\mathbb{Q}/\mathbb{Z}$:*

$$(n+1)\,\theta_p(n+1) \;=\; \theta_p(n) \quad \text{in } \mathbb{Q}/\mathbb{Z} \qquad (\text{with } \theta_p(n) := 0 \text{ if } p = n+1). \tag{8.3}$$

Iterating from n_1 to n_2 ($4 \leq n_1 < n_2$) with $P(n_1, n_2) = n_2!/n_1!$ gives $P(n_1, n_2)\theta_p(n_2) = \theta_p(n_1)$ in $\mathbb{Q}/\mathbb{Z}$ for all $p \leq n_2$, projecting the level-n_2 torus sum onto level n_1: $\sum_{p \leq n_1} \epsilon_p(n_2)\theta_p(n_1) \equiv \frac{m_{n_2}}{M_{n_1}} \pmod 1$.

(iii) ***Zero-Dimensional Locked-Torus Collapse:*** *Factor $M_n = M_n^{\mathrm{locked}} M_n^{+}$, where*

$$M_n^{\mathrm{locked}} \;:=\; 2^{v_2(n!)-1} \prod_{p \in \mathcal{P}_{\mathrm{odd}}(n) \setminus \{2\}} p^{v_p(n!)} \prod_{q \in \mathcal{P}^{-}_{\mathrm{even}}(n)} q^{v_q(n!)}, \qquad M_n^{+} \;:=\; \prod_{q \in \mathcal{P}^{+}_{\mathrm{even}}(n)} q^{v_q(n!)}. \tag{8.4}$$

*Then $\log\big(\frac{M_n^{\mathrm{locked}}}{\sqrt{n!}}\big) = \frac{1}{2}\big(1 + \Sigma^{*}_{\mathrm{parity}}(n)\big)n + S^{-}_{\mathrm{even}}(n) + O\big(\frac{n}{\log n}\big)$. Whenever $M_n^{\mathrm{locked}} > \sqrt{n! + 1}$, each Rédei kernel vector $(\mathbf{x}, \epsilon_2)$ uniquely determines a single integer $r_{\mathrm{locked}}(\mathbf{x}, \epsilon_2) \in \{0, 1, \dots, M_n^{\mathrm{locked}} - 1\}$ via the locked congruences on $\{2\} \cup \mathcal{P}_{\mathrm{odd}}(n) \cup \mathcal{P}^{-}_{\mathrm{even}}(n)$, collapsing the number of free coordinates on $\mathcal{P}^{+}_{\mathrm{even}}(n)$ to **zero** and requiring the exact integer equality $m = r_{\mathrm{locked}}(\mathbf{x}, \epsilon_2) \leq \sqrt{n! + 1}$ together with $r_{\mathrm{locked}}(\mathbf{x}, \epsilon_2)^2 \equiv 1 \pmod{M_n^{+}}$.*

Proof. (i) Multiplying $\Sigma_n(\boldsymbol{\epsilon})$ by M_n gives $x_{\boldsymbol{\epsilon}} := \sum_{p \leq n} \epsilon_p a_p(n)(M_n/q_p(n)) \equiv \epsilon_p \pmod{q_p(n)}$ for all $p \leq n$. By the Chinese Remainder Theorem, $\boldsymbol{\epsilon} \mapsto x_{\boldsymbol{\epsilon}} \bmod M_n$ is a bijection onto $\{m \in \mathbb{Z}/M_n\mathbb{Z} : m^2 \equiv 1 \pmod{2M_n}\}$. For $\boldsymbol{\epsilon} = +\mathbf{1}$, $x_{+\mathbf{1}} \equiv 1 \pmod{q_p(n)}$ for all $p \leq n$, so $x_{+\mathbf{1}} \equiv 1 \pmod{M_n}$, proving (8.2). For $\boldsymbol{\epsilon} \notin \{\pm\mathbf{1}\}$, $m_{\boldsymbol{\epsilon}} := M_n \|\Sigma_n(\boldsymbol{\epsilon})\|_{\mathbb{T}} \in \{2, \dots, \lfloor M_n/2 \rfloor\}$ satisfies $m_{\boldsymbol{\epsilon}}^2 \equiv 1 \pmod{n!}$ and $m_{\boldsymbol{\epsilon}}^2 - 1 > 0$, so $m_{\boldsymbol{\epsilon}}^2 - 1 = k_{\boldsymbol{\epsilon}} n!$ for a unique integer $k_{\boldsymbol{\epsilon}} \geq 1$.

(ii) For any prime $p \le n+1$, $M_{n+1} = (n+1)M_n$, so $w_p(n+1) = w_p(n) + v_p(n+1)$. Writing $n+1 = p^{v_p(n+1)}u_p$ with $p \nmid u_p$, we have $\frac{M_{n+1}}{q_p(n+1)} = u_p \frac{M_n}{q_p(n)}$. Reducing $u_p \frac{M_n}{q_p(n)} a_p(n+1) \equiv 1 \pmod{q_p(n+1)}$ modulo $q_p(n) \mid q_p(n+1)$ and multiplying by $a_p(n)$ gives $u_p a_p(n+1) \equiv a_p(n) \pmod{q_p(n)}$. Multiplying by $p^{v_p(n+1)}$ and dividing by $q_p(n+1) = p^{v_p(n+1)} q_p(n)$ yields (8.3).

(iii) Since $\log(n!) = v_2(n!)\log 2 + S_{\mathrm{odd}}(n) + S^{+}_{\mathrm{even}}(n) + S^{-}_{\mathrm{even}}(n)$ and $\log M_n^{\mathrm{locked}} = (v_2(n!)-1)\log 2 + S_{\mathrm{odd}}(n) + S^{-}_{\mathrm{even}}(n)$, subtracting $\frac{1}{2}\log(n!)$ and applying Theorem 2.5(i) gives

$$\begin{aligned} \log\left(\frac{M_n^{\mathrm{locked}}}{\sqrt{n!}}\right) &= \frac{1}{2}v_2(n!)\log 2 - \frac{1}{2}\Delta_{\mathrm{parity}}(n) + S^{-}_{\mathrm{even}}(n) - \log 2 \\ &= \frac{1}{2}(1 + \Sigma^{*}_{\mathrm{parity}}(n))n + S^{-}_{\mathrm{even}}(n) + O\left(\frac{n}{\log n}\right), \end{aligned}$$

since $\log 2 - (\log 2 - 1) = 1$. By Theorem 5.5(ii), on each Rédei kernel point $(\mathbf{x}, \epsilon_2)$ all signs ϵ_p on $\{2\} \cup \mathcal{P}_{\mathrm{odd}}(n) \cup \mathcal{P}^{-}_{\mathrm{even}}(n)$ are deterministically locked, uniquely fixing $m \equiv r_{\mathrm{locked}}(\mathbf{x}, \epsilon_2) \pmod{M_n^{\mathrm{locked}}}$. When $M_n^{\mathrm{locked}} > \sqrt{n!+1} = m > 1$, this congruence forces the exact integer equality $m = r_{\mathrm{locked}}(\mathbf{x}, \epsilon_2)$ in $\mathbb{Z}$. □

8.2. **Exact Resonance-Free L^{2k} Parseval Identities for the Riesz Product and p-Adic Binomial Lifting.** Let $\mu_n := 2^{-\pi(n)} \sum_{\boldsymbol{\epsilon} \in \{-1,1\}^{\pi(n)}} \delta_{\Sigma_n(\boldsymbol{\epsilon})}$ be the uniform signed-sum measure on $\mathbb{T}$, whose Fourier transform is the Riesz cosine product

$$\widehat{\mu}_n(h) = \prod_{p \le n} \cos(2\pi h\, \theta_p(n)) \qquad (h \in \mathbb{Z}). \tag{8.5}$$

Theorem 8.2 (Exact Resonance-Free L^{2k} Parseval Identities and Simultaneous p-Adic Binomial Series Lifting)**.** *Let $n \ge 5$ be an integer, and define $P_{\min}(n) := \min_{p \le n} p^{w_p(n)}$.*

(i) ***Exact Resonance-Free L^{2k} Moments of the Riesz Product:*** *We have $P_{\min}(n) > \frac{n}{2}$. For every integer $k \ge 1$,*

$$\frac{1}{M_n} \sum_{h=0}^{M_n - 1} \left|\widehat{\mu}_n(h)\right|^{2k} = \prod_{p \le n} \left(2^{-2k} \sum_{\substack{0 \le j \le 2k \\ 2(j-k) \equiv 0 \pmod{p^{w_p(n)}}}} \binom{2k}{j} \right), \tag{8.6}$$

*which for every integer k in the **resonance-free range** $1 \le k < \frac{1}{2}P_{\min}(n)$ (including all $1 \le k \le \lfloor n/4 \rfloor$) simplifies identically to the independent-phase central binomial moment*

$$\frac{1}{M_n} \sum_{h=0}^{M_n - 1} \left|\widehat{\mu}_n(h)\right|^{2k} = \left(2^{-2k} \binom{2k}{k} \right)^{\pi(n)} \qquad \forall\, 1 \le k < \frac{1}{2}P_{\min}(n). \tag{8.7}$$

(ii) ***Simultaneous p-Adic Binomial Series Lift on the CRT Torus:*** *For every prime $p \le n$, the p-adic binomial series $\mathcal{B}(n!) := \sum_{j=0}^{\infty} \binom{1/2}{j} (n!)^j$ converges in $\mathbb{Z}_p$ to the unique 1-unit square root of $1+n!$, so any solution (n, m) to $n!+1 = m^2$ satisfies $m = \epsilon_p(n)\, \mathcal{B}(n!)$ in $\mathbb{Z}_p$ for all $p \le n$ simultaneously. Consequently, for every integer $R \ge 1$, if $X_R(\boldsymbol{\epsilon}(n)) \in (-M_{n,R}/2, M_{n,R}/2]$ denotes the centered CRT lift of $\boldsymbol{\epsilon}(n)$ modulo $M_{n,R} := (n!/4)^R$, then*

$$m \equiv X_R(\boldsymbol{\epsilon}(n)) \cdot \sum_{j=0}^{R-1} \binom{1/2}{j} (n!)^j \pmod{\left(\frac{n!}{4}\right)^R}, \tag{8.8}$$

and at $R = 2$, $X_2(\boldsymbol{\epsilon}(n)) = m(1 - \frac{1}{2}n!)$ holds as an exact integer identity in $\mathbb{Z}$ for all $n \ge 6$.

Proof. (i) For $p = 2$, $2^{v_2(n!)-1} \ge 2^{n - \lfloor \log_2 n \rfloor - 2} > n/2$ for $n \ge 5$; for odd $p \le n$ with $j = v_p(n!) \ge 1$, if $j = 1$ then $p > n/2$, while if $j \ge 2$ then $p^j > \max((n/(j+1))^j, 3^{\sqrt{n}}) > n/2$, so $P_{\min}(n) > n/2$. Since $M_n = \prod_{p \le n} q_p(n)$ with $q_p(n) = p^{w_p(n)}$ pairwise coprime and $\gcd(a_p(n), q_p(n)) = 1$, the Chinese Remainder Theorem implies that as h runs over $\mathbb{Z}/M_n\mathbb{Z}$, the residues $r_p := h a_p(n) \bmod q_p(n)$ run independently over $\prod_{p \le n} (\mathbb{Z}/q_p(n)\mathbb{Z})$. Expanding $\cos^{2k}(2\pi r_p/q_p(n)) = 2^{-2k} \sum_{j=0}^{2k} \binom{2k}{j} e^{2\pi i (2j-2k) r_p / q_p(n)}$

and averaging over $r_p \in \mathbb{Z}/q_p(n)\mathbb{Z}$ yields (8.6). When $1 \le k < \frac{1}{2}P_{\min}(n)$, every $0 \le j \le 2k$ satisfies $|2(j-k)| \le 2k < P_{\min}(n) \le q_p(n)$, so $2(j-k) \equiv 0 \pmod{q_p(n)} \iff j = k$, giving (8.7).

(ii) For every $j \ge 1$, Legendre's formula gives $v_2(\binom{1/2}{j}) \ge -(2j-1)$ and $v_p(\binom{1/2}{j}) \ge 0$ for odd p. Since $v_2(n!) \ge 3$ for $n \ge 4$ and $v_p(n!) \ge 1$ for odd $p \le n$, we have $v_p(\binom{1/2}{j}(n!)^j) \ge j\, v_p(n!/4) \to \infty$ as $j \to \infty$ for every $p \le n$. Thus $\mathcal{B}(n!)$ converges in $\mathbb{Z}_p$ to the unique square root of $1+n!$ congruent to $+1 \pmod{p^{w_p(n)}}$. Since $(\epsilon_p(n)m)^2 = 1+n!$ and $\epsilon_p(n)m \equiv +1 \pmod{p^{w_p(n)}}$, Hensel's Lemma gives $m = \epsilon_p(n)\mathcal{B}(n!)$ in $\mathbb{Z}_p$. Reducing modulo $p^{Rv_p(n!/4)}$ and combining over $p \le n$ by the Chinese Remainder Theorem gives (8.8). At $R = 2$, $(1+\frac{1}{2}n!)^{-1} \equiv 1 - \frac{1}{2}n! \pmod{(n!/4)^2}$, and for $n \ge 6$, $|m(1-\frac{1}{2}n!)| < \frac{1}{2}(n!/4)^2$, forcing $X_2(\boldsymbol{\epsilon}(n)) = m(1-\frac{1}{2}n!)$ in $\mathbb{Z}$. □

8.3. **Subgaussian Concentration, Winding-Branch Decomposition, and the Harmonic Uncertainty Barrier.** Let $\tilde{\theta}_p(n) \in (-1/2, 1/2]$ denote the centered real lift of $\theta_p(n) \in \mathbb{T}$, and let $S_n(\boldsymbol{\epsilon}) := \sum_{p \le n} \epsilon_p \tilde{\theta}_p(n) \in [-\frac{1}{2}\pi(n), \frac{1}{2}\pi(n)]$ be the real signed sum, with variance $\sigma_n^2 := \sum_{p \le n} \tilde{\theta}_p(n)^2$ and fourth moment $M_{4,n} := \sum_{p \le n} \tilde{\theta}_p(n)^4 \ge \sigma_n^4/\pi(n)$.

Theorem 8.3 (Fourth-Order Subgaussian Tail, Winding-Branch Decomposition, and the Harmonic Uncertainty Barrier)**.** *Let $n \ge 5$ be an integer, and let $\delta_n = \frac{2\sqrt{n!}+1}{n!}$.*

(i) ***Fourth-Order Log-Cosh and Log-Cosine Subgaussian Bounds:*** *The function $\kappa_4(u) := u^{-4}(\frac{1}{2}u^2 - \log\cosh u)$ is strictly positive and strictly decreasing on $(0,\infty)$, with $\kappa_4(0^+) = \frac{1}{12}$. For every $\lambda \in \mathbb{R}$ and $|\xi| < \pi$, under uniform $\boldsymbol{\epsilon} \in \{-1,1\}^{\pi(n)}$,*

$$\begin{aligned} \mathbb{E}_{\boldsymbol{\epsilon}}[e^{\lambda S_n(\boldsymbol{\epsilon})}] &= \prod_{p \le n} \cosh(\lambda\tilde{\theta}_p(n)) \le \exp\Big(\frac{1}{2}\sigma_n^2\lambda^2 - \kappa_4(\lambda/2)\, M_{4,n}\lambda^4\Big), \\ |\mathbb{E}_{\boldsymbol{\epsilon}}[e^{i\xi S_n(\boldsymbol{\epsilon})}]| &= \prod_{p \le n} |\cos(\xi\tilde{\theta}_p(n))| \le \exp\Big(-\frac{1}{2}\sigma_n^2\xi^2 - \frac{1}{12}M_{4,n}\xi^4\Big). \end{aligned} \tag{8.9}$$

Consequently, in the integer winding-number decomposition of $\mathcal{N}_{\mathrm{arc}}(n;\delta_n)$ across $z \in [-L_n, L_n]$ ($L_n := \lfloor \sum_{p \le n} |\tilde{\theta}_p(n)| \rfloor$), the outer winding branches $|z| \ge Z_0 + \delta_n$ satisfy

$$\sum_{|z| \ge Z_0 + \delta_n} \mathbb{P}_{\boldsymbol{\epsilon}}(|S_n(\boldsymbol{\epsilon}) - z| \le \delta_n) \le 2\exp\Big(-\frac{Z_0^2}{2\sigma_n^2} - \kappa_4\Big(\frac{Z_0}{2\sigma_n^2}\Big)\frac{M_{4,n}Z_0^4}{\sigma_n^8}\Big). \tag{8.10}$$

(ii) ***Beurling–Selberg–Hölder L^{2k} Arc Bound and the Trivial-Solution Barrier:*** *For every $H \ge 1$ and $1 \le k < \frac{1}{2}P_{\min}(n)$, Vaaler's Beurling–Selberg majorant of $\mathbf{1}_{[-\delta_n,\delta_n]}$ and Hölder's inequality yield*

$$\mathcal{N}_{\mathrm{arc}}(n;\delta_n) \le 2^{\pi(n)}\Big(2\delta_n + \frac{1}{H+1}\Big) + 2\Big(2\delta_n + \frac{1}{H+1}\Big) H^{1-\frac{1}{2k}} M_n^{\frac{1}{2k}} \left(\binom{2k}{k}^{\frac{1}{2k}}\right)^{\pi(n)}. \tag{8.11}$$

Proving $\mathcal{N}_{\mathrm{arc}}(n;\delta_n) < 4$ via upper-bounding $|\widehat{\mu}_n(h)|$ requires $H \ge 2^{\pi(n)-1}$ and $\frac{1}{H}\sum_{h=1}^{H} |\widehat{\mu}_n(h)| < 2^{-\pi(n)}$, whereas the root-mean-square of $|\widehat{\mu}_n(h)|$ is $2^{-\pi(n)/2}$ by (8.7) because the two trivial vectors $\boldsymbol{\epsilon} = \pm\mathbf{1}$ always lie inside $[-\delta_n, \delta_n]$ and contribute $2^{1-\pi(n)}\cos(4\pi h/n!)$ to $\widehat{\mu}_n(h)$.

Proof. (i) For $u > 0$, $f(u) := \frac{1}{2}u^2 - \log\cosh u$ and $g(u) := u^4$ satisfy $f(0) = g(0) = 0$ and $\frac{f'(u)}{g'(u)} = \frac{u - \tanh u}{4u^3}$, whose derivative ratio is $\frac{1-\operatorname{sech}^2 u}{12u^2} = \frac{1}{12}\big(\frac{\tanh u}{u}\big)^2$, which is strictly decreasing on $(0,\infty)$. By the monotone l'Hôpital rule, $\kappa_4(u)$ is strictly decreasing on $(0,\infty)$. Since $|\tilde{\theta}_p(n)| \le 1/2$, we have $|\lambda\tilde{\theta}_p(n)| \le |\lambda|/2$, so $\log\cosh(\lambda\tilde{\theta}_p(n)) \le \frac{1}{2}\lambda^2\tilde{\theta}_p(n)^2 - \kappa_4(\lambda/2)\lambda^4\tilde{\theta}_p(n)^4$. Summing over $p \le n$ and combining with $\log|\cos u| \le -\frac{1}{2}u^2 - \frac{1}{12}u^4$ ($|u| < \pi/2$) yields (8.9), and Markov's inequality at $\lambda = Z_0/\sigma_n^2$ gives (8.10).

(ii) By Vaaler [60, Theorem 19], there exists a degree-H trigonometric polynomial $I_H^+(x) = \sum_{|h| \le H} \hat{I}_H^+(h)e(hx) \ge \mathbf{1}_{[-\delta_n,\delta_n]}(x)$ with $0 \le |\hat{I}_H^+(h)| \le \hat{I}_H^+(0) = 2\delta_n + \frac{1}{H+1}$. Integrating against $2^{\pi(n)}\mu_n$

and applying Hölder's inequality $\sum_{h=1}^{H}|\widehat{\mu}_n(h)| \le H^{1-\frac{1}{2k}}\big(\sum_{h=0}^{M_n-1}|\widehat{\mu}_n(h)|^{2k}\big)^{\frac{1}{2k}}$ together with (8.7) cancels $2^{\pi(n)}$ against $2^{-\pi(n)}$ in the L^{2k} norm and yields (8.11). □

Theorem 8.4 (External Wilson Quadratic-Residue Sieve at Primes $q > n$ and Macroscopic Sieve Independence)**.** *Let (n, m) be a positive integer solution to $n! + 1 = m^2$.*

(i) ***External Wilson Sieve at Primes*** *$q > n$: For every prime $q = n + d + 1 > n$ ($d \ge 0$), the integer $W_d := d!\,(d! - (-1)^d)$ is a quadratic residue (or zero) modulo q:*

$$\left(\frac{d!\,(d! - (-1)^d)}{q}\right) \in \{0, +1\} \qquad \forall \text{ primes } q = n + d + 1 > n. \tag{8.12}$$

In particular: if $q = n + 1$ is prime ($d = 0$), then $(n + 1)^2 \mid (n! + 1)$; if $q = n + 2$ is prime ($d = 1$), then $\left(\frac{2}{q}\right) = +1 \implies n \equiv 5, 7 \pmod 8$; if $q = n + 3$ is prime ($d = 2$), then $\left(\frac{2}{q}\right) = +1 \implies n \equiv 4, 6 \pmod 8$; and if $q = n + 4$ is prime ($d = 3$), then $\left(\frac{42}{q}\right) \in \{0, +1\}$.

(ii) ***Macroscopic Independence of the Archimedean Taylor-Remainder and Primorial Sieves:*** *For fixed $A \ne 0$, $k \ge 2$, let $\eta_k := \max(2^{1-k}, \frac{1}{2k(k-1)})$ and $\eta_k' := \frac{\eta_k}{1+\eta_k} = \max(\frac{1}{2^{k-1}+1}, \frac{1}{2k(k-1)+1})$, and fix $\delta \in (0, \eta_k'/2)$. Let $Q(n) = \prod_{p \le n} p$. Every solution to $n! + A = m^k$ satisfies $m^k \equiv A \pmod{Q(n)}$ and $\{m^k e - Ae\} = R_n^{(e)} := \sum_{j=n+1}^{\infty} \frac{n!}{j!} \in (\frac{1}{n+1}, \frac{1}{n})$. There exists a constant $C_k > 0$ such that for all $X \ge Q(n)^{C_k/\delta}$, the joint counting function $\mathcal{S}_{A,k}(X; n) := \#\{1 \le m \le X : m^k \equiv A \pmod{Q(n)} \text{ and } \{m^k e - Ae\} \in (\frac{1}{n+1}, \frac{1}{n})\}$ satisfies*

$$\mathcal{S}_{A,k}(X; n) \;=\; X \cdot \rho_{A,k}(n) \cdot \frac{1}{n(n+1)}\Big(1 + O_{k,\delta}\big(n^2 X^{-\eta_k'+\delta}\big)\Big), \qquad \eta_k' \;=\; \frac{\eta_k}{1+\eta_k}, \tag{8.13}$$

where $\rho_{A,k}(n) = Q(n)^{-1}\#\{r \bmod Q(n) : r^k \equiv A \pmod{Q(n)}\}$.

Proof. (i) For any prime $q = n + d + 1 > n$, Wilson's theorem gives $-1 \equiv (q-1)! = n!\prod_{j=1}^{d}(q-j) \equiv (-1)^d d!\,n! \pmod q$, so $d!\,n! \equiv -(-1)^d \pmod q$. Multiplying $m^2 = n! + 1$ by $(d!)^2$ gives $(d!\,m)^2 = d!(d!\,n!) + (d!)^2 \equiv d!(d! - (-1)^d) = W_d \pmod q$, proving (8.12).

(ii) Multiplying $m^k - A = n!$ by $e = \sum_{j=0}^{\infty}\frac{1}{j!}$ gives $m^k e - Ae = M_n^{(e)} + R_n^{(e)}$ with $M_n^{(e)} = \sum_{j=0}^{n}\frac{n!}{j!} \in \mathbb{Z}$ and $\frac{1}{n+1} < R_n^{(e)} < \frac{1}{n}$. Parameterizing $m = Q(n)y + r$ ($0 \le y < Y_n := \lfloor X/Q(n)\rfloor$) for each valid residue $r \bmod Q(n)$ yields the polynomial $f_r(y) = (Q(n)^k e)y^k + \cdots + (r^k - A)e$. Since the irrationality measure of e is $\mu(e) = 2$ [19], $\|hk!\,Q(n)^k e\| \gg_\varepsilon h^{-1-\varepsilon}Q(n)^{-k(1+\varepsilon)}$ for all $h \ge 1$. Applying Weyl's differencing method [62] ($2 \le k \le 5$) and the Bourgain–Demeter–Guth Vinogradov mean-value theorem [3] ($k \ge 6$) bounds the normalized exponential sum at frequency $1 \le h \le H$ by $|Y_n^{-1}\sum_{y=1}^{Y_n} e(hf_r(y))| \ll_{\varepsilon_1} (h/Y_n)^{\eta_k} Y_n^{\varepsilon_1} Q(n)^{c_k}$. Substituting this into the Erdős–Turán inequality [22] $D_{Y_n}(r) \ll H^{-1} + \sum_{h=1}^{H} h^{-1}|Y_n^{-1}\sum_{y=1}^{Y_n} e(hf_r(y))|$ and choosing the optimal harmonic cutoff $H := \lfloor Y_n^{\eta_k/(1+\eta_k)}\rfloor$ balances H^{-1} and $H^{\eta_k}Y_n^{-\eta_k}$ to yield $D_{Y_n}(r) \ll_{\varepsilon_1} Y_n^{-\eta_k'+\varepsilon_1}Q(n)^{c_k} \ll X^{-\eta_k'+\varepsilon_1}Q(n)^{\eta_k'+c_k}$ with $\eta_k' = \frac{\eta_k}{1+\eta_k}$. Choosing $C_k := 2(\eta_k' + c_k)$ and $\varepsilon_1 = \delta/2$, the condition $X \ge Q(n)^{C_k/\delta}$ ensures $Q(n)^{\eta_k'+c_k} \le X^{\delta/2}$ and $D_{Y_n}(r) \ll_{k,\delta} X^{-\eta_k'+\delta}$. Summing $Y_n\frac{1}{n(n+1)} + O(Y_n D_{Y_n}(r))$ over the $Q(n)\rho_{A,k}(n)$ residue classes yields (8.13). □

9. Frey–Hellegouarch Elliptic Curves, Class Group Explosions, and Analytic Thresholds

In this final section we construct the Frey–Hellegouarch elliptic curves attached to (1.1), prove the effective Tatuzawa class number and narrow 2^J-rank bounds for $K_n = \mathbb{Q}(\sqrt{q_n})$, record the five conditional finiteness criteria (C1)–(C5) complementing Theorem 1.2, and establish the exact analytic thresholds governing Ribet's level-lowering theorem, Matveev's and Yu's theorems on linear forms in logarithms, and short character sums.

9.1. Frey–Hellegouarch Elliptic Curves, Base-Change Unit Factorizations, and Level-2 CM Reduction.

Theorem 9.1 (First-Power Factorial Frey Curve $\mathcal{E}_n$, Quadratic Base-Change Unit Factorization, and Level-2 CM Reduction)**.** *Let (n, m) be a solution to $n! + 1 = m^2$ with $n \geq 8$, so $k \in \{1, 2\}$ by Theorem 1.1.*

(i) *The elliptic curve $\mathcal{E}_n : Y^2 = X^3 + 2mX^2 + X$ over $\mathbb{Q}$ is modular and minimal and semistable at all odd primes $p \leq n$, with Weierstrass discriminant $\Delta(\mathcal{E}_n) = 64\, n!$ (minimal at $p = 2$ up to a 2^{12} Weierstrass coordinate scaling $X = 4x + 1, Y = 8y + 4x$ when $m \equiv -1 \pmod{32}$), conductor $N(\mathcal{E}_n) = 2^{\alpha} \prod_{3 \leq p \leq n} p$ ($\alpha \in \{0, 1, 2, 3\}$), and Frobenius traces $a_p(\mathcal{E}_n) = \left(\frac{-m}{p}\right)$ for all $3 \leq p \leq n$. Over $K_n = \mathbb{Q}(\sqrt{q_n})$, $\mathcal{E}_n$ acquires full K_n-rational 2-torsion and factors into units in $\mathcal{O}_{K_n}^{\times}$:*

$$\mathcal{E}_n/K_n \;:\; Y^2 \;=\; X(X + \varepsilon_n^k)(X + \varepsilon_n^{-k}). \tag{9.1}$$

(ii) *In the $k = 2$ regime, the Level-2 Frey curve $F_n^{(2)} : Y^2 = X(X - 2(X_1(q_n) - 1))(X + 4)$ has Weierstrass discriminant $\Delta(F_n^{(2)}) = 2^{12} q_n^2 Y_1(q_n)^4$ (minimal at all odd primes p, and at $p = 2$ up to a 2^{12} coordinate scaling), slashed conductor $N(F_n^{(2)}) = 2^{\alpha'} \frac{\prod_{p \leq n} p}{\mathrm{rad}_{\mathrm{odd}}(X_1(q_n))}$ ($\alpha' \in \{0, 1, 2, 3\}$), surjective mod-$\ell$ Galois representation $\bar{\rho}_{F_n^{(2)}, \ell} : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_2(\mathbb{F}_\ell)$ for every prime $\ell \geq 5$, and at every odd prime $p \mid X_1(q_n)$ reduces modulo p to the fixed CM curve $E_{\mathrm{CM}} : Y^2 = V^3 - 4V$ of conductor 32 with $j(F_n^{(2)}) \equiv 1728 \pmod{X_1(q_n)^2}$. Moreover, the coprime triple $(A_n, B_n, C_n) = (\frac{X_1(q_n)-1}{c_0}, \frac{2}{c_0}, \frac{X_1(q_n)+1}{c_0})$ has abc quality $q(A_n, B_n, C_n) \geq (\frac{1}{4 \log 2} - o(1)) \log n \to \infty$.*

Proof. (i) The Weierstrass equation $\mathcal{E}_n : Y^2 = X^3 + 2mX^2 + X$ has invariants $b_2 = 8m$, $b_4 = 2$, $b_6 = 0$, $b_8 = -1$, so

$$c_4(\mathcal{E}_n) \;=\; b_2^2 - 24 b_4 \;=\; 16(4m^2 - 3) \;=\; 16(4n! + 1), \qquad \Delta(\mathcal{E}_n) \;=\; -b_2^2 b_8 - 8 b_4^3 \;=\; 64(m^2 - 1) \;=\; 64\, n!.$$

For every odd prime $p \leq n$, we have $p \mid n!$ and $c_4(\mathcal{E}_n) \equiv 16 \not\equiv 0 \pmod{p}$, so $\mathcal{E}_n$ is minimal at p with multiplicative reduction of exact first-power factorial valuation $v_p(\Delta(\mathcal{E}_n)) = v_p(n!)$. Reducing modulo p, where $m \equiv \varepsilon_p \in \{\pm 1\} \pmod{p}$, gives $Y^2 \equiv X(X + \varepsilon_p)^2 \pmod{p}$ with node at $X \equiv -\varepsilon_p \pmod{p}$ and tangent cone $Y^2 \equiv -\varepsilon_p(X + \varepsilon_p)^2 \pmod{p}$, so $a_p(\mathcal{E}_n) = \left(\frac{-\varepsilon_p}{p}\right) = \left(\frac{-m}{p}\right)$. Moreover, $\mathcal{E}_n$ is $\mathbb{Q}$-rationally 2-isogenous via $x = Y^2/X^2, y = Y(1 - X^2)/X^2$ to $\widehat{\mathcal{E}}_n : y^2 = x(x - 2(m-1))(x - 2(m+1))$ of discriminant $2^{12}(n!)^2$, whose quadratic twist by -1 is $E_n^{(2)} : Y^2 = X(X - 2(m-1))(X + 4)$, which satisfies $a_p(E_n^{(2)}) = +1$ for all primes $p \leq n$ with $p \equiv 1 \pmod{4}$. Over $K_n = \mathbb{Q}(\sqrt{q_n})$, since $2m = \varepsilon_n^k + \varepsilon_n^{-k}$ and $\varepsilon_n^k \cdot \varepsilon_n^{-k} = 1$, the cubic factors as $X^3 + (\varepsilon_n^k + \varepsilon_n^{-k})X^2 + X = X(X + \varepsilon_n^k)(X + \varepsilon_n^{-k})$, proving (9.1).

(ii) In the $k = 2$ regime, shifting $V = X + 2$ in $F_n^{(2)} : Y^2 = X(X - 2(X_1(q_n) - 1))(X + 4)$ gives the model

$$F_n^{(2)} \;:\; Y^2 \;=\; (V^2 - 4)(V - 2X_1(q_n)) \;=\; V^3 - 2X_1(q_n)V^2 - 4V + 8X_1(q_n), \tag{9.2}$$

with root differences 4, $2(X_1(q_n) - 1)$, and $2(X_1(q_n) + 1)$, discriminant $\Delta(F_n^{(2)}) = 2^{12}(X_1(q_n)^2 - 1)^2 = 2^{12} q_n^2 Y_1(q_n)^4$, and $c_4(F_n^{(2)}) = 64(X_1(q_n)^2 + 3)$. At every odd prime $p \mid X_1(q_n)$, we have $p \nmid q_n Y_1(q_n)$, so

$F_n^{(2)}$ has good reduction at p (slashing $\mathrm{rad}_{\mathrm{odd}}(X_1(q_n))$ from the conductor $N(F_n^{(2)})$), and reducing (9.2) modulo $p \mid X_1(q_n)$ yields $Y^2 \equiv V^3 - 4V \pmod p$, which is the fixed CM elliptic curve E_{CM} of conductor 32 and j-invariant 1728; indeed,

$$j(F_n^{(2)}) - 1728 \;=\; \frac{64\,X_1(q_n)^2\,\big(X_1(q_n)^2-9\big)^2}{\big(X_1(q_n)^2-1\big)^2} \;\equiv\; 0 \pmod{X_1(q_n)^2}. \tag{9.3}$$

By the modularity theorem [63, 2], $\mathcal{E}_n$, $E_n^{(2)}$, and $F_n^{(2)}$ are modular. For any prime $p \in I_1 = (n/2, n]$, $v_p(j(F_n^{(2)})) = -2 < 0$, so $j(F_n^{(2)}) \notin \mathbb{Z}$. Since $F_n^{(2)}(\mathbb{Q})[2] \simeq (\mathbb{Z}/2\mathbb{Z})^2$, Mazur's and Kenku's rational isogeny theorems [39, 31] imply that $\bar\rho_{F_n^{(2)},\ell}$ is irreducible for every prime $\ell \geq 5$, and Tate's p-adic uniformization at $p \in I_1$ shows that inertia I_p acts via a non-trivial transvection of order ℓ, forcing $\mathrm{Im}(\bar\rho_{F_n^{(2)},\ell}) = \mathrm{GL}_2(\mathbb{F}_\ell)$ by Serre's open image theorem [51]. Furthermore, for $(A_n, B_n, C_n) = (\frac{X_1(q_n)-1}{c_0}, \frac{2}{c_0}, \frac{X_1(q_n)+1}{c_0})$, Theorem 4.2(i) gives $\log \mathrm{rad}(A_nB_nC_n) = \log \mathrm{rad}(2q_nY_1(q_n)) \leq (\log 2 + o(1))n$ while $\log C_n = (\frac14 + o(1))n\log n$, yielding $q(A_n,B_n,C_n) \geq (\frac{1}{4\log 2} - o(1))\log n \to \infty$, while writing $n! = u_n^3 v_n$ (v_n cube-free) and setting $x_n = u_nv_n, y_n = mv_n$ gives $0 < y_n^2 - x_n^3 = v_n^2 \leq x_n^{\frac{12+o(1)}{\log n}}$. □

9.2. **Effective Class Group Explosions and Conditional Finiteness under (C1)–(C5).**

Theorem 9.2 (Effective Tatuzawa Class Number Explosion and Conditional Finiteness under (C1)–(C5))**.** *Let $\varepsilon \in (0, 1/8)$.*

(i) ***Effective Tatuzawa Class Number Explosion and Narrow 2^J-Rank Dichotomy:*** *With at most one exceptional integer $n_* \in \mathbb{N}$, every solution $n \geq 8$ to $n! + 1 = m^2$ satisfies the effective non-genus class number lower bound*

$$h_{\text{non-gen}}(q_n) \;:=\; \frac{h(q_n)}{2^{\omega(D_n)-2}} \;>\; \frac{0.655\,\varepsilon\,k\,q_n^{1/2-\varepsilon}}{n\log n\cdot 2^{\pi(n)}} \;=\; q_n^{1/2-\varepsilon-o(1)}, \tag{9.4}$$

and in the narrow class group $\mathrm{Cl}^+(q_n) \simeq \mathcal{A}_2(q_n) \oplus \mathrm{Cl}^+_{\mathrm{odd}}(q_n)$, either $|\mathrm{Cl}^+_{\mathrm{odd}}(q_n)| \geq q_n^{1/4-\varepsilon}$, or $\exp(\mathcal{A}_2(q_n)) \geq n^{\frac{1}{4\log 2}-\varepsilon}$ and $\mathrm{rank}_{2^J}(\mathcal{A}_2(q_n)) \geq (\frac18 - \varepsilon)\frac{n}{\log n}$ for all $1 \leq J \leq \frac{\log n}{8\log 2}$.

(ii) ***Conditional Finiteness under (C1)–(C5):*** *In addition to Greenberg's conjecture (Theorem 1.2), the equation $n! + 1 = m^2$ has only finitely many solutions (n, m) if, for all sufficiently large n, at least one of the following five standard hypotheses holds:*

- ***(C1)*** *($\mathbb{F}_2$ Rédei–Artin–Weight / 2-adic Laplacian inconsistency): $\ker_{\mathbb{F}_2}(M_n^{(0)})$ contains no vector passing the Level-1 system $M_n^{(1)} y_{\mathcal{L}_1^+(n)} = \mathbf{b}_n^{(1)}$ and $R_n^2 = Q_n$;*
- ***(C2)*** *(Short character sum cancellation): $\sum_{p\in(n/2,n]} \psi_{\min}(p)\log p < (\frac{1-\log 2}{2} - \delta)n$ or non-vanishing of the octic Artin indicator $\Psi_n(p) := \big(1 + \left(\frac{2}{p}\right)\big)\big(1 - \left(\frac{-\delta_1^{\mathrm{mid}}(n)}{p}\right)\big)\big(1 - \left(\frac{\delta_2^{\mathrm{mid}}(n)}{p}\right)\big)$ on $3 \leq p \leq n$ [26];*
- ***(C3)*** *(Rank-of-apparition growth): $\max_{p\in(n/3,n/2]} \alpha_p(q_n) \geq 3$ [47];*
- ***(C4)*** *(Cohen–Lenstra–Gerth–Smith non-genus bound or CF period growth): $h_{\text{non-gen}}(q_n) \leq q_n^{1/2-\delta}$ or $L(q_n) \gg n(\log n)^{1+\delta}$ [15, 25, 55];*
- ***(C5)*** *(Modular trace exclusion, Szpiro's conjecture, Weak abc, or Weak Hall conjecture) [42, 57, 28].*

Proof. (i) By Dirichlet's analytic class number formula [18, Chapter 6], $h(q_n)R(q_n) = \frac12\sqrt{D_n}L(1,\chi_{D_n})$. By Tatuzawa's effective refinement of Siegel's theorem [52, 58], for any $\varepsilon \in (0, 1/8)$, $L(1,\chi_D) > 0.655\,\varepsilon\,D^{-\varepsilon}$ holds for all fundamental discriminants $D > 0$ with at most one exception D_*. By Theorem 1.1, distinct solutions $n_1 < n_2$ have distinct square-free cores $q_{n_1} \neq q_{n_2}$, so $D_n = D_*$ can occur for at most one solution $n_* \in \mathbb{N}$. Substituting $\sqrt{D_n} \geq \sqrt{q_n}$, $L(1,\chi_{D_n}) > 0.655\,\varepsilon\,(4q_n)^{-\varepsilon}$, and $R(q_n) \leq \log \varepsilon_n \leq \frac{1}{2k} n\log n$ yields (9.4). In $\mathrm{Cl}^+(q_n) \simeq \mathcal{A}_2(q_n) \oplus \mathrm{Cl}^+_{\mathrm{odd}}(q_n)$, Gauss's genus theory gives $\mathrm{rank}_2(\mathcal{A}_2(q_n)) = \omega(D_n) - 1 = (\log 2 + o(1))\frac{n}{\log n}$; thus if $|\mathrm{Cl}^+_{\mathrm{odd}}(q_n)| < q_n^{1/4-\varepsilon}$, then $\log_2 |\mathcal{A}_2(q_n)| \geq (\frac14 -$

$\frac{\varepsilon}{2})\log_2 q_n = (\frac{\log 2}{4} - \frac{\varepsilon}{2} - o(1))n$, which forces both $\log_2 \exp(\mathcal{A}_2(q_n)) \geq \frac{\log_2 |\mathcal{A}_2(q_n)|}{\operatorname{rank}_2(\mathcal{A}_2(q_n))} \geq (\frac{1}{4\log 2} - \varepsilon)\log n$ and $\operatorname{rank}_{2^J}(\mathcal{A}_2(q_n)) \geq (\frac{1}{8} - \varepsilon)\frac{n}{\log n}$ for all $1 \leq J \leq \frac{\log n}{8\log 2}$.

(ii) For large n, (C1) directly negates Lemma 5.3(i) and Theorem 4.2(i); (C2) directly negates Proposition 5.1(ii); (C3) directly negates Proposition 3.4; (C4) directly negates (9.4) or Theorem 1.1; and (C5) directly negates Theorem 9.1. □

9.3. **Exact Analysis of the Ribet Level-Lowering, Matveev–Yu Linear Forms in Logarithms, and Short Character Sum Thresholds.** We conclude by quantifying the exact analytic thresholds separating our unconditional structural theorems from an unconditional proof of (C1)–(C5).

Proposition 9.3 (The Ribet Level-Lowering, Matveev–Yu Logarithmic Rank, and Short Character Sum Thresholds)**.** *Let (n, m) be a large solution to $n! + 1 = m^2$ ($k \in \{1, 2\}$).*

(i) ***Ribet's Level-Lowering Valuation Barrier on*** $I_1 = (n/2, n]$***:*** *Across the Hellegouarch curve $E : Y^2 = X(X - q_n Y_1^2)(X + 1)$, its three 2-isogenous curves $\mathcal{E}_n = E'_a$ ($\Delta = 64\, n!$), E'_b ($\Delta = 256 q_n Y_1^2 X_1^8$), and E'_c ($\Delta = 256 X_1^2 q_n^4 Y_1^8$), as well as $E_n^{(2)}$ ($\Delta = 2^{c_1}(n!)^2$) and $F_n^{(2)}$ ($\Delta = 2^{c_2} q_n^2 Y_1^4$), every prime $p \in I_1 = (n/2, n]$ satisfies $v_p(n!) = v_p(q_n) = 1$ and $v_p(X_1) = v_p(Y_1) = 0$, so*

$$v_p(\Delta_{\min}) \ \in \ \{1, 2, 4\} \qquad \forall\, p \in I_1 = (n/2, n]. \tag{9.5}$$

Because $v_p(\Delta_{\min}) \in \{1, 2, 4\}$ is coprime to every odd prime $\ell \geq 3$ (while for $\ell = 2$ full rational 2-torsion makes $\bar{\rho}_{E,2}$ reducible), Ribet's level-lowering theorem [46] cannot strip any prime $p \in I_1 = (n/2, n]$ from the conductor of $\bar{\rho}_{E,\ell}$ ($\ell \geq 5$), forcing the lowered modular level N_ℓ to satisfy

$$N_\ell \ \geq \ \frac{1}{\ell} \prod_{n/2 < p \leq n} p \ = \ \exp\!\left(\left(\frac{1}{2} - o(1)\right) n\right). \tag{9.6}$$

(ii) ***The Matveev–Yu Linear Forms in Logarithms Rank Barrier and Rank-1 Collapse:*** *In the rational linear form $\Lambda_1 := \log \frac{m+1}{m-1} = \frac{2}{m-1} + O(m^{-3})$ ($\log |\Lambda_1|^{-1} = (\frac{1}{2} + o(1))n \log n$), the prime factors of $m - 1$ and $m + 1$ generate a multiplicative subgroup of $\mathbb{Q}^\times$ of rank*

$$N \ = \ \omega(m-1) + \omega(m+1) - 1 \ \geq \ n^{1/2 - o(1)} \tag{9.7}$$

by Theorem 5.2(i) (with explicit Mertens lower bound $N \geq (6e^{\gamma - 1/2} - o(1))\frac{\sqrt{n}}{\log n}$), so the rank factor $C^N N^N \prod_{i=1}^N \log p_i \geq \exp(n^{1/2 - o(1)})$ in Matveev's archimedean theorem [38] and Yu's p-adic theorem [64] exceeds $\log |\Lambda_1|^{-1} \asymp \frac{1}{2} n \log n$. Passing to $\mathbb{Z}[\sqrt{q_n}]^\times = \langle -1, \varepsilon_n \rangle$ collapses the multiplicative rank to 1, where $m + s_n \sqrt{q_n} = \varepsilon_n^k$ ($k \in \{1, 2\}$) holds identically, while in the $k = 2$ rank-2 rational form $\Lambda_2 = \log(A_n/B_n) - 4\log Z = \log(1 + \frac{1}{B_n Z^2}) \asymp \frac{2}{\sqrt{n!}}$, the coprime rational number $\alpha_n = A_n/B_n$ has logarithmic height $h(\alpha_n) = \max(\log A_n, \log B_n) = (\frac{1}{2} + o(1))n \log n \asymp \log |\Lambda_2|^{-1}$.

(iii) ***The Short Character Sum Logarithmic Scale*** $x = n \asymp \log Q_{\min}$ ***vs. Consecutive*** $Q \leq 4n(n+1) \ll x^2$***:*** *For a single index n, the character $\psi_{\min}$ in Proposition 5.1(ii) has conductor $Q_{\min} < 8\sqrt{q_n} = 2^{\frac{1}{2}n + o(n)}$, so the prime sieve endpoint $x = n \asymp \frac{2}{\log 2} \log Q_{\min}$ lies at the logarithmic scale relative to $Q_{\min}$ (below the unconditional Burgess threshold $Q_{\min}^{1/4+\varepsilon}$ [13] and the GRH Lagarias–Odlyzko threshold $(\log Q_{\min})^2 \asymp n^2$ [34]). By contrast, for consecutive solutions $(n, n+1)$, the four characters $\chi_X, \chi_Y, \chi_Z, \chi_W$ of Theorem 6.1(iii) have polynomial conductor $Q_i \leq 4n(n+1) \ll x^2$.*

Proof. For (i), Vélu's 2-isogeny formula applied to the three rational 2-torsion points of $E : Y^2 = X(X - A)(X + B)$ ($A + B = C$) gives $\Delta(E') = -256ABC^4$, yielding the discriminants of E'_a, E'_b, E'_c upon permuting $(q_n Y_1^2, 1, X_1^2)$. On $I_1 = (n/2, n]$, $v_p(q_n) = 1$ and $v_p(X_1) = v_p(Y_1) = 0$, giving (9.5), and Ribet's theorem [46] requires $\ell \mid v_p(\Delta_{\min})$ to lower the level at a prime p of multiplicative

reduction, proving (9.6). Parts (ii) and (iii) follow directly from Theorem 5.2(i), Theorem 4.2(i), Proposition 5.1(ii), and Theorem 6.1(iii). □

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